Same Shock, Separate Channels: House Prices and Firm Performance in the Great Recession

by

G. Jacob Blackwood
Amherst College

CES 26-03 **January 2026**

The research program of the Center for Economic Studies (CES) produces a wide range of economic analyses to improve the statistical programs of the U.S. Census Bureau. Many of these analyses take the form of CES research papers. The papers have not undergone the review accorded Census Bureau publications and no endorsement should be inferred. Any opinions and conclusions expressed herein are those of the author(s) and do not represent the views of the U.S. Census Bureau. All results have been reviewed to ensure that no confidential information is disclosed. Republication in whole or part must be cleared with the authors.

Papers in the series are posted on www.census.gov/library/working-papers/series/ces-wp.html. For information about the series, contact Danielle H. Sandler, Editor, Discussion Papers, U.S. Census Bureau, Center for Economic Studies, 4600 Silver Hill Road, Washington, DC 20233, CES.Working.Papers@census.gov.

Abstract

Combining confidential business-level microdata with housing and banking data, I document large and persistent effects of local house prices on employment at small businesses, and particularly young businesses, during the Great Recession. I show that the effect on entry is important for explaining the disproportionate effect on young businesses, while young firm exit is also disproportionately affected. I then explore the channels through which house prices affect business outcomes. I use survey data to show that reliance on either personal assets or home equity is associated with increased sensitivity to house prices. I then use local bank balance sheet information to show both young and old firms are sensitive to local credit shocks, with some evidence of a larger effect on young businesses. I develop a macroeconomic model that is consistent with these findings where house prices work through two channels: a bank credit supply channel and a housing collateral channel.

Keyword: Firm Dynamics, House Prices, Credit Supply, Business Cycles

JEL Classification: E44, D22, D25, R31

* Department of Economics, Amherst College, Amherst, MA, 01002. Email: jblackwood@amherst.edu. I am indebted to John Haltiwanger, Sebnem Kalemli- Ozcan, and Boragan Aruoba for helpful guidance and support. I also thank the anonymous referees, Luminita Stevens, John Shea, and participants at the University of Maryland macroeconomics seminar, Spring 2017 Midwest Macro Meetings, 2017 FSRDC Conference, and the U.S. Census Bureau CES seminar. All errors are mine. Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

1 Introduction

House prices in the United States collapsed by roughly 25 percent from 2007 to 2012, precipitating the largest financial crisis since the Great Depression. Recent literature has shown the impact of the ensuing recession fell disproportionately on young firms (Fort et al., 2013), but the reasons for this differential effect are still largely undetermined (Davis and Haltiwanger, 2024). This paper sheds further light on the differential effect of house prices across the age/size distribution, and provides new evidence on the mechanisms through which house prices impacted different firm types. I find substantial causal effects of house prices on firm employment that are very persistent, and disproportionate effects on young firms that are somewhat less persistent. I document an important role for both firm entry and exit. I also provide new evidence that firm reliance on personal finance, common among young firms, amplifies sensitivity to house prices. Furthermore, I show firms are sensitive to local bank exposure to house prices, capturing a previously underexplored local credit supply effect. I then quantify the effect of house prices on aggregate outcomes in a model where house prices impact firms through two channels–a credit supply and a collateral channel.

My empirical work extends previous literature documenting the excess sensitivity of young firms to the business cycle in general (Fort et al., 2013; Sedlàček and Sterk, 2017) and local house prices during the Great Recession in particular (Davis and Haltiwanger, 2024). While these papers primarily use the publicly available data, I combine confidential data in the Longitudinal Business Database (LBD) from the US Census Bureau with house price data from Freddie Mac (2007-2012). I am thus able to control for additional micro-level heterogeneity for the near-universe of firms, which is not possible with publicly available data. I find that house prices have persistent effects on both young (less than 5 years old) and old/small firms (at least 5 years old and less than 500 employees), with cumulative effects lasting beyond 5 years.[1] As discussed in Davis and Haltiwanger (2024), young firms are particularly sensitive to house price shocks, although this excess sensitivity dissipates after 3 years at most. Excluding entrants from my sample reduces the effect for young firms substantially, suggesting much of the excess sensitivity of young firms can be attributed to

[1] Young businesses are overwhelmingly small, so I do not split them by size.

entry, consistent with Sedlàček and Sterk (2017). On the other hand, exit probability is also sensitive to house prices, and particularly for young firms.

I then extend this analysis by exploring channels through which local house prices affect firms. I focus on two channels: collateral and credit supply. The first channel captures how lower home values can affect an entrepreneur's entry and hiring decisions if housing and personal assets can be used as collateral to finance a business. Evidence for this collateral mechanism includes Lastrapes, Schmutte, and Watson (2022) and Kerr, Kerr, and Nanda (2022), who explore the importance of housing collateral for entrepreneurship and small businesses, finding modest but potentially lasting effects.[2] This paper contributes uniquely to this the literature by providing evidence of firm sensitivity to house prices in the Great Recession based on their history of reliance on home equity and personal savings. To my knowledge, there has been little empirical work exploring the relevance of this mechanism during the crisis. I use data from the Survey of Business Owners (SBO) from the US Census Bureau (2007,2012), documenting substantial reliance on housing and personal assets for collateral and financing among young firms. Businesses that relied on these forms of credit were excessively sensitive to house prices, suggesting the collateral mechanism potentially played a key role in the propagation of the house price decline during the Great Recession.

House price shocks can also impact local credit supply—the second channel. Recent literature documents the effect of credit supply restriction on firm outcomes. Chodorow-Reich (2014) and Greenstone, Mas, and Nguyen (2020) demonstrate the impact of banking market variation for business outcomes. Small firms are generally found to be susceptible to financial conditions during the Great Recession (Chodorow-Reich, 2014; Siemer, 2019), particularly young firms (Chodorow-Reich, 2014; Siemer, 2019; Davis and Haltiwanger, 2024). However, the role of smaller banks in propagating the crisis through credit supply remains relatively unexplored relative to large banks (Greenstone, Mas, and Nguyen, 2020) and syndicated loan markets (Chodorow-Reich, 2014). This paper fills that gap by exploring the role of local banks.[3] These banks are particularly exposed to local housing conditions through their

[2]Evidence from Harding and Rosenthal (2017), Corradin and Popov (2015), Fairlie and Krashinsky (2012), and Schmalz, Sraer, and Thesmar (2016) all corroborate an important role for housing collateral.

[3]Several studies show business formation does depend on local banking markets outside of the financial crisis (Black and Strahan, 2002; Cetorelli and Strahan, 2002; and Kerr and Nanda, 2009). With respect to

balance sheets. As housing market conditions deteriorate, bank balance sheets also deteriorate, leading to credit restrictions for businesses relying on local banks. To capture this local bank credit supply channel, I integrate publicly available data on banks from FDIC call reports into my data (Federal Deposit Insurance Corporation, 2005-2012). I focus on house price interactions with *pre*-crisis bank balance sheets in the state, which proxy for local bank "health" or "preparedness" prior to the crisis and capture how local house price shocks are amplified or mitigated by local banking conditions. I find meaningful effects of this credit supply channel on old/small and young businesses, with some evidence of a stronger effect on young businesses. However, this effect does not appear to influence entry disproportionately.

I then construct a model that is consistent with these empirical findings, capturing the importance of credit supply as well as collateral, allowing for impacts on entry and exit margins. The model has two main channels through which house prices affect firms: a *credit supply* channel, where bank exposure to housing generates higher credit spreads, and a *collateral channel*, as firms can borrow against housing. As identified in my empirical work, bank balance sheets interact with house prices to generate tighter credit conditions–an increase in interest rates and/or spreads–which in turn affects firms' ability to invest and hire.[4] Likewise, I incorporate a channel that limits firm credit demand through housing collateral values (similar to Decker, 2015). Models of financial frictions often feature collateral constraints that depend on capital collateral values (e.g. Kiyotaki and Moore, 1997; Cooley, Marimon, and Quadrini, 2004). In my model, both capital and housing are used as collateral.

The collateral channel is important for explaining the differential impact of a housing crisis on entrants and young firms in the model, while the credit supply channel matters for both young and old businesses. Young firms tend to be constrained, with constraints either binding or close to binding, and thus are more sensitive to tightening in the constraint due to falling house prices. The effect of both channels on aggregate outcomes is substantial, and the

the Great Recession, Davis and Haltiwanger (2024) note the influence of house prices on young firms could be (in part) due to "local economic fortunes affect[ing] the lending capacity of local banks", which they are not able to capture with their empirical approach.

[4]My paper closely follows Gertler and Kiyotaki (2015) by imposing an internal incentive constraint on banks. However, rather than expose banks to capital quality or productivity shocks through equity claims on capital, banks have housing equity on their balance sheets.

interaction of the two channels generates an even more severe recession. For policymakers, the differntial impact of these two channels could matter for the design of interventions.

In Section 2, I present the empirical framework and results. I detail the model in 3, discuss calibration in 4, present results and counterfactuals in 5, and conclude in Section 6.

2 Empirical Analysis

In my empirical work, I first document the disproportionate effect of local house prices on young firm employment, consistent with previous literature. I build on these findings to show that the disproportionate effect on young firms persists somewhat, but does not worsen, in subsequent periods. I then document the substantial role of entry and the impact of house prices on firm exit at various time horizons.

With these results in hand, I then seek to disentangle channels hypothesized in previous literature. In particular, I explore the role of personal collateral, including home equity lines of credit, on firm sensitivity to house prices. I interact house price growth with indicators of sources of startup and continuing financing constructed from the SBO. While there certainly could be selection effects, these results provide evidence for the bank credit supply channel and the housing collateral channel, which I capture in my quantitative model.

Finally, I aim to identify whether *local* house price changes influence *local* businesses through *local* banks, to see whether the outsize effect of house prices on young businesses is partly attributable to local banking conditions.[5] At the heart of the strategy are interactions of annual house price changes with *pre-crisis* indicators of bank balance sheets. By focusing on balance sheets prior to the crisis, I am relying on an assumption that bank balance sheet structure, and bank health in general, prior to 2007 did not anticipate the subsequent declines in housing prices over the Great Recession. I discuss this assumption more in Section 2.2.1. Additionally, there must be disproportionate exposure of local banks to local house prices, which in turn has a significant impact on bank health. This leads banks to restrict their

[5]This approach relies on geographic segmentation in lending markets. Survey evidence indicates small businesses borrow from nearby banks (Petersen and Rajan, 2002; Brevoort, Holmes, and Wolken, 2009), consistent with theoretical work on the acquisition of soft information (Boot and Thakor, 1994; Agarwal and Hauswald, 2010). Further, the evidence of banking relationships during expansions and recessions (Petersen and Rajan, 1994; Chodorow-Reich, 2014), suggest that such segmentation is important.

lending. Then, the interaction of house prices with the pre-crisis bank balance sheets would serve as a proxy for changes in local bank credit supply.

2.1 Data

I combine the Longitudinal Business Database (LBD) from the US Census Bureau with public data on bank balance sheets from FDIC call reports, house price data from the Federal Home Loan Mortgage Corporation (Freddie Mac), and indicators of sources of financing from the Survey of Business Owners (SBO). The LBD is an annual dataset derived from administrative records that covers the near-universe of establishments in the US economy. It includes a measure of employees as of March 12 of the year and annual revenue. Furthermore, the dataset contains data on location, industry, multi-unit status, and age.[6] I focus on employment growth at the local level within a firm. That is, the unit of observation in each specification is a firm's establishments within a metro area and industry. This allows me to focus on the response of the firm to local shocks impacting its establishments, without overweighting firms with multiple establishments in a metro/industry bin. To simplify exposition, I will refer to the unit of observation as an establishment. I explore heterogeneity by firm demographic groups: young, old, old/small, and old/large. Age and size are defined at the *firm* level and assigned to all establishments at the firm regardless of location or industry. Following Davis and Haltiwanger (2024), I drop sectors dominated by and nonprofits (NAICS 61, 622, 624, an 813) and sectors directly related to housing and financial markets (NAICS 52 and 53). I consider additional industry restrictions in Section 2.5.

I obtain metro house prices from Freddie Mac (2007-2012) and unemployment from the Bureau of Labor Statistics (2007-2012). I begin my sample at the onset of the recession in 2008. Since my focus is on house prices, which continued to decline until the second quarter of 2012, I end my sample in 2012.[7] I use year-over-year changes in March to line up with LBD employment timing.[8]

[6]Further detail on the data construction process for the LBD, including cleaning, can be found in the data appendix of Chow et al. (2021). I use the 2022 vintage.

[7]Additionally, bank exits, a key variable in my analysis, rose sharply beginning in 2008 and continued at an elevated pace until at least 2012 as well.

[8]Previous drafts used FHFA county-level house prices along with Saiz (2010) elasticities, which could be appropriate if house prices affect businesses at a more granular level. However, Freddie Mac house prices

To capture the role of business owner assets, I construct a sample with information on sources of finance by merging firms in each of the years in 2008-2012 from the LBD with the 2007 and 2012 SBO. The SBO captured data from over 2.3 million employer firms in 2007 and 1.75 million employer firms in 2012, according to the US Census Bureau (2007,2012). In my analysis, I include establishments of firms that show up in either year, prioritizing the matches from the 2007 SBO, which better indicates pre-crisis financial conditions. Comparing sample counts, I match about 25% of the main sample. All results pertaining to the SBO sample are sample-weighted to account for selection into the survey frame. I document the difference in responsiveness to house prices in this subsample relative to the main sample, and find both display similar estimates.

Most of the previous literature on credit supply effects in the Great Recession cover large and publicly traded firms.[9] To develop a measure of financial conditions in the local market of the establishment that applies for a broader set of firms, I use data from the Federal Deposit Insurance Corporation (2005-2012) call report database, providing detailed information on assets, liabilities, locations, and ownership history. I take annual averages of these quarterly data and roll up banks to the ownership level, or bank holding company level. Consistent with the theory of financial market segmentation mentioned above, I restrict to banks that only have branches in one state and bank holding companies with one bank. These banks range from several million dollars in assets to large community banks of well over a billion dollars. Figure OA1 in the online appendix illustrates their importance, as they accounted for roughly 10% of business loans in the 2000s and over 30% of small business loans, a focus of this paper.[10] I roll each variable up by state, creating a representative "state bank

are preferred to FHFA house prices because the montly (rather than annual) frequency allows for better synchonization with the March 12 date of measurement in the LBD. Furthermore, he supply elasticities were developed at the metro level and were created using this house prices series. Nonetheless, I include the tables from the previous analysis in the online appendix, which is largely consistent with the main results.

[9]Dinlersöz et al. (2018) uses the ORBIS dataset to cover a larger portion of small and young firms. Chodorow-Reich (2014) documents banking relationships among larger firms in the syndicated loan market.

[10]I restrict to banks, further removing large banks, such as credit card companies, that only have one reported branch, but are obviously not local state banks. Due to the small number of state banks in Delaware, the District of Columbia, and New Hampshire, I merge these states with Pennsylvania, Maryland, and Vermont, respectively.

balance sheet" to characterize variation in balance sheet structure across states, focusing on mortgage shares of total assets.

2.2 Empirical Framework

In line with previous literature, I regress the establishment $e's$ employment growth $\Delta Y_{e,t}$ as calculated in Davis, Haltiwanger, and Schuh (1996), henceforth DHS growth rates, from time $t-1$ to t on the log change in metro area m house prices $\Delta HPI_{m,t}$ from $t-1$ to t:[11] Furthermore, I consider "local projections" of growth in subsequent h periods.

$$\Delta Y_{e,t+h} = \beta_{\mathbf{h}} \mathbf{\Delta HPI_{m,t}} + \alpha_{m,h} + \delta_{r,h} \times \delta_{t,h} + \tau_{i,h} + \mu_h size_{e,t} + \epsilon_{e,t+h}. \tag{1}$$

I include time effects $\delta_{t,h}$ interacted with Census-defined region r effects $\delta_{r,h}$, and metro area fixed effects $\alpha_{m,h}$. I also include detailed (6-digit NAICS) industry fixed effects $\tau_{i,h}$, and the log of the denominator of the DHS growth rate as a size control. I cluster at the metro area. Due to the apparent nonlinearities in age and size, as stressed by Fort et al. (2013) and Dinlersöz et al. (2018), the results are broken out by four firm categories: young (less than 5 years old), old (5 years or older), old/small (at least 5 years old and less than 500 employees), and old/large (at least 5 years old and at least 500 employees), where age and size are defined at the firm level.[12]

House prices are endogenous, since positive firm outcomes are likely to positively impact local housing markets. So, I implement the instrumental variable approach using the housing supply elasticities from Guren et al. (2021). Conceptually, the predicted value can be thought of as the predicted value from the following regression:[13]

$$\Delta HPI_{m,t} = \Gamma \Delta HPI_{US,t} \eta_m + a_m + d_r \times d_t + \varepsilon_{m,t}, \tag{2}$$

[11] Note this measure is inclusive of entry and exit, and is calculated as the change in employment over the average of employment in the two periods, $\Delta Y_t = \frac{Y_t - Y_{t-1}}{(Y_t + Y_{t-1})/2}$. I use the denominator as the control for size.

[12] I report a fully interacted alternative in Table OA3 of the online appendix.

[13] Note this is run at the establishment level. I omit the establishment and subscripts regressors in (2) (which are included in the first stage) for simplicity since house prices are invariant within metro. Advantages of this instrument, including additional power and robustness to the critique in Davidoff (2016) are discussed at length in Guren et al. (2021). In addition to this approach, I also document in online Appendix D.5 previous analysis using the Saiz (2010) instrument, which yields similar results.

where η_m is a metro-level measure of housing supply elasticities calculated in Guren et al. (2021), a_m is a metro area fixed effect, and $d_r \times d_t$ is the appropriate interaction of region and time effects discussed in Guren et al. (2021).The identification assumption is that predicted house price changes $\widehat{\Delta HPI}_{m,t} = \widehat{\Gamma}\Delta HPI_{US,t} * \eta_{m,t}$ are orthogonal to the error term in the second stage: $\epsilon_{e,t}$. The predicted house price changes are then the explanatory variables in the second stage (1).

I capture potential collateral effects by merging the dataset with the SBO from the US Census Bureau. I modify equation (1) with interactions of house prices with indicators of entrepreneur credit access. Specifically, I interact house prices with two indicators. First, I include an indicator of whether home equity loans (HELOCs) were used to finance at startup or in the reference year for either the 2007 or 2012 survey. Second, I include a similarly defined indicator for personal savings and assets. Personal wealth and housing collateral are often used to secure financing for enterprises (see Table 2), and thus a shock to housing values can affect business outcomes directly.[14] Formally:

$$\Delta Y_{e,t+h} = \beta_{\mathbf{q,h}}\mathbf{\Delta HPI_{m,t}} + \beta_{\mathbf{HLC,h}}\mathbf{\Delta HPI_{m,t}} * \mathbf{HELOC_e} + \beta_{\mathbf{SVG,h}}\mathbf{\Delta HPI_{m,t}} * \mathbf{SAVING_e} \tag{3}$$

$$+HELOC_e + SAVING_e + \alpha_{m,h} + \delta_{r,h} \times \delta_{t,h} + \tau_{i,h} + \mu_h size_{e,t} + \epsilon_{e,t+h}.$$

Here, focus in on the marginal effect of relying on either home equity loans or personal savings/assets at each horizon h after a house price shock, indicated by $\beta_{HLC,h}$ and $\beta_{SVG,h}$, respectively. Positive values of these coefficients indicate that establishments of firms that are reliant on these forms of finance are more sensitive to house prices. As before, house prices are instrumented using the Guren et al. (2021) instrument, and observations are weighted by SBO sample weights to correct for the nonrepresentative sample.

Finally, following Cole and White (2012), who show the predictive power of a set of bank balance sheet ratios on bank failure during this period, I interact house prices with an indicator of state bank balance sheets in 2005. The use of a pre-crisis snapshot of bank

[14] In this sense, a house price shock impacts total assets, which may restrict borrowing capacity even if the firm is not directly borrowing against their house.

balance sheets captures bank preparedness, and thus their tendency to adjust credit supply in response to house price shocks. I use a simple framework that directly includes the interaction between house prices and the mortgage share of bank balance sheets within the state. The specification adds to (1) the interaction between house prices and the share of mortgages on state bank balance sheets:

$$\Delta Y_{e,t+h} = \beta_{\mathbf{q,h}} \mathbf{\Delta HPI_{m,t}} + \beta_{\mathbf{m,h}} \mathbf{\Delta HPI_{m,t}} * \mathbf{MTG_{s,2005}} + \alpha_{m,h} + \delta_{r,h} \times \delta_{t,h} + \tau_{i,h} + \mu_h size_{e,t} + \epsilon_{e,t+h}. \tag{4}$$

I instrument for house prices as before, and further instrument the interaction of house prices and mortgage shares with interactions between housing price elasticities, national house prices, and the state mortgage share.[15]

2.2.1 Identification Discussion

The key challenges to my identification strategy are the endogeneity of house prices and the plausibility of the assumption the financial indicators are independent of other factors related to firm performance in the Great Recession. To address house price endogeneity, I use the approach in Guren et al. (2021), and include some additional robustness checks using the Saiz (2010) instrument in the online Appendix D.5. This approach requires that predicted local house price growth from national house price growth interacted with local housing supply elasticities be uncorrelated with the error term in the second stage. In subsequent regressions, whenever house prices are interacted with pre-crisis bank balance sheets in OLS regressions, IV specifications likewise use similar interactions in the first stage.[16]

The second challenge is that pre-crisis bank balance sheets are correlated with other determinants of employment outcomes during the crisis. My identification strategy relies on geographic segmentation–that businesses tend to borrow from banks in their area. Survey evidence indicates this is the case for small businesses (Petersen and Rajan, 2002; Brevoort,

[15] Additionally, in online Appendix D.1 I consider a two-stage setup where bank health is proxied by bank exit. The results are qualitatively similar, but less precise, so I leave discussion to the appendix.

[16] The instrument is not without criticism, and Howard and Liebersohn (2025) show that the instrument is correlated with income, opening the door for the instrument to potentially pick up differential effects on locations based on income.

Holmes, and Wolken, 2009), consistent with theoretical work on the acquisition of soft information (Boot and Thakor, 1994; Agarwal and Hauswald, 2010). However, one would not expect large businesses to be as constrained to their local banks, and most would have access to broader financial markets. My results are consistent with this interpretation, since I find that results for old businesses are driven by old/small businesses, and large businesses are generally not sensitive to local credit conditions.

Another concern is that pre-crisis mortgage shares are correlated with "bubbly" housing markets: those states with high mortgage shares are also those with the largest runup and subsequent fall in house prices. Figure OA1 (b) in the online appendix shows this is not the case. House price declines in the 2007-2012 period are not significantly correlated with the pre-crisis mortgage share. Additionally, it could be that states with banks that held a large share of mortgages generally make poor lending decisions. In that case, banks with high mortgage shares would also experience higher default and delinquency rates on *commercial* loans during the Great Recession. Figure OA1 (c) in the appendix demonstrates this is not the case–average state delinquency rates over the Great Recession are uncorrelated with mortgage shares among banks in my sample.[17]

A related concern for the mortgage approach is that many banks were engaged in securitization. The mortgage variable here is individual single-family mortgages held on bank balance sheets, not packaged security products of multiple mortgages. It is possible banks planned to sell these mortgages, but an assumption that these banks were not involved in securitization is not necessary. It is sufficient to assume banks maintained similar shares of individual mortgages *that were disproportionately exposed to local house prices* between 2005 and the onset of the recession. This is possible if, for example, banks are replacing mortgage sales with newly issued mortgages.

Finally, reliance on personal assets and home equity loans could be correlated with excess house price sensitivity due to selection, as businesses that engage in activities or business models that vary more with housing cycles might be screened from external financing. Although some of the selection issues can be captured by industry and location effects, as well

[17]Similarly, Cole and White (2012) generally find mortgages to be negatively associated with bank failure, suggesting that it is not a measure that typically captures poor bank management.

as the fixed effects for home equity and personal finance included in the regression, it is likely some of the marginal effect captured by $\beta_{HLC,h}$ and $\beta_{SVG,h}$ is due to selection. Still, the estimates of these coefficients are large, and at a minimum suggest that reliance on home equity and personal finance are a potential channel through which house prices affected business outcomes in the Great Recession.

2.3 Results

In what follows, I establish the correlation between housing values and firm performance briefly before discussing the alternatives for identifying a local credit channel. In each table, I present results by young, old, old/small, and old/large.

2.3.1 House Prices and Employment Growth

I begin with baseline regressions of establishment (within-firm/industry/metro) employment growth on house price growth, as in equations (1) and (2). Results show a strongly significant relationship between house prices and employment growth for establishments of young, old, and old/small firms. The smaller coefficient on old/large businesses suggests they are less sensitive to local conditions, as expected. The IV estimates, at the bottom of Table 1, show an increased point estimate on the house price coefficient that is still significant for both young and old. The point estimate for old businesses is driven by small firms. Overall, the results are economically meaningful for old/small and young firms, with a 1 percent increase in house prices leading to a roughly 0.27-0.29 percentage point increase in young employment growth and a 0.14-0.16 percentage point increase in old/small employment growth. In the pooled sample, the elasticity is between 0.2 and 0.21–somewhat higher than the estimates between 0.05 and 0.16 for retail employment in Guren et al. (2021). These point estimates suggest that the national decline of 25 percent can account for 5.4-6.4 percentage points of the decline in young employment (of a roughly 20 percent decline) and about 3.3 percentage points of the decline in old/small business employment.

Table 1: Employment Growth Regressions on House Prices by Firm Age/Size

		Total	**Young**	**Old**	**Old/Small**	**Old/Large**
OLS	$\triangle$ HPI	0.205	0.269	0.162	0.164	0.0719
		(0.0181)	(0.0268)	(0.0152)	(0.0170)	(0.0137)
	Adj. R^2	0.0414	0.0108	0.0312	0.0349	0.0106
IV	$\triangle\widehat{HPI}$	0.209	0.289	0.149	0.142	0.0505
		(0.0244)	(0.0423)	(0.0173)	(0.0202)	(0.0259)
First Stage F-stat		213.4	289.5	186.4	179.1	297.9
Cent.R^2		0.00442	0.00204	0.0232	0.0271	0.000946
Obs.		22110000	6760000	15350000	13880000	1470000

Displays β_0 in (1). Dependent variable is the DHS employment growth rate regressed on log changes in metro house price growth. Note: Young < 5yo, Old>=5 yo, Small <500 employees, Large >=500 employees; controls include log(DHS denominator), metro area effects, industry effects, and (region × year) effects. $\triangle\widehat{HPI}$ indicates house prices instrumented by changes in national house prices interacted with metro housing supply elasticities from Guren et al. (2021). First Stage Kleibergen-Paap rk Wald and Angrist Pischke F-statistics are reported for IV regressions. Standard error clustered at the metro level.

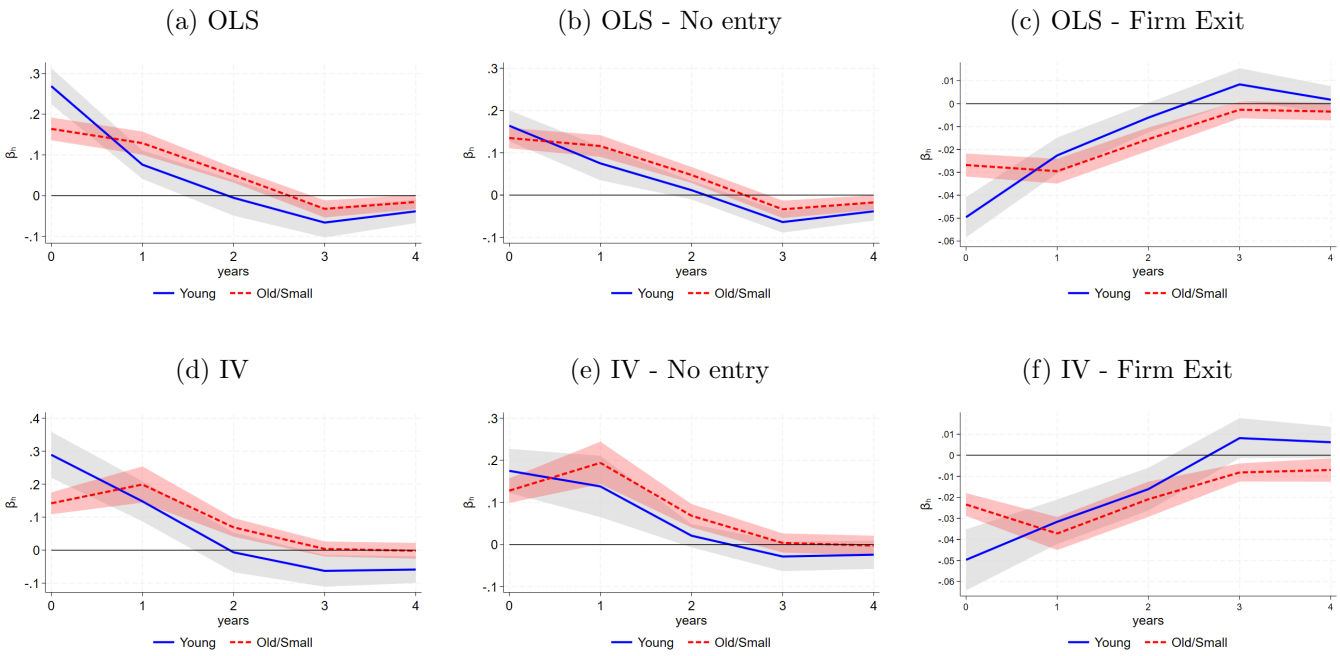
2.3.2 House Prices and Employment Growth Dynamics

Next, I consider the persistence of these effects by exploring a linear projection of house prices onto subsequent employment growth at the establishment level. I plot the coefficient at each time horizon $h = 0, 1, 2, 3, 4$ along with 90 percent confidence intervals. Entry is incorporated by adding entering firms in subsequent years to the left hand side for horizons $h > 0$. Figure 1 displays the results (note that the first period is equivalent to the first column of Table 1). Since the left-hand side variable is a growth rate, note that positive coefficients in subsequent years mean employment continues to expand (fall) in response to a house price increase (decrease) two years after the shock. Magnitudes differ across OLS and IV specifications, but the persistence is similar.[18]

How do the effects differ by firm age? I focus on young and old/small firms, which are the most sensitive to house prices on impact in Table 1. Firms are categorized as young/old at the time of the shock, and likewise, firm size is determined at the time of the shock. Figure

[18]For all local projections figures, I present the results in table form in online Appendix C, including observation counts, standard errors, R^2 statistics, and F-statistics for IV regressions.

Figure 1: House Prices and Employment Growth Dynamics - Age Comparison



Displays β_h in (1)- semi-elasticity of employment at year $t+h$ on house prices in t. Dependent variable is the DHS employment growth rate in (a)-(d) and a firm exit indicator in (e) and (f), regressed on log changes in metro house price growth. Note: Young $<$ 5yo, Old$>$=5 yo, Small $<$500 employees; controls include log(DHS denominator), metro area effects, industry effects, and (region $\times$ year) effects. IV indicates house prices instrumented by changes in national house prices interacted with metro housing supply elasticities from Guren et al. (2021). Area bands indicate $\pm 1.645 * SE$. Includes entrants in each year $t+h$, where post-shock entrants are given the initial size for the DHS denominator. See online Appendix C for table with observations, R^2 statistics, and F-statistics. Standard error clustered at the metro level.

1 displays the results split along these dimensions. In the aggregate, house prices impact young firms more in the first period. However, while the effect on growth is persistent for both groups, young firm growth is not persistently impacted *relative to old/small firms*. In fact, for young firms, the effect on growth is statistically insignificant at horizon $t+2$, in contrast to the effect on establishments at old/small firms, and the cumulative effect is larger for young firms only until horizon $t+2$.[19] This could partly be due to selection effects sharply forcing young firms to consolidate or exit on impact, and those that survive or enter in the wake of the shock are disproportionately productive and high growth.

[19]See Figure OA2 in Appendix B.2.

2.3.3 Entry and Exit Margins

How important are the entry and exit margins in explaining the dynamics observed in Figure 1? I explore the role of entry by restricting to continuing firms and comparing to the results of the full sample. Panels (c) and (d) of Figure 1 explore these effects for firms of all ages and young firms, since there is no significant difference at any horizon between young firms and old/small firms once entrants are excluded.

For the exit margin, I run a linear probability model with exit on the left hand side and the same regressors as in equation (1) on the right hand side. Exit is defined at the firm level–the indicator is 1 only if an establishment is part of a *firm* that exits at time $t + h$, and not only an exiting establishment. This is primarily to focus on firm, rather than establishment, dynamics. Panels (e) and (f) of Figure 1 display the results. I find exit probability was significantly impacted by house prices, and effects differed by firm age. A decline in house prices of 1 percent yields a .05 and 0.025 percentage point increase in exit rates for young firms and old firms, respectively. The national drop of 25 percent in house prices would imply an increase in the exit probability of one percentage point for young firms (with a typical exit rate for all firms of around 10 percent).[20] However, there is some impact of entry here as well, since lower entry contributes to a lower estimate of β_h.

These results suggest the entry and exit margins were a key part of the firm dynamics story that unfolded during the Great Recession. In the model I present in Section 3, I incorporate entry and exit margins that respond to house prices in light of these findings.

2.4 Mechanisms

2.4.1 Entrepreneurial Collateral

Table 2 displays evidence for the prevalence of personal finance and home equity in starting and maintaining a business. I match the SBO to the LBD sample used in the previous analysis, and provide summary statistics for 2008 and 2012.[21] In 2008, I find that 9.5 percent of firms use home equity financing, and almost half use some form of personal assets (but

[20] Since this is an establishment-level regression, this is close to an establishment-weighted exit probability.

[21] The survey was conducted in 2007 and 2012, but 2008 is the first year of my analysis. Overall counts are similar to the published data.

Table 2: SBO - Startup/Continuing Financing Sources

	2008			2012		
	Home Equity	Personal Saving/ Assets	None/ Other	Home Equity	Personal Saving/ Assets	None/ Other
Total	0.0914	0.471	0.438	0.0742	0.429	0.497
Old	0.0759	0.4553	0.4679	0.0783	0.4419	0.4802
Young	0.1303	0.5106	0.3592	0.0535	0.3612	0.5830

Source: Survey of Business Owners. Merged with LBD for 2008 and 2012 (must appear in either the 2007 or 2012 survey). Sample-weighted. See Table OA2 for detail and standard errors. "Home Equity" indicates home equity loans provided startup or continuing capital during the survey year. "Personal Saving" indicates personal assets or savings provided startup or continuing capital. Standard error clustered at the metro level.

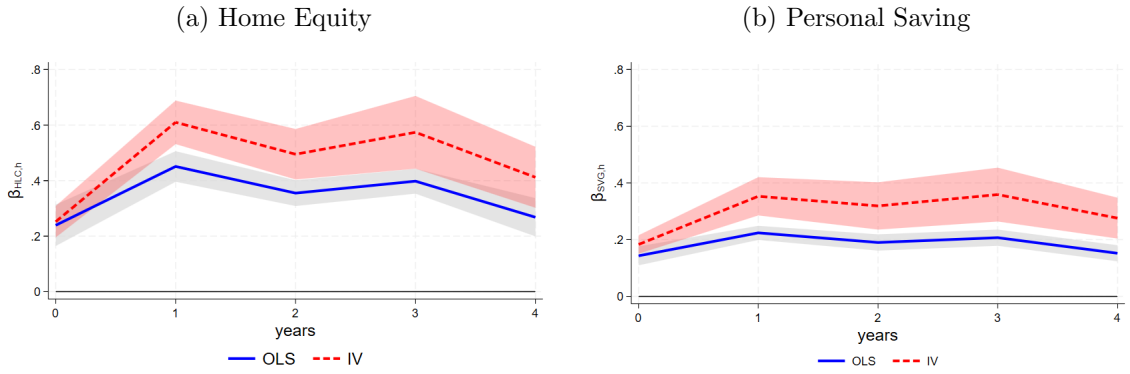
not home equity). Young businesses are particularly reliant on these forms of financing, with 14 percent relying on home equity and over 50 percent relying on personal assets. However, 2012 shows a much different story. Young firms are notably less reliant on personal assets than old firms. This could capture a lack of access to credit using home equity among young firms during the housing crisis, as well as some selection effects as the firms that are able to enter are more likely to have access to external credit.

Figure 2 displays the interaction terms in equation (3) (main effects are documented in Figure OA12 in the online appendix). The results demonstrate that reliance on these types of financing implies substantially more sensitivity to house prices. In fact, an establishment at a firm that relies on personal saving is twice as sensitive to house prices as a firm that does not rely on either home equity or personal savings, and an establishment at a firm that relies on home equity is three times as sensitive on impact.[22] Furthermore, these effects are very persistent, and even increase at later horizons. Comparing the effect of personal saving and home equity, reliance on home equity implies a higher sensitivity to house prices.[23]

[22] See Table OA9 for table format.

[23] One concern with focusing on the sample represented in Table 2 is that the sample is curtailed relative to the near-universe captured in the LBD. This is partially addressed by sample-weighting, and Figure OA4 in online Appendix B.2 presents a comparison of the results of estimating equation (1) on the SBO sample relative to the overall sample in Figure 1, with similar results.

Figure 2: House Prices and Entrepreneurial Collateral



Displays $\beta_{f,h}$ in (3) - marginal impact on $\beta_{q,h}$ of indicator for home equity/personal saving financing ($f \in SVG, HLC$). Dependent variable is the DHS employment growth rate at year $t+h$, regressed on log changes in metro HPI growth and interaction of HPI growth at year t with finance indicators. Note: controls include log(DHS denominator), metro area effects, industry effects, and (region $\times$ year) effects. IV regressions indicate HPI instrumented by changes in national HPI interacted with metro housing supply elasticities from Guren et al. (2021). Area bands indicate $1.645 * SE$ of the estimated effect. Includes entrants in each year $t+h$, where post-shock entrants are given their initial size for the DHS denominator. See online Appendix C for observation, R^2, and F-statistic tables. Standard errors clustered at the metro level.

2.4.2 Local Credit Markets - Mortgage Concentration Approach

Table 3 presents the results for Equation (4). I normalize the mortgage share to have mean zero, so main effects can be interpreted as the effect of house prices in the state with the mean mortgage share. The interaction with mortgage shares is significant for young firms in the OLS specification and IV specification.[24] Likewise, there is a significant response of employment among establishments of old/small firms (at the one percent level) in response to house prices and their interaction with mortgage shares in both specifications. This supports the notion that a higher mortgage share among state banks impacts old/small and young firms through local banks that are more sensitive to house price shocks.

To put the effect on young and old/small firms in context, a one standard deviation increase in mortgage share would lead to a roughly 25-50 percent increase in the elasticity. Focusing on the IV results, the elasticity of young firm employment for a state with a

[24]The interaction coefficient in the OLS and IV specifications is significant at the 5 percent level.

Table 3: Emp. Growth Regressions by Firm Age/Size: Mtg. Share

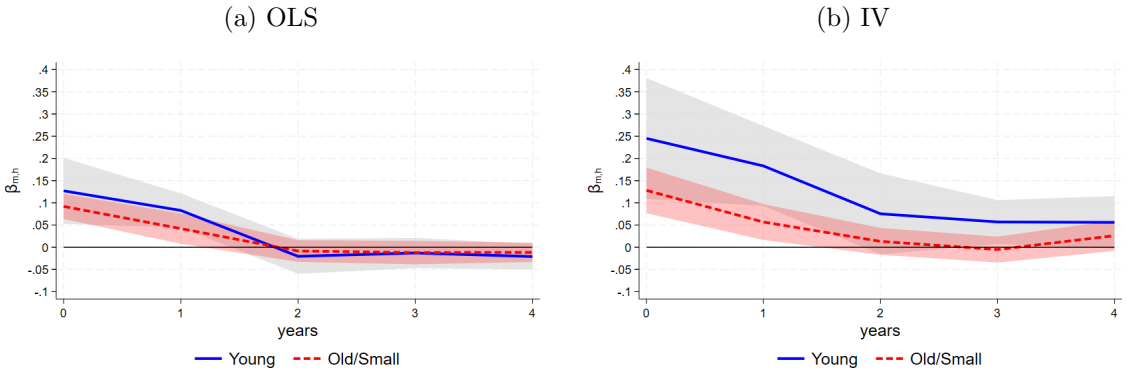
	Young (OLS)	**Old/Small (OLS)**	**Young (IV)**	**Old/Small (IV)**
$\triangle$ HPI	0.351	0.220	0.421	0.209
	(0.0495)	(0.0187)	(0.0748)	(0.0282)
$\triangle$ HPI*	0.127	0.0920	0.245	0.128
MTG_RATIO	(0.0451)	(0.0172)	(0.0826)	(0.0312)
R^2	0.0108	0.0349	0.00204	0.0271
K-P F-stat			34.51	38.17
HPI A-P F-stat			161.9	96.71
$HPI * MTG_RATIO$ A-P F-stat			42.59	41.22
Obs.	6760000	13880000	6760000	13880000

Displays $\beta_{q,0}$ and $\beta_{m,0}$ in (4). Dependent variable is the DHS employment growth rate, regressed on log changes in metro house price growth and interaction of house prices with standardized state-bank mortgage share in 2005. Note: Young < 5yo, Old>=5 yo, Small <500 employees; includes log(DHS denominator), metro area effects, industry effects, and (region × year) effects. IV regression instruments for house prices and interaction with mortgage share, house prices instrumented by changes in national house prices interacted with metro housing supply elasticities from Guren et al. (2021). First Stage Kleibergen-Paap rk Wald and Angrist Pischke F-statistics are reported for IV regressions. Standard error clustered at the metro level.

mortgage share one standard deviation above the mean would be 0.51 relative to 0.34 at the mean, and 0.3 for old/small firms relative to 0.19 at the mean. Thus a 25 percent decline in house prices would result in an additional 4.25 percentage point decline in growth rates for establishments at young firms in a state one standard deviation above the mean, and a 2.75 percentage point decline for old/small firms. Chodorow-Reich (2014), by comparison, finds a one-standard deviation increase in exposure to credit supply shocks leads to a 1-3 percentage point decline in the growth rate.

Figure 3 presents the dynamic response of the interaction coefficient at later time horizons. Panels (a) and (b) show that the effect of local bank mortgage shares persists somewhat to period $t+1$, but is not as persistent as the effect of house prices themselves. Again, this does not mean there is not persistent *cumulative* effect beyond $t+1$, but the effects on growth dissipate relatively quickly. Panels (c) and (d) display the same results while dropping en-

Figure 3: House Prices and Local Bank Balance Sheets



Displays $\beta_{m,h}$ in (4) - marginal impact on $\beta_{q,h}$ of a 1-standard deviation increase in 2005 mortgage share. Dependent variable is the DHS employment growth rate at year $t+h$, regressed on log changes in metro HPI growth and interaction of HPI at time t with standardized state-bank mortgage share in 2005. Note: controls include log(DHS denominator), metro area effects, industry effects, and (region $\times$ year) effects. IV indicates house prices instrumented by changes in national HPI interacted with metro housing supply elasticities from Guren et al. (2021). Area bands indicate $\pm 1.645 * SE$. Includes entrants in year $t+h$, where post-shock entrants are given their initial size for the DHS denominator. See online Appendix C for table with observations, R^2 statistics, and F-statistics. Standard error clustered at the metro level.

trants among young firms, and there is no significant effect of ignoring the entry margin for this coefficient. This suggests that while the entry margin might be impacted, it is not disproportionately impacted relative to growth at establishments of continuing firms.

2.5 Extensions

In addition to the identification concerns addressed in Section 2.2.1, I explore alternative mechanisms and general robustness in the online Appendix. One concern is these results pick up demand effects. I address this concern in two ways. First, I present results in online Appendix D.3.1 that includes metro-area unemployment as an additional control, as in Davis and Haltiwanger (2024). Effects are attenuated somewhat, but conclusions are robust. Second, I explore a specification that limits the analysis to "tradable" sectors as in Mian and Sufi (2014) in online Appendix D.3.2. As might be expected, this restrictive set of industries exhibits very different and less precise quantitative results, but the qualitative effects are very similar, especially for the mechanisms discussed in 2.4.1 and 2.4.2. To see how

much is driven by construction in particular, I exclude construction in online Appendix D.4. Results are somewhat attenuated, but similar qualitatively and in most cases quantitatively.

I consider an alternative outcome in payroll, rather than employment alone, in online Appendix D.2. This is a relatively underexplored variable that captures both wage differentials and intensive margin adjustments that employment alone misses. I find similar results, although point estimates are generally larger. Leaving aside formal tests (the samples are potentially different given the underlying data), the higher point estimates could suggest that an intensive margin is adjusting–hours and therefore pay is cut more sharply than employment. Alternatively, it could mean that higher paid workers are disproportionately laid off. Either way, it could suggest that welfare effects are potentially larger than the main results indicate if highly paid workers tend to work at productive establishments.

Finally, I consider whether the effect of house prices on firms was different during the Great Recession period than other points in the housing cycle. I run the baseline regression in (1) in the real house price boom period (1996-2006), and in the recovery from the Great Recession to the onset of pandemic (2013-2019). Table A2 presents the results. The OLS point estimates are similar for young firms, and somewhat smaller for old/small firms, suggesting that the disproportionate effect on young firms is mildly weaker in the Great Recession. However, this result does not hold up under the instrumental variable specification. Effects on both young and old/small firms are weaker in the IV specification during the boom and recovery than they are in the Great Recession. In the boom period, the effect on old/small firms is insignificant (and the point estimate is negative). While the disproportionate effect on young firms is still significant at the one percent level in the boom, the effect on young firms is not statistically significant from the effect on old firms in the recovery.

3 Model Outline

Having grounded the credit supply and collateral channels in the data, I specify a model which captures these channels and key dimensions of my empirics: effects by firm age, entry and exit margins, and dynamic effects. I can then quantify the significance of each channel in the aggregate, as well as how impacts vary by age, affect entry and exit, and change at different horizons. Time is continuous. There are four sectors: a representative household, an

entrepreneurial producer, banks, and housing construction. The household works, saves, and consumes nondurables and housing services both from owner-occupied and rental housing. To preview the main experiment, house prices fall due to a shift in household preferences.[25]

The key assumptions lie in the entrepreneurial and financial sectors. Entrepreneurs produce using labor and capital. Capital accumulation and hiring of labor are both impeded by financial frictions. In particular, a collateral requirement potentially limits their borrowing. The constraint depends on assets, since they borrow against their own capital and, crucially, household equity in housing. The dependence on aggregate housing is a simplifying assumption that captures an unmodeled connection between households and entrepreneurs, who after all, are also part of a household. This means household choices over housing do not take into account the effects on entrepreneurial outcomes. In other words, I assume the prospect of using housing as business collateral has a neglible impact on housing choices.[26] Additionally, the model features life cycle dynamics that enable analysis of entry. Although the empirics also highlighted varied effects by firm size, this was primarily due to the focus on local housing and banking markets. So, I restrict attention to firm age in the model.

The banking sector also propogates shifts in housing preferences. Banks take deposits and hold three assets: mortgages, business loans, and rental housing. The ownership of rental housing is similar to equity claims on capital in Gertler and Kiyotaki (2015), except the claims are now on housing. Just as capital prices lower bank net worth through their equity claims in Gertler and Kiyotaki (2015), shocks to house prices lower net worth for a bank holding rental housing. An internal incentive compatibility constraint limits the extent to which banks can leverage their net worth, so lower net worth leads to a tighter constraint,

[25]The specific nature of the shock is not critical–the key is that that shock lies outside of the production and financial sectors. Producer reliance on housing collateral and bank exposure to housing then transmits and amplifies the shock to the production sector. I choose a shock to the household which changes house prices. Previous literature has considered preference shocks in the boom/bust cycle–a shock to preferences, or more specifically beliefs about future preferences, is studied in Kaplan, Mitman, and Violante (2020).

[26]In online Appendix F.2 I consider an alternative where entrepreneurs have a portfolio allocation choice at entry when housing is chosen, but is constant thereafter. The results are mostly similar, but subtle differences suggest exploration of the relationship between housing choice and entrepreneurship is an area for future research.

which in turn leads to tighter lending conditions for the household and businesses.

3.1 Household Problem

The household maximizes utility over consumption C, leisure l, owner-occupied housing H, and rental housing H_R. It provides labor $L = 1 - l$ to producers at wage w. The household also saves by depositing D at deposit rate r in the banking sector. I require that each household must borrow through mortgages M to finance housing expenditures. Formally, the household problem is the discounted (at rate ρ) flow of utility:

$$\max_{\{C,L,H_{new},m\}_{t=0}^{\infty}} \int_{t=0}^{\infty} e^{-\rho t} U(C, 1-L, H, H_R) \tag{5}$$

$$s.t. \quad \dot{D} = wL + rD + \Pi - R_h H_R - d_m - \frac{\phi}{2}\frac{d_m^2}{M} - C - W_b, \tag{6}$$

$$\dot{H} = H_{new} - \delta_h H \quad , \quad \dot{M} = q_h H_{new} + r_m M - d_m \qquad (M \geq 0)\,, \tag{7}$$

where d_m are mortgage payments, q_h is the owner-occupied house price, R_h is the rental rate, and Π is profit from the final goods producer. New housing H_{new} is financed by mortgages M. The flow of mortgage payments is inhibited by a quadratic portfolio rebalancing cost with scale parameter ϕ, which permits a spread between deposit rates r and mortgage rate r_m. I also fund bank endowments through lump sum injections W_b. First order and envelope conditions imply (suppressing arguments):

$$w = \frac{U_L}{U_C}, \quad R_h = \frac{U_{H_R}}{U_C}, \quad (\rho - r)\, U_C = \frac{d}{dt} U_C, \tag{8}$$

$$q_h\,(\rho + \delta_h)\left(1 + \phi\frac{d_m}{M}\right) U_C = U_H + \frac{d}{dt} q_h \left(1 + \phi\frac{d_m}{M}\right) U_C, \tag{9}$$

$$\left(\rho\left(1 + \phi\frac{d_m}{M}\right) - r_m\right) U_C = \frac{d}{dt}\left(1 + \phi\frac{d_m}{M}\right) U_C. \tag{10}$$

Equation (8) gives standard static expressions of prices as marginal rates of substitution and the standard intertemporal Euler equation. Equation (9) governs the choice between present consumption and durable owner-occupied housing. Equation (10) gives the intertemporal Euler using mortgage rates, which can deviate from the deposit rate due to the portfolio rebalancing cost. Imposing steady state (where time derivatives are equal to zero), gives the

usual condition $r = \rho$, and the steady state relationship between spreads and rebalancing costs $r_m = r + \rho\phi d_m$. In steady state we have:

$$q_h = \frac{U_H}{(\rho + \delta_H)(1 + \phi d_m) U_C}. \tag{11}$$

Equation (11) equates the house price to the marginal rate of substitution between housing and consumption. The steady state rent to price ratio is: $\frac{R_h}{q_h} = (\rho + \delta_h) \frac{U_{H_R}}{U_H}$.

3.2 Production

The entrepreneurial sector combines a modified Hopenhayn (1992) model of firm entry and exit, similar to Shaker Akhtekhane (2017), with a model of firms whose borrowing and operations are inhibited by credit market frictions. The model extends work by Kaplan, Moll, and Violante (2018) to a producer's problem where borrowing constraints are endogenous to producer assets and housing collateral. Firms are able to borrow, but only up to a certain fraction of their owned capital plus some collateral from the aggregate housing stock.

3.2.1 Final Goods Sector

Entrepreneurs are intermediate goods producers, where goods are differentiated according to a monopolistic framework with the following aggregator:

$$Y = \int_i y_i^\gamma di, \tag{12}$$

where y_i is intermediate good firm i's physical output. A final goods producer aggregates and sells the final output Y, which is the numeraire. Payments to producers are made in the numeraire good as well. This yields the pricing function: $p_i = \gamma y_i^{\gamma-1}$. Profits sent to the household are then $\Pi = Y - \int_i p_i y_i d_i = (1 - \gamma) Y$.

3.2.2 Intermediate Goods Producers: Continuing Firms

Continuing intermediate firms have liquid assets/borrowing b, and illiquid productive physical capital k. They hire labor each period at wage rate w, which combines with capital and plant-specific productivity z to produce physical output. The state of the production sector can be described at any time t by the joint distribution $\mu_t(dz, dk, db)$. Liquid assets bear

interest rate r when the entrepreneur is saving, and higher rate r_ℓ when borrowing. This borrowing rate also affects the cost of labor, but only for those who do not have liquid saving b to fund payments, while those firms with higher wage bills than liquid saving must use financial services that bear a cost I assume is equal to the borrowing rate. Upon producing, earning interest/paying interest on liquid assets, and paying workers, the entrepreneur can choose to invest profits or consume. Investment is subject to convex adjustment costs $f(i,k) = \frac{\chi}{2}\frac{(i-\delta k)^2}{k}$, and capital depreciates at rate δ. Log productivity follows a Ornstein-Uhlenbeck process with parameters μ_z and σ_z. Their problem is defined as follows:

$$V(z,k,b) = \max_{c,h,i} \int_0^\infty e^{-(\rho+\zeta)t} u(c_t) dt \tag{13}$$

$$s.t. \quad \dot{b} = \gamma z^\gamma k^{\gamma\alpha} h^{\gamma(1-\alpha)} + r(b)b - wh - r_\ell \min\left\{\max\left\{wh-b,0\right\}, wh\right\} - i - f(i,k) - c,$$

$$\dot{k} = i - \delta k, \quad dlog\left(z\right) = -\mu_z log\left(z\right) dt + \sigma(z_t) dW_t,$$

$$-b + wh + r_\ell \min\left\{\max\left\{wh-b,0\right\}, wh\right\} \leq \xi k + \xi_h q_h H, \quad r\left(b\right) = \begin{cases} r, & b > 0 \\ r_\ell, & b < 0 \end{cases}.$$

To build intuiton for the results, note house prices directly affect firms by tightening collateral constraints $\xi_h q_h$, and as we will see, by influencing borrowing rates via the banking sector.[27]

3.2.3 Intermediate Goods Producers: Entry and Exit

Existing firms exit the economy in two ways: random exogenous exit arriving at rate ζ or hitting the borrowing limit without the ability to generate non-negative savings (and therefore abide by the constraint). Some borrowing levels b are feasible for firms in the upper part of the productivity distribution, but firms would be "underwater"–unable to generate positive cash flow even if consumption is 0–if they had sufficiently low productivity. These firms must either get lucky with a positive productivity shock that allows them to improve their balance sheet, or they hit the boundary and are forced to liquidate. In this case, the ex-entrepreneur is assumed to receive constant payoff of c_{exit} in the limbo state unless hit by the death shock, and so the value function for such firms is $v(z,k,b) = \frac{c_{exit}}{\rho+\zeta}$.[28]

[27] The influence of finance costs and the collateral constraint on the labor margin creates labor wedges to which, as discussed in Chari, Kehoe, and McGrattan (2007), aggregate outcomes are sensitive.

[28] In the event of a firm liquidating, I assume that entrepreneur capital and profits are used to repay debt obligations, and any remaining assets are discarded.

There is a continuum of potential entrants on the interval $[0, 1]$, and the realized flow of entrants m is governed by a modified free-entry condition:

$$m = \overline{m} exp\left(\eta\left(\int_0^1 v\left(z, k_{ent}, b_{ent}\right)\psi(z)dz - c_e\right)\right), \tag{14}$$

where η governs the elasticity of supply of entry with respect to deviations of the free entry condition from zero, and $\overline{m}$ is a parameter that relates the mass of the entry distribution to the mass of continuing firms. Compared to the Hopenhayn (1992) model, the supply of entrants is governed by η and is not infinitely elastic. Entrants are endowed with a pre-defined amount of equity to invest in capital, κ_e. They can then borrow up to an exogenous limit $b \leq \xi\overline{q_h}\overline{H}$, where $\overline{q_h}$ and $\overline{H}$ are the steady state values for house prices and owner occupied housing. I assume the constraint binds, so that $k_{ent} = \kappa_e + \xi_h\overline{q_h}\overline{H}$.[29]

3.3 Banking Sector

This model is a continuous time extension of Gertler and Karadi (2011) and Gertler and Kiyotaki (2015), where banks are subject to an internal enforcement constraint which generates a spread between the lending rate and the deposit rate. In these models, intermediaries are exposed to risk by holding equity claims, rather than debt claims with default risk. The value of equity claims on capital fluctuates with capital prices, and therefore impact bank net worth. Similarly, here banks are exposed to house price fluctuations through equity claims on housing. Banks take deposits from households to fund their assets, but their ability to lend is subject to an incentive constraint that requires some equity or "skin in the game." That is, the bank is limited in its ability to leverage its initial wealth. I model this as a leverage constraint, where lending capacity depends on the bank's net worth. This constraint generates a positive spread between the lending rate and the deposit rate.

In the real world, assets like mortgages and MBS can depreciate in response to house prices, reducing bank net worth and lending capacity. In this model, the exposure to rental housing prices represents exposure to house price fluctuations through mortgages and other lending related to owner-occupied homes as well as commercial properties.

[29] I assume a constant initial constraint to prevent capital grids from moving endogenously over time. Following techniques discussed in Kaplan, Moll, and Violante (2018), as well as related papers, I set up the Bellman equation as shown in online Appendix E.2 and solve the model numerically.

3.3.1 Bank Problem

Banks make choices over a portfolio of assets. They lend to businesses via loans ℓ, lend to housing markets via mortgages m, and own rental housing H_R which yields rental return R_h and is valued at price q_h.[30] All assets are subject to a flow cost ψ that is proportional to the asset's value. Bank net worth is the value of assets less deposits, $n = q_h H_R + \ell + m$, and banks have linear preferences in end of life net worth, discounting it by σ due to exogenous death. The enforcement constraint requires the bank franchise value be greater than some fraction θ of assets. The bank's objective is thus given by:

$$V(l, m, H_R) = \max_{\dot{l}, \dot{m}, \dot{H}_R} \int_0^\infty \left(1 - e^{-\sigma}\right) e^{-(\rho+\sigma)t} n \, dt \tag{15}$$

$$s.t. \quad \int_0^\infty \left(1 - e^{-\sigma}\right) e^{-(\rho+\sigma)t} n \, dt \geq \theta a,$$

$$\dot{n} = \left(\frac{\dot{q}_h + R_h - \delta_h}{q_h} - \psi\right) q_h H_R + (r_\ell - \psi)\,\ell + (r_m - \psi)\, m - rd,$$

$$n = a - d, \qquad a = q_h H_R + \ell + m.$$

In online Appendix E.4, I walk through details of solving this problem. The evolution of the bank leverage ratio is given by a first order differential equation:

$$\dot{\varphi}_t = \left(\rho + \sigma - r - \left(R_a - \psi - r\right)\varphi_t\right)\varphi_t - \frac{\left(1 - e^{-\sigma}\right)}{\theta}. \tag{16}$$

Equation (16) shows that changes in bank leverage do not depend on bank size, but depend negatively on the spread. I show in online Appendix E.4 that, for a stationary distribution where $\dot{\varphi}_t = 0$, steady state leverage is the solution to a quadratic equation, and under relevant parameterizations it increases with credit spreads.

3.3.2 Bank Aggregation

Since bank net worth always grows, bank exit and entry are needed to ensure stationary aggregates. Total net worth and assets in the banking sector are defined, respectively, as:[31]

$$N = \int_0^1 n_j d_j \quad \text{and} \quad A = \int_0^1 a_j d_j.$$

[30] While businesses do face a cost equivalent to the lending rate associated with bank facilitation of wage payments, these payments are not an asset for the bank.

[31] Likewise, denote $H_R = \int_0^1 h_{R,i} d_i$, $M = \int_0^1 m_i di$ and $\boldsymbol{\ell} = \int_0^1 \ell_i di$.

Defining $\Phi = \frac{A}{N}$ as the leverage ratio in the banking sector as a whole, with W_b parameterizing the amount of wealth endowed to entering banks and $w_b = \frac{W_b}{N}$ as the share of entering bank net worth to total net worth, the growth rate for total bank net worth is:

$$\frac{\dot{N}}{N} = e^{-\sigma}\left[\left(R_a - \psi - r\right)\Phi + r\right] + \left(1 - e^{-\sigma}\right)\left(w_b - 1\right). \tag{17}$$

Combining with the individual bank solution problem, this law of motion is key for defining an equilibrium and determining interest rate spreads.

3.4 Housing Production

New housing is produced by a competitive sector of identical housing producers. The profit maximization problem is static, and given by:

$$\pi_H = \max_{H_{new}} q_H H_{new} - C\left(H_{new}, H\right).$$

Here, profits depend on the quantity of new housing and also the level of the existing housing stock. Consider a cost function of the form:

$$C\left(H_{new}, H\right) = \frac{\kappa_H}{\Upsilon} H_{new}^{\Upsilon} + \gamma_H abs\left(H_{new} - \delta H\right).$$

Consider three cases: $H_{new} > \delta_H H$, $H_{new} = \delta_H H$, and $H_{new} = \delta_H H$. The first order condition gives:

$$q_H = \begin{cases} \kappa_H H_{new}^{\Upsilon - 1}, & H_{new} = \delta_H H \\ \kappa_H H_{new}^{\Upsilon - 1} - \gamma_H, & H_{new} \neq \delta_H H \end{cases}. \tag{18}$$

In the main analysis, I consider a case where $\kappa_H = 0$ and $\gamma_H = \infty$, so supply is perfectly inelastic. Alternatively, in online Appendix F.3 I consider a case where $\gamma_H = 0$ and Υ is calibrated to capture housing stock dynamics during the Great Recession.

3.5 Equilibrium

An equilibrium for this economy is a path of the production sector distribution $\mu_t(dz, dk, db)$, household aggregates $\{D(t), H(t), H_R(t), L(t), C(t), d_m(t), M(t)\}$, corresponding bank aggregates, and prices $\{w(t), r(t), r_m(t), r_\ell(t), q_h(t)\}$ such that the decisions described by optimal conditions and constraints for the household, producers, and banks hold, and markets clear, with the exception of the market for deposits. In the main analysis, this market clears in

steady state, but along a transition path I assume the economy is open, so the deposit rate is perfectly elastic. Thus, all adjustments to interest rate spreads directly affect borrowing costs. This assumption captures, in a reduced-form way, the effect of the zero-lower bound on borrowing costs deposit rates in models with nominal rigidities. Nonetheless, I consider a model with a flexible deposit rate in online Appendix F.4. I describe market clearing conditions explicitly in online Appendix E.3.

3.6 Transition

To represent a crisis, I explore a temporary exogenous shock to housing preference parameters, completely unanticipated by agents. Upon realization of the shock, agents know with certainty the future path the economy will take. The shock itself, displayed graphically in online appendix E.1, is a sharp decline in both the preference for housing in general (governed by parameter ν_h) and owner-occupied housing in particular (given by parameter ϵ_h), and a gradual recovery in both parameters. Such a shock broadly captures the decline and slow recovery of house prices and home ownership rates. Crucially, there is no shock to collateral constraints ξ or financial intermediary constraint θ, or to aggregate TFP as in Gertler and Kiyotaki (2015). To be sure, a shock to intermediation parameters or TFP is likely in the context of the Great Recession. However, this paper shows the potential for the same shock from housing markets to be transmitted and amplified through the separate channels of collateral values and endogenous credit supply.

4 Preferences and Calibration

I target moments described in Appendix Table A3 for calibration, noting whether parameters are jointly determined or taken from the literature. All rates are annual, with long-run averages used for smoothing. For the household, I use a utility function that is additively separable in consumption/labor and housing. The consumption/labor preferences are as in Greenwood, Hercowitz, and Huffman (1988) (specifically log) with linear disutility of labor, and CES housing preferences with housing taste ν_h and owner-occupied housing taste ϵ_h:

$$U(C, 1-L, H, H_R) = ln\left(C + \nu_l\left(1-L\right)\right) + \nu_h\left(\epsilon_h H^{1-\sigma_h} + \left(1-\epsilon_h\right)H_R^{1-\sigma_h}\right)^{\frac{1}{1-\sigma_h}}. \quad (19)$$

These preferences imply that the wage will be pinned down by the disutility of labor parameter $w = \nu_l$, and does not change. Thus, the model will have some similarities to a sticky-wage framework, perhaps more appropriate in the short run. Crucially, in the Hopenhayn (1992) model, the wage is key for determining entry rates in steady state, so ν_l is chosen to match the entry rate in the US in 2005 (about 10 percent according to US Census Bureau, 2005-2012). The parameters related to housing and mortgage costs target the owner-occupied share, house price-to-rent ratio, and mortgage debt-to-housing value ratio. The discount rate ρ is standard and δ_h is consistent with BEA data on housing depreciation.

The chosen persistence of firm productivity μ_z is central to the range of potential estimates in Asker, Collard-Wexler, and Loecker (2014). Likewise with depreciation δ, capital share α, markups γ, housing depreciation δ_h, and discount rate ρ, I use standard values from the literature.[32] I calibrate the standard deviation of productivity shocks to target dispersion in TFPR from Cunningham et al. (2023), and the adjustment cost parameter targets the standard deviation of investment rates in Cooper and Haltiwanger (2006). The collateral constraint ξ targets data on corporate debt to output, entry mass $\overline{m}$ pins down entrepreneurs as a fraction of the population, and entry level endowment κ_e to match employment-weighted entry rates from the BDS. I calibrate the exit flow value of consumption c_{exit} to the average size of firm in the economy, and entry cost c_e to match the employment to population ratio.

The elasticity of entry η governs the response of entry to shocks. Entry falls to 7.7 percent in the model, compared to 7.6 percent in the data. The calibrated value of 0.725, is clearly low compared to the infinite elasticity in Hopenhayn (1992). A model that explores determinants of entry, as in Jovanovic (1982), would be of interest for future work.

Finally, I use financial sector parameters to support a leverage ratio of 10, which is in the range discussed in Gertler and Kiyotaki (2015), and an intermediation cost of 180 basis points (Phillipon, 2015). I select the fraction of bank net worth re-distributed to new banks which implies the size of a new bank is 10 percent of the average bank. Consistent with BDS data, I set the exogenous bank death rate to 8 percent. I then calculate the constraint parameter θ consistent with a target leverage ratio.

[32] I specifically target entrepreneurial sector capital share and markups.

5 Results

5.1 Aggregate Results: Housing Crisis

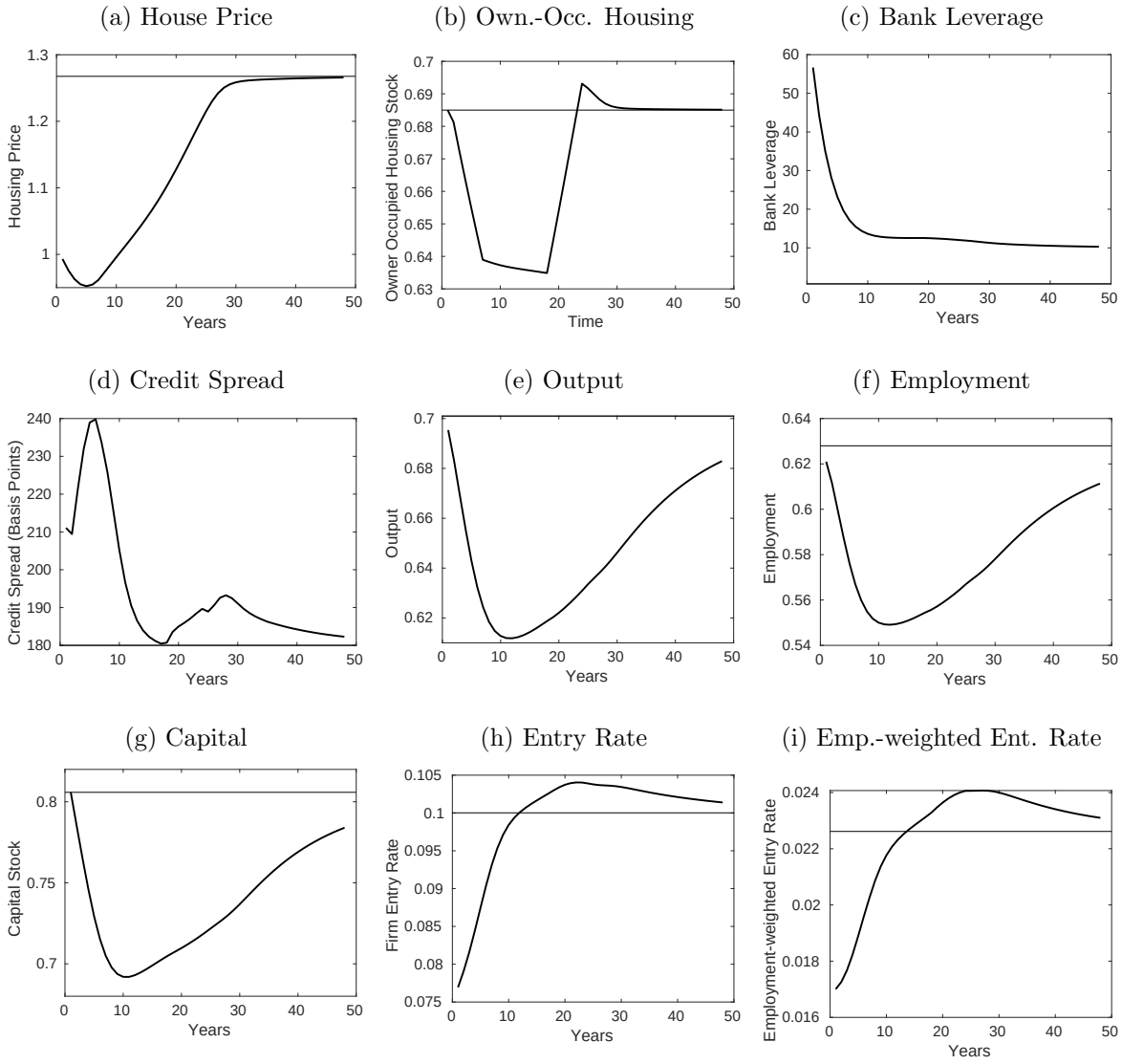
I now turn to analysis of the response of the calibrated economy described above to a housing collapse. Specifically, I focus on a change in demand for overall housing in ν_h and owner-occupied housing ϵ_h. Results for key aggregates are shown in Figure 4. The preference shock results in a house price decline of about 25 percent five years out, roughly in line with observed house prices in the Freddie Mac (2007-2012) data during the Great Recession. Owner-occupied housing falls as well, again in line with what we see in homeownership.[33] On the financial side, bank leverage rises sharply. The resulting increase in credit spreads is about 60 basis points. This is somewhat low compared to the data, as credit spreads rose between 300 and 500 basis points.

Overall, the shock to housing preferences leads to a housing crisis with falling housing collateral values and tightening credit conditions, consistent with the experience of the U.S. in the Great Recession. This crisis has real effects in terms of output and employment, as tightening credit conditions and lower house prices lead to severe contractions in output, employment, and the capital stock. On impact, employment falls sharply, as the entry rate collapses by 2.3 percentage points, before eventually recovering by the end of the first decade after the crisis. Output and employment persistently deviate from long-run steady state. Output falls about 13 percent below its steady state (equivalent to trend output in the data) with the employment to population ratio falling just below 55 percent. This is somewhat sharper than in the data, when EPOP fell to 58 percent.

Turning to the effect on firms by age, I display the decline in young (<5 years) vs. old firm employment from the model relative to the steady state in Figure 5. Here, young business employment drops about 2 percent on impact, and further falls by about 27 percent over five years in part due to the cumulative loss of young firms through lower entry, as seen in panel (h) of Figure 4. This is a bit larger than the the decline in the data, as young firm employment fell by 22 percent according to the BDS. While entry is an important driver

[33] Board of Governors of the Federal Reserve System (US) (2004-2006b) data show owner-occupied rates falling from a high of roughly 69 percent in 2004 to 63 percent by 2016.

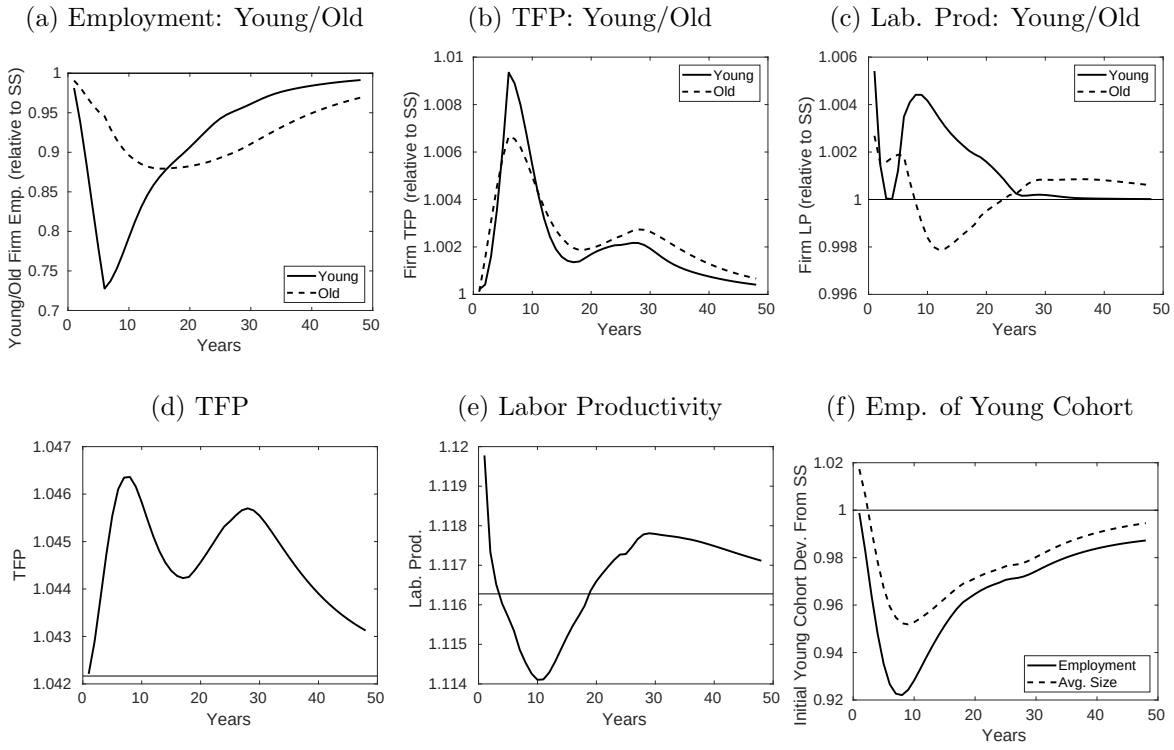
Figure 4: Housing Crisis: Aggregate Impulse Responses



of aggregate results, existing young firms also suffer, which I demonstrate by simulating a "young" cohort at the time of the shock (so it does not incorporate entry) relative to a counterfactual cohort in the steady state, displayed in panel (f) of Figure 5. This shows a persistent and large impact on this cohort as continuing young firms cut employment and the exit rate increases, leading to lower long-run contributions of the cohort. This effect is due to both the exit and average size of continuing firms in this cohort, despite an initial positive selection that drives average size up briefly on impact.

For older firms, employment falls mildly on impact, but eventually declines by about 12 percent after more than two decades, to some degree reflecting the cumulative effect

Figure 5: Housing Crisis: Age Profile and Productivity Responses



Note: Panels (a)-(c) are measured relative to steady state values . Panel (f) tracks a cohort who is young at the time of the initial shock relative to a steady state cohort.

on initially young cohorts as they age. Consistent with the empirics, younger firms are more susceptible to house price declines, at least initially. However, after house prices begin to recover, panel (a) of Figure 5 shows young firms rebound more quickly than old firms, consistent with the results in Figure 1.

Productivity rises for young and old firms, displayed in Figures 5(b) and (c), despite lower aggregate TFP. This is in part because tighter collateral constraints and higher interest rates are associated with higher labor productivity, as discussed in Gopinath et al. (2017) and Blackwood (2025). Further, selection within firm types could lead to cleansing effects, as lower productivity firms exit. On the other hand, if relatively productive firms are forced to cut employment more (or exit), then a sullying effect would lead to lower aggregate productivity. Under this calibration, the cleansing effect largely dominates. Interestingly, in the alternative model with an initial housing choice in online Appendix F.2, higher reliance on collateral leads to an initially dominant sullying effect, although the effect is small, with

productivity remaining within one percent of steady state.

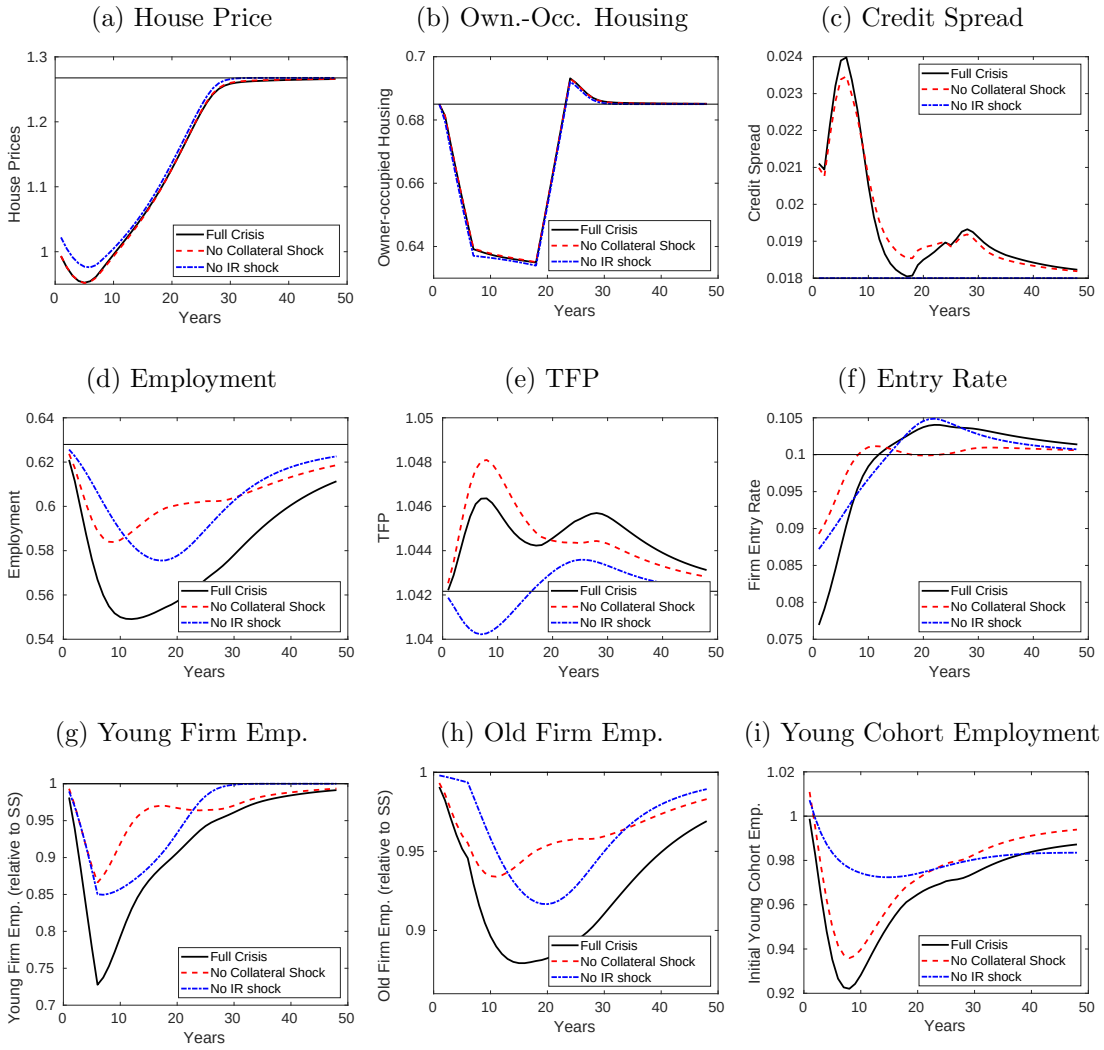
5.2 Exploring Channels and Policy via Counterfactuals

The aggregate outcomes observed are the result of house prices working through both the collateral channel and the interest rate channel. Whether each channel is relevant for aggregate outcomes, and which firms are affected by each channel, are key questions for policymakers. To gain insights on these questions, I consider counterfactuals where I shut down each channel in turn, then re-calculate relevant equilibrium prices. First, I fix the value of housing at its steady state level in the firm's collateral constraint and feed in the same preference shock as in the full crisis, computing the interest rate path and house price path that is consistent with the household and bank problem. I call this the "no collateral shock" counterfactual. Likewise, I fix the lending rate in the "no interest rate shock" counterfactual, and find house prices that are consistent with the household problem. Results are displayed in Figure 6.

These experiments provide two insights. First, they illustrate the relevance of each channel for aggregate outcomes. Second, they can be thought of as a reduced-form representation of policy interventions. The counterfactual with no interst rate effect represents policy aimed at shoring up bank balance sheets (and thus preventing interest spreads from increasing), similar to the bank support undertaken by policymakers in the Great Recession. The counterfactual with no collateral effect represents a more obscure policy where government guarantees provide stable credit-worthiness for borrowers, but can capture the effect of broader policies, either through government spending or intervention into housing markets, that support borrower collateral values.

The top row of Figure 6 shows that shutting down the collateral channel does little to change house price paths or the share of owner-occupied housing, and credit spreads are not as volatile. Shutting down the interest rate channel dampens the effect of the shock on house prices marginally. The muted differences across counterfactuals in housing and financial markets translates to similar results in the absence of general equilibrium effects (see online Appendix F.1). In terms of production and employment, both channels contribute to the overall decline in determining aggregate outcomes. Initially, the interest rate channel is more active, as the no collateral counterfactual falls more sharply. This highlights the

Figure 6: Housing Crisis: Endogenous House Price Experiments



Note: Counterfactual crises displayed. "No Collateral Shock" corresponds to fixed collateral values counterfac5ual, "No IR Shock" corresponds to fixed interest rate counterfactual. Counterfactual prices and housing measures displayed in the top row. Measures are relative to steady state values in (g)-(i).

importance of bank support to keep spreads low, especially early on in the crisis. On the other hand, eliminating credit spreads would not eliminate the recession entirely, and a severe and protracted recession still ensues even with no interest rate channel. The implication here is that supporting lenders alone is potentially insufficent to address the crisis.

Turning to impacts by firm age, young firms are similarly sensitive to both channels in panels (f) and (g), consistent with the empirics in Section 2.4. The initial cohort of young firms is more drastically impacted by the interest rate channel, but persistent effects

of the collateral channel are evident over the long run in panel (i). Old firms are likewise disproportionately affected by the interest rate channel in panel (h). As emphasized by Gopinath et al. (2017) and Blackwood (2025), firms at or near optimal scale are more sensitive to interest rates, and in this context mature firms are generally closer to optimal scale.

This leads to a striking result for productivity. TFP falls initially in the counterfactual with no interest rate shock in (e), but *rises* in the counterfactual with no collateral shock. Interest rate shocks tend to reduce the "distortions" (to use the language of Hsieh and Klenow, 2009) by moving firms, especially older firms who are near optimal scale, away from collateral constraints, as discussed in Gopinath et al. (2017). A less distorted economy has higher TFP, all things being equal. However, tighter constraints increase distortions, and young firms are more likely to be constrained. On balance, TFP rises modestly in the short run in the full crisis, as large firms contribute disproportionately to aggregate productivity.

The results are consistent with the empirical evidence: both credit supply and collateral channels matter for firms, in particular young firms. At the aggregate level, the full extent of the crisis requires both channels, although the collateral channel appears to be somewhat more important in the medium term. For policymakers addressing financial recessions, the model suggests the source of financial pressure can be important for policy. Banking conditions are important in the empirics and the model, but are not the full story. Thus, addressing credit conditions by supporting financial institutions might not be enough. Collapsing collateral constraints hit young businesses relatively hard, and contribute substantially to aggregate outcomes in the model. As a result, programs focusing on securing credit for young and small businesses more directly could fill the gap.

6 Conclusion

This paper presents evidence on the impact of house prices working through various channels to affect firm growth in the Great Recession, and constructs a macroeconomic model that is consistent with the evidence. In the model and the data, young firms are disproportionately impacted by house prices, but the disproportionate effect on young businesses is not as persistent as the overall effect of the shock. In the model and in the data, bank credit matters for all firms, and the collateral channel also contributes substantially to firm outcomes.

In the model this is particularly true for young firms, and in the data, young firms were disproportionately reliant on these forms of finance on the eve of the Great Recession.

Overall, the model replicates key moments in the empirical work, and describes the aggregate implications for overall employment and welfare. Both credit supply and collateral channels have important aggregate effects, with the collateral channel in particular impacting entry and activity at young firms. These findings are relevant for policy makers seeking to effectively combat downturns, as targeting only one channel may leave the job half done.

References

Agarwal, Sumit and Robert Hauswald (2010). "Distance and Private Information Lending". In: *Review of Financial Studies* 23.7, pp. 2256–2303.

Albouy, David, Gabriel Ehrlich, and Yingyi Liu (2016). "Housing Demand, Cost-of-Living Inequality, and the Affordability Crisis." In: *NBER Working Paper Series* 22816.

Asker, John, Allan Collard-Wexler, and Jan De Loecker (2014). "Dynamic Inputs and Resource (Mis)Allocation." In: *Journal of Political Economy* 122.5, pp. 1013–1063.

Barlevy, Gadi (2002). "The Sullying Effect of Recessions". In: *Review of Economic Studies* 69.1, pp. 65–96.

Basu, Susanto and John G. Fernald (1997). "Returns to Scale in US Production: Estimates and Implications". In: *Journal of Political Economy* 105.2.

Black, Sandra E. and Philip E. Strahan (2002). "Entrepreneurship and Bank Credit Availability". In: *Journal of Finance* 57.6, pp. 2807–2833.

Blackwood, G. Jacob (2025). "Financial markets and misallocation: the long and short of leverage and productivity dispersion". In: *Macroeconomic Dynamics* 29, pp. 1–27.

Blackwood, G. Jacob, Lucia S. Foster, Cheryl A. Grim, John Haltiwanger, and Zoltan Wolf (2021). "Macro and Micro Dynamics of Productivity: From Devilish Details to Insights." In: *American Economic Journal: Macroeconomics* 13.3, pp. 142–172.

Board of Governors of the Federal Reserve System (US) (2004-2006a). *Nonfinancial Corporate Business; Debt Securities and Loans; Liability, Level [BCNSDODNS]*. Accessed October 27, 2024 via FRED, Federal Reserve Bank of St. Louis. URL: https://fred.stlouisfed.org/series/BCNSDODNS.

Board of Governors of the Federal Reserve System (US) (2004-2006b). *Owners' Equity in Real Estate, Level [OEHRENWBSHNO]*. Accessed October 27, 2024 via FRED, Federal Reserve Bank of St. Louis. URL: https://fred.stlouisfed.org/series/OEHRENWBSHNO.

Boot, Arnoud W. A. and Anjan V. Thakor (1994). "Moral Hazard and Secured Lending in an Infinitely Repeated Credit Market Game". In: *International Economic Review* 35.4, pp. 899–920.

Brevoort, Kenneth P., John A. Holmes, and John D. Wolken (2009). "Distance Still Matters: The Information Revolution in Small Business Lending and the Persitent Role of Location, 1993-2003". In: *Finance and Economics Discussion Series, Division of Research & Statistics and Monetary Affairs, Federal Reserve Board*.

Bureau of Labor Statistics (2007-2012). *Local Area Unemployment Statistics*. Accessed May 6, 2024. URL: https://download.bls.gov/pub/time.series/la/la.data.60.Metro.

Cetorelli, Nicola and Philip E. Strahan (2002). "Finance as a Barrier to Entry: Bank Competition and Industry Structure in Local U.S. Markets". In: *Journal of Finance* 57.6, pp. 2807–2833.

Chari, V.V., Patrick J. Kehoe, and Ellen R. McGrattan (2007). "Business Cycle Accounting". In: *Econometrica* 75.3, pp. 781–836.

Chodorow-Reich, Gabriel (2014). "The Employment Effects of Credit Market Disruptions: Firm-Level Evidence from the 2008-9 Financial Crisis." In: *Quarterly Journal of Economics* 129.1, pp. 1–59.

Chow, Melissa, Teresa C. Fort, Christopher Goetz, Nathan Goldschlag, James Lawrence, Elisabeth Ruth Perlman, Martha Stinson, and T. Kirk White (2021). "Re-designing the Longitudinal Business Database". In: *CES Working Paper Series* CES-21-08.

Cole, Rebel A. and Lawrence J. White (2012). "Dèjà Vu All Over Again: The Causes of US Commercial Bank Failures This Time Around". In: *Journal of Financial Services Research* 42.1, pp. 5–29.

Cooley, Thomas, Ramon Marimon, and Vincenzo Quadrini (2004). "Aggregate Consequences of Limited Constract Enforceability." In: *Journal of Political Economy* 112.4, pp. 817–847.

Cooper, Russell W. and John C. Haltiwanger (July 2006). "On the Nature of Capital Adjustment Costs". In: *Review of Economic Studies* 73.3, pp. 611–633.

Corradin, Stefano and Alexander Popov (2015). "House Prices, Home Equity Borrowing, and Entrepreneurship". In: *Review of Financial Studies* 28.8, pp. 2399–2428.

Cunningham, Cindy, Lucia S. Foster, Cheryl A. Grim, John Haltiwanger, Jay Stewart, and Zoltan Wolf (2023). "Dispersion in Dispersion: Measuring Establishment-level Differences in Productivity." In: *The Review of Income and Wealth* 69.4, pp. 999–1032.

Davidoff, Thomas (2016). "Supply Constraints Are Not Valid Instrumental Variables for Home Prices Because They Are Correlated With Many Demand Factors". In: *Critical Finance Literature* 5, pp. 177–206.

Davis, Steven J. and John Haltiwanger (2024). "Dynamism Diminished: The Role of Housing Markets and Credit Conditions". In: *American Economic Journal: Macroeconomics* 16.2, pp. 29–61.

Davis, Steven J., John C. Haltiwanger, and Scott Schuh (1996). *Job Creation and Destruction*. Cambridge, MA: MIT Press.

Decker, Ryan A. (2015). "Collateral Damage: Housing, Entrepreneurship, and Job Creation". In: *mimeo*.

Dinlersöz, Emin, Henry Hyatt, Şebnem Kalemli-Özcan, and Veronika Penciakova (2018). "Leverage over the Life Cycle of the U.S. Firms." In: *NBER Working Paper Series* 25226.

Fairlie, Robert W. and Harry A. Krashinsky (2012). "Liquidity Constraints, Household Wealth, and Entrepreneurship Revisited". In: *Review of Income and Wealth* 58.2, pp. 197–394.

Federal Deposit Insurance Corporation (2005-2012). *Statistics on Depository Institutions*. Accessed August 24, 2016. URL: https://www7.fdic.gov/sdi/download%5C_large%5C_list%5C_outside.asp.

Fort, Teresa C., John Haltiwanger, Ron S. Jarmin, and Jaview Miranda (2013). "How Firms Respond to Business Cycles: The Role of Firm Age and Firm Size". In: *IMF Economic Review* 61.3, pp. 520–559.

Foster, Lucia, Cheryl Grim, and John Haltiwanger (2016). "Reallocation in he Great Recession: Cleansing or Not?" In: *Journal of Labor Economics* 34.S1, S293–S331.

Freddie Mac (2007-2012). *Freddie Mac House Price Index*. Accessed February 8, 2024. Federal Home Loan Mortgage Corporation. URL: https://www.freddiemac.com/fmac-resources/research/docs/fmhpi_master_file.csv.

Garner, Thesia I. and Randal Verbrugge (2009). "Reconciling User Costs and Rental Equivalence: Evidence from the US Consumer Expenditure Survey". In: *Journal of Housing Economics* 18.3, pp. 172–192.

Gertler, Mark and Peter Karadi (2011). "A Model of Unconventional Monetary Policy". In: *Journal of Monetary Economics* 58.1, pp. 17–34.

Gertler, Mark and Nobuhiro Kiyotaki (2015). "Banking, Liquidity, and Bank Runs in an Infinite Horizon Economy". In: *American Economic Review* 105.7, pp. 2011–2043.

Gopinath, Gita, Şebnem Kalemli-Özcan, Loukas Karabarbounis, and Carolina Villegas-Sanchez (2017). "Capital Allocation and Productivity in Southern Europe." In: *Quarterly Journal of Economics* 132.4, pp. 1915–1967.

Greenstone, Michael, Alexandre Mas, and Hoai-Luu Nguyen (2020). "Do Credit Market Shocks affect the Real Economy? Quasi-Experimental Evidence from the Great Recession and 'Normal' Economic Times". In: *American Economic Journal: Economic Policy* 12.1, pp. 200–225.

Greenwood, Jeremy, Zvi Hercowitz, and Gregory W. Huffman (1988). "Investment, Capacity Utilization, and the Real Business Cycle". In: *American Economic Review* 78.3, pp. 402–417.

Guren, Adam M., Alisdair McKay, Emi Nakamura, and Jòn Steinsson (2021). "Housing Wealth Effects: The Long View". In: *Review of Economic Studies* 88.2, pp. 669–707.

Harding, John P. and Stuart S. Rosenthal (2017). "Homeownership, Housing Capital Gains and Self-Employment". In: *Journal of Urban Economics* 99, pp. 120–135.

Headd, Brian (2003). "Redefining Business Success: Distinguishing Between Closure and Failure". In: *Small Business Economics* 21, pp. 51–61.

Hopenhayn, Hugo A. (1992). "Entry, Exit, and Firm Dynamics in Long Run Equilibrium". In: *Econometrica* 60.5, pp. 1127–1150.

Howard, Greg and Jack Liebersohn (Aug. 2025). "How Regional Inequality and Migration Drive Housing Prices and Rents". In: *Journal of Economic Perspectives* 39.3, pp. 3–26.

DOI: 10.1257/jep.20241426. URL: https://www.aeaweb.org/articles?id=10.1257/jep.20241426.

Hsieh, Chang-Tai and Peter J. Klenow (2009). "Misallocation and Manufacturing TFP in China and India." In: *Quarterly Journal of Economics* 124.4, pp. 1403–1448.

Hurst, Erik G and Benjamin W Pugsley (Oct. 2015). *Wealth, Tastes, and Entrepreneurial Choice*. Working Paper 21644. National Bureau of Economic Research. DOI: 10.3386/w21644. URL: http://www.nber.org/papers/w21644.

ICE Data Indices, LLC (2004-2006a). *ICE BofA AAA US Corporate Index Effective Yield [BAMLC0A1CAAAEY]*. Accessed October 27, 2024 via FRED, Federal Reserve Bank of St. Louis. URL: https://fred.stlouisfed.org/series/BAMLC0A1CAAAEY.

— (2004-2006b). *ICE BofA BBB US Corporate Index Effective Yield [BAMLC0A4CBBBEY]*. Accessed October 27, 2024 via FRED, Federal Reserve Bank of St. Louis. URL: https://fred.stlouisfed.org/series/BAMLC0A4CBBBEY.

Jovanovic, Boyan (1982). "Selection and the Evolution of Industry". In: *Econometrica* 50.3, pp. 649–670.

Kaplan, Greg, Kurt Mitman, and Giovanni L. Violante (2020). "The Housing Boom and Bust: Model Meets Evidence". In: *Journal of Political Economy* 128.9, pp. 3285–3345.

Kaplan, Greg, Benjamin Moll, and Giovanni L. Violante (2018). "Monetary Policy According to HANK". In: *American Economic Review* 108.3, pp. 697–743.

Kerr, Sari Pekkala, William R. Kerr, and Ramana Nanda (2022). "House prices, home equity and entrepreneurship: Evidence from U.S. census micro data". In: *Journal of Monetary Economics* 130, pp. 103–119. ISSN: 0304-3932. DOI: https://doi.org/10.1016/j.jmoneco.2022.06.002. URL: https://www.sciencedirect.com/science/article/pii/S030439322200085X.

Kerr, William R. and Ramana Nanda (2009). "Democratizing Entry: Banking Deregulations, Financing Constraints, and Entrepreneurship". In: *Journal of Financial Economics* 94, pp. 124–149.

Kiyotaki, Nobuhiro and John Moore (1997). "Credit Cycles". In: *Journal of Political Economy* 105.2, pp. 211–248.

Lastrapes, William D., Ian Schmutte, and Thor Watson (2022). "Home Equity Lending, Credit Constraints and Small Business in the United States". In: *Economic Inquiry* 60.1, pp. 43–64.

Mian, Atif and Amir Sufi (2014). "What Explains the 2007-2009 Drop in Employment?" In: *Econometrica* 82, pp. 2197–2223.

Petersen, Mitchell A. and Raghuram G. Rajan (1994). "The Benefits of Lending Relationships: Evidence from Small Business Data." In: *Journal of Finance* 49.1, pp. 3–37.

— (2002). "Does Distance Still Matter? The Information Revolution and Lending". In: *Journal of Finance* 57.6, pp. 2533–2570.

Phillipon, Thomas (2015). "Has the US Finance Industry Become Less Efficient?" In: *American Economic Review* 105.4, pp. 1408–1438.

Saiz, Albert (2010). "The Geographic Determinants of Housing Supply Elasticity". In: *Quarterly Journal of Economics* 125.3, pp. 1253–1296.

Schmalz, Martin C., David A. Sraer, and David Thesmar (2016). "Housing Collateral and Entrepreneurship". In: *Journal of Finance* 72.1, pp. 99–132.

Sedlàček, Petr and Vincent Sterk (2017). "The Growth Potential of Startups over the Business Cycle". In: *American Economic Review* 107.10, pp. 3182–3210.

Shaker Akhtekhane, Saeed (2017). "Firm Entry and Exit in Continuous Time". In: *Ohio State University Working Paper*.

Siemer, Michael (2019). "Employment Effects of Financial Constraints during the Great Recesssion". In: *Review of Economics and Statistics* 101.1, pp. 16–29.

U.S. Bureau of Economic Analysis (2004-2006). *Gross Domestic Product [GDP]*. Accessed October 27, 2024 via FRED, Federal Reserve Bank of St. Louis. URL: https://fred.stlouisfed.org/series/GDP.

US Census Bureau (2005-2012). *Business Dynamics Statistics*. Accessed August 21, 2024. URL: https://www.census.gov/programs-surveys/bds.html.

— (2007,2012). *Survey of Business Owners*. Accessed 2024. URL: https://www.census.gov/programs-surveys/sbo.html.

A Additional Figures and Tables

A.1 Additional Tables and Figures

Table A1: Emp. Growth Regressions by Firm Age/Size: Fully Interacted

	No ΔUR (OLS)	**With ΔUR (OLS)**	**No ΔUR (IV)**	**With ΔUR (IV)**
$\triangle$ HPI	0.135	0.164	0.128	0.142
	(0.0147)	(0.017)	(0.0177)	(0.0202)
Young $\times\triangle$ HPI	0.0931	0.104	0.141	0.147
	(0.0177)	(0.019)	(0.0263)	(0.0296)
Old/Large $\times\triangle$ HPI	-0.0810	-0.0926	-0.0899	-0.0916
	(0.0199)	(0.0194)	(0.0286)	(0.0255)
Obs.	22110000	22110000	22110000	22110000
R^2	0.0501	0.05	0.0118	0.0118
First-stage F-stat			97.66	99.34

Displays β_h and interactions with age categories in a fully interacted version of equation (1). Dependent variable is the DHS employment growth rate regressed on log changes in metro house price growth. Note: Young < 5yo, Old>=5 yo, Small <500 employees; includes log(DHS denominator), metro area effects, industry effects, and (region $\times$ year) effects. IV regression instruments for both house prices and interaction with mortgage share, house prices instrumented by changes in national house prices interacted with metro housing supply elasticities from Guren et al. (2021). First Stage Kleibergen-Paap rk Wald and Angrist Pischke F-statistics are reported for IV regressions. Standard error clustered at the metro level.

Table A2: Emp. Growth Regressions by Firm Age/Size: Comparison to Boom and Recovery

		Young (OLS)	**Old/Small (OLS)**	**Young (IV)**	**Old/Small (IV)**
Boom (1997-2006)	△ HPI	0.262	0.0965	0.135	-0.0459
		(0.0327)	(0.0147)	(0.0542)	(0.0598)
R^2		0.00673	0.0312	0.00445	0.0277
First-Stage F-stat				182.5	136.0
Obs.		14230000	25790000	14230000	25790000
Recovery (2013-2019)	△ HPI	0.315	0.103	0.209	0.121
		(0.0441)	(0.0204)	(0.0802)	(0.0377)
R^2		0.00779	0.0272	0.00569	0.0244
First-Stage F-stat				100.1	96.79
Obs.		9140000	20130000	9140000	20130000

Displays β_h in equation (1). Dependent variable is the DHS employment growth rate regressed on log changes in metro house price growth in the boom (1997-2006) and recovery (2013-2019) periods. Note: Young < 5yo, Old>=5 yo, Small <500 employees; includes log(DHS denominator), metro area effects, industry effects, and (region × year) effects. IV regression instruments for both house prices and interaction with mortgage share, house prices instrumented by changes in national house prices interacted with metro housing supply elasticities from Guren et al. (2021). First Stage Kleibergen-Paap rk Wald and Angrist Pischke F-statistics are reported for IV regressions. Standard error clustered at the metro level.

Table A3: Calibration

Parameter	Value	Target Description	Data	Model	Source
Household					
Targeted					
ν_l	0.61	Entry Rate	0.1	0.1	BDS (2004-2006 average)
ϕ	2.89	Mtg. debt-to-housing	0.39	0.42	FF (2004-2006 average)
ν_h	0.34	Owner-occupied share	0.685	0.685	Census (2002-2007 average)
ϵ_h	0.792	q_h/R_h	12	12	Garner and Verbrugge (2009)
Selected					
ρ	0.04	Deposit Rate	4%	4%	–
δ_h	0.03	Housing Depreciation	–	–	BEA
Production					
Targeted					
α	0.275	Entrep. Sector Labor Share	0.61	0.6	BEA (2004-2006 average)
γ	0.91	Markups	0.1	0.1	Basu and Fernald (1997)
σ_z	0.035	IQR of TFPR	0.5	0.47	Chow et al. (2021)
χ	0.49	S.D. of (I/K)	0.337	0.372	Cooper and Haltiwanger (2006)
ξ	1.26	Corporate Debt-to-Entrep. Production	0.418	0.424	FF/ BEA (2004-2006 average)
κ_e	3.11	Emp. Entry Rate	0.024	0.023	BDS (2004-2006 average)
c_{exit}	0.253	Mean Emp.	22.3	21.9	BDS (2005)
c_{ent}	6.54	EPOP	0.628	0.628	BLS (2004-2006 average)
η	0.725	Entry rate decline	0.076	0.077	Min. in GR
Selected					
$\overline{m}$	0.05	Entrep. to Pop. Share	0.05	0.05	GEM (2004-2006 average)
μ_z	0.151	Prod. Persistence	0.151	0.151	Asker, Collard-Wexler, and Loecker (2014)
δ	0.069	Capital Depreciation	0.069	0.069	Cooper and Haltiwanger (2006)
ζ	0.067	Nondistressed Exit	0.67	0.67	Headd (2003)
σ_h	1.5	Elasticity of Housing to price	2/3	2/3	Albouy, Ehrlich, and Liu (2016)
Banking					
Targeted					
θ	0.165	Leverage	10	10	FDIC (2005)
ψ	0.0147	Credit Spread	1%-4%	1.8%	ICE BofA
Selected					
σ	0.08	Death Rate	8%	8%	BDS (2005-2006 average)
ω_b	0.1	Entrant size/Mean Size	–	–	–

Note: Data sourced from U.S. Bureau of Economic Analysis (2004-2006), Board of Governors of the Federal Reserve System (US) (2004-2006a), Board of Governors of the Federal Reserve System (US) (2004-2006b), US Census Bureau (2005-2012), and Bureau of Labor Statistics (2007-2012) are denoted as such. Data from the BEA, FF, and BLS were accessed through FRED. FDIC targets are based on my tabulations from state bank data Federal Deposit Insurance Corporation, 2005-2012. ICE Data Indices, LLC (2004-2006a) and ICE Data Indices, LLC (2004-2006b) data from FRED displays range from "Investment Grade" options-adjusted spreads to "high-yield." Flow of Funds mortgages reflect all single-family mortgages, and housing from the flow of funds includes all household owner-occupied real estate.

ONLINE APPENDIX for "Same Shock, Separate Channel: House Prices and Firm Performance in the Great Recession"

B Additional Figures and Tables

B.1 Expanded Tables

Table OA1: Emp. Growth Regressions by Firm Age/Size: Mtg. Share

	Young	**Old**	**Old/Small**	**Old/Large**
OLS				
$\triangle$ HPI	0.351	0.215	0.220	0.0835
	(0.0495)	(0.0157)	(0.0187)	(0.0184)
$\triangle$ HPI*	0.127	0.0877	0.0920	0.0181
MTG_RATIO	(0.0451)	(0.0150)	(0.0172)	(0.0155)
Obs.	6760000	15350000	13880000	1470000
Adj. R^2	0.0108	0.0312	0.0349	0.0106
IV				
$\triangle\widehat{HPI}$	0.421	0.208	0.209	0.0666
	(0.0748)	(0.0245)	(0.0282)	(0.0305)
$\triangle\widehat{HPI}$*	0.245	0.114	0.128	0.0302
MTG_RATIO	(0.0826)	(0.0267)	(0.0312)	(0.0258)
Obs.	6760000	15350000	13880000	1470000
K-P F-stat	34.51	39.13	38.17	72.31
HPI A-P F-stat	161.9	100.5	96.71	149.8
$HPI * MTG_RATIO$ A-P F-stat	42.59	42.09	41.22	72.31
Cent. R^2	0.00204	0.0232	0.0271	0.000946

Displays $\beta_{q,0}$ and $\beta_{m,0}$ in equation (4). Dependent variable is the DHS employment growth rate regressed on log changes in metro house price growth and interaction of house prices with standardized state-bank mortgage share in 2005. Note: Young < 5yo, Old>=5 yo, Small <500 employees, Large >=500 employees; includes log(DHS denominator), metro area effects, industry effects, and (region × year) effects. IV regression instruments for both house prices and interaction with mortgage share, house prices instrumented by changes in national house prices interacted with metro housing supply elasticities from Guren et al. (2021). First Stage Kleibergen-Paap rk Wald and A-P F-statistics are reported for IV regressions. Clustered at metro level. Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

Table OA2: SBO - Startup/Continuing Financing Sources (Expanded)

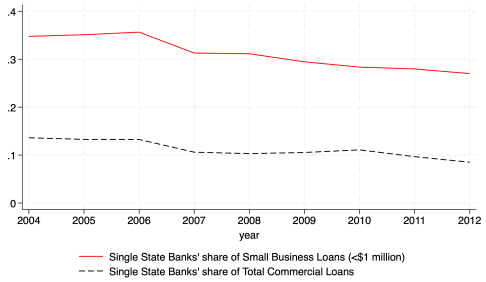
	2008			
	None/Other	Home Equity	Personal Saving	Total
Old	0.326	0.0589	0.343	0.727
	(0.000702)	(0.000358)	(0.00072)	(0.000694)
Young	0.0961	0.036	0.141	0.273
	(0.000465)	(0.000296)	(0.000544)	(0.000694)
Total	0.422	0.0949	0.483	1
	(0.000748)	(0.000453)	(0.00076)	
	2012			
	None/Other	Home Equity	Personal Saving	Total
Old	0.386	0.0714	0.391	0.848
	(0.000757)	(0.000403)	(0.000762)	(0.00058)
Young	0.087	0.00867	0.0562	0.152
	(0.000457)	(0.000152)	(0.000374)	(0.00058)
Total	0.473	0.08	0.447	1
	(0.00078)	(0.000427)	(0.000777)	

Source: Survey of Business Owners. Merged with sample underlying Table 1 for 2008 and 2012 (must appear in either the 2007 or 2012 survey). Sample-weighted, standard errors in parentheses. "Home Equity" indicates home equity loans provided either startup capital or continuing capital during the survey year. "Personal Saving" indicates personal assets or savings provided startup or continuing capital. Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

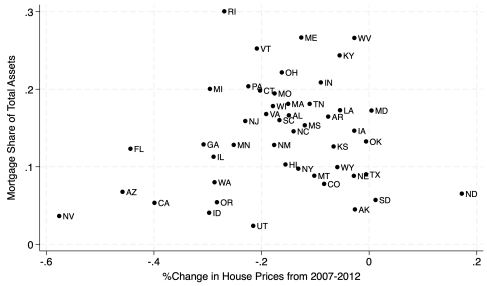
B.2 Additional Figures

Figure OA1: Single-State Banks and Single-Bank/Single-State BHC's Share of Lending and House Prices

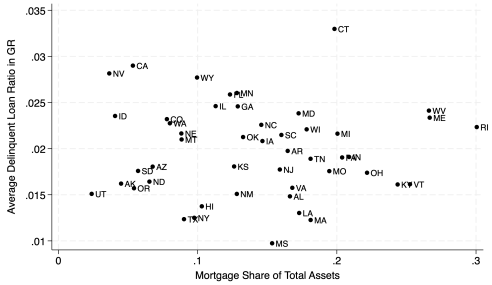
(a) Lending Share (percent of total valuation)



(b) Mortgage Share and House Price Changes

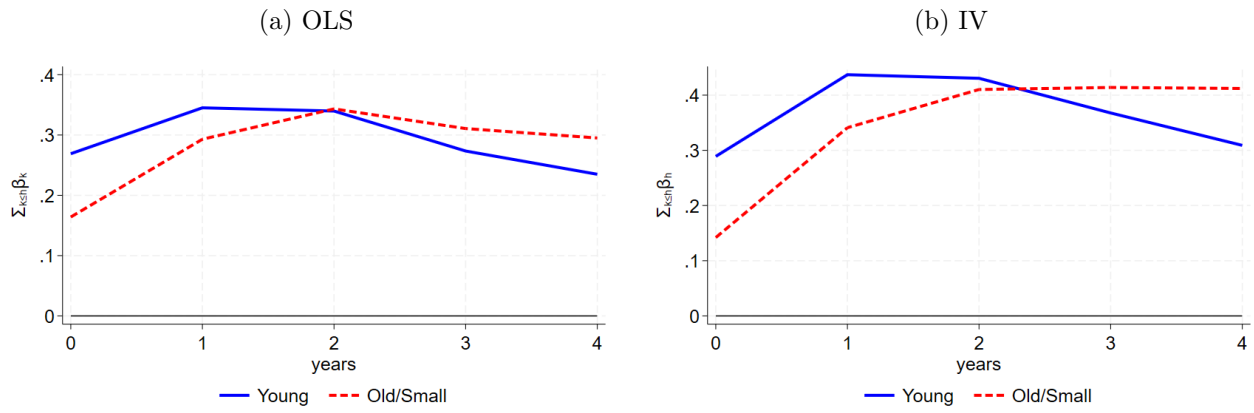


(c) Pre-crisis Mortgage Share and Subsequent Delinquency Rates for Commercial Loans



Author's Calculations from the FDIC Call Report Data and Freddie Mac HPI data. Mortgage Shares are calculated as an average of all single-state banks in 2005. Delinquency rates are calculated as the average annual delinquency ratio of delinquent commercial loans (noncurrent and 90 days past due) to total commercial loans in a state during the years 2007-2012.

Figure OA2: House Prices and Cumulative Employment Growth Dynamics - Age Comparison

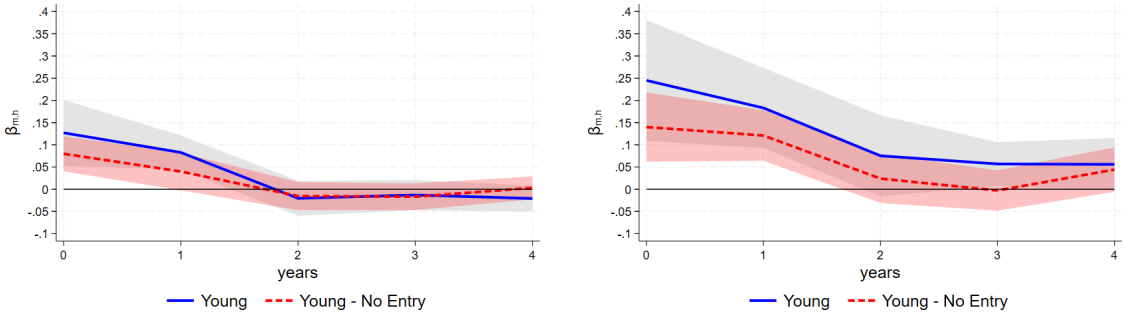


Displays $\sum_i^h \beta_i$ in equation (1)- semi-elasticity of employment at $t+h$ on house prices in t. Dependent variable is the DHS employment growth rate regressed on log changes in metro house price growth. Note: Young < 5yo, Old>=5 yo, Small <500 employees; controls include log(DHS denominator), metro area effects, industry effects, and (region × year) effects. IV regressions indicate house prices instrumented by changes in national house prices interacted with metro housing supply elasticities from Guren et al. (2021). Includes entrants in each year $t+h$, where post-shock entrants are given their initial size for the DHS denominator. Standard errors clustered at the metro level. Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

Figure OA3: House Prices and Local Bank Balance Sheets - No entry

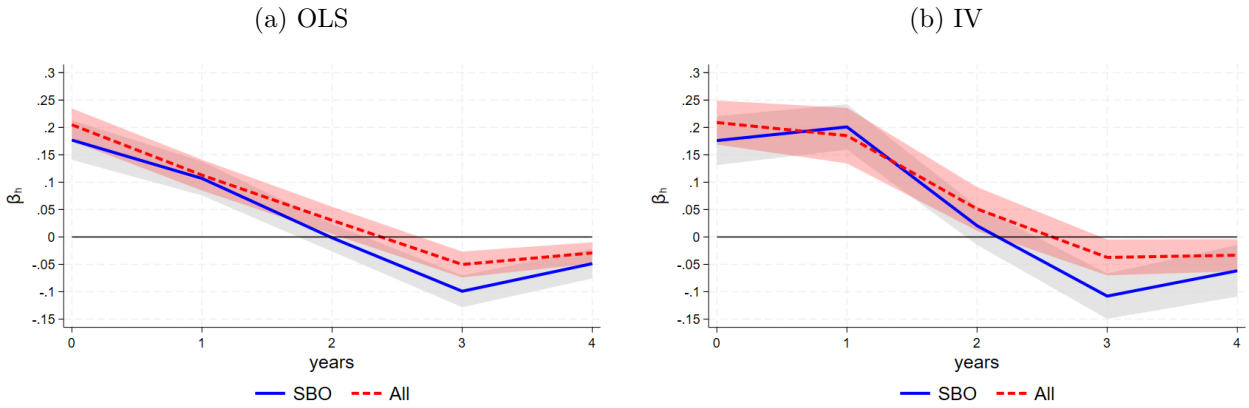
(a) OLS - Entry Comp.

(b) IV -Entry Comp.



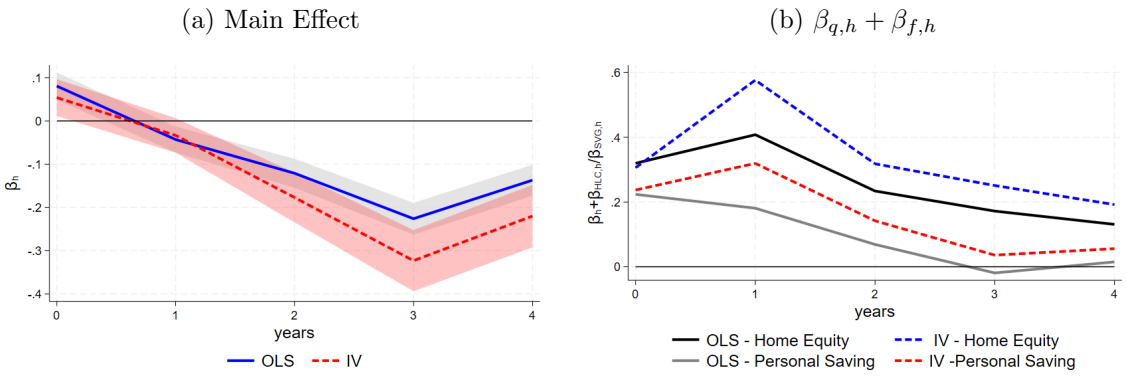
Displays $\beta_{m,h}$ in equation (4) - marginal impact on $\beta_{q,h}$ of a 1-standard deviation increase in 2005 mortgage share. Dependent variable is the DHS employment growth rate at time $t+h$ regressed on log changes in metro house price growth and interaction of house prices at time t with standardized state-bank mortgage share in 2005. Note: controls include log(DHS denominator), metro area effects, industry effects, and (region $\times$ year) effects. IV regressions indicate house prices instrumented by changes in national house prices interacted with metro housing supply elasticities from Guren et al. (2021). Area bands indicate $1.645 * SE$ of the estimated effect. Includes entrants in each year $t + h$, where post-shock entrants are given their initial size for the DHS denominator. See Appendix C for table format with observation counts, r-squared statistics, and F-statistics. Standard errors clustered at the metro level. Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

Figure OA4: House Prices and Employment Growth Dynamics - SBO (weighted) and Full Sample Comparison



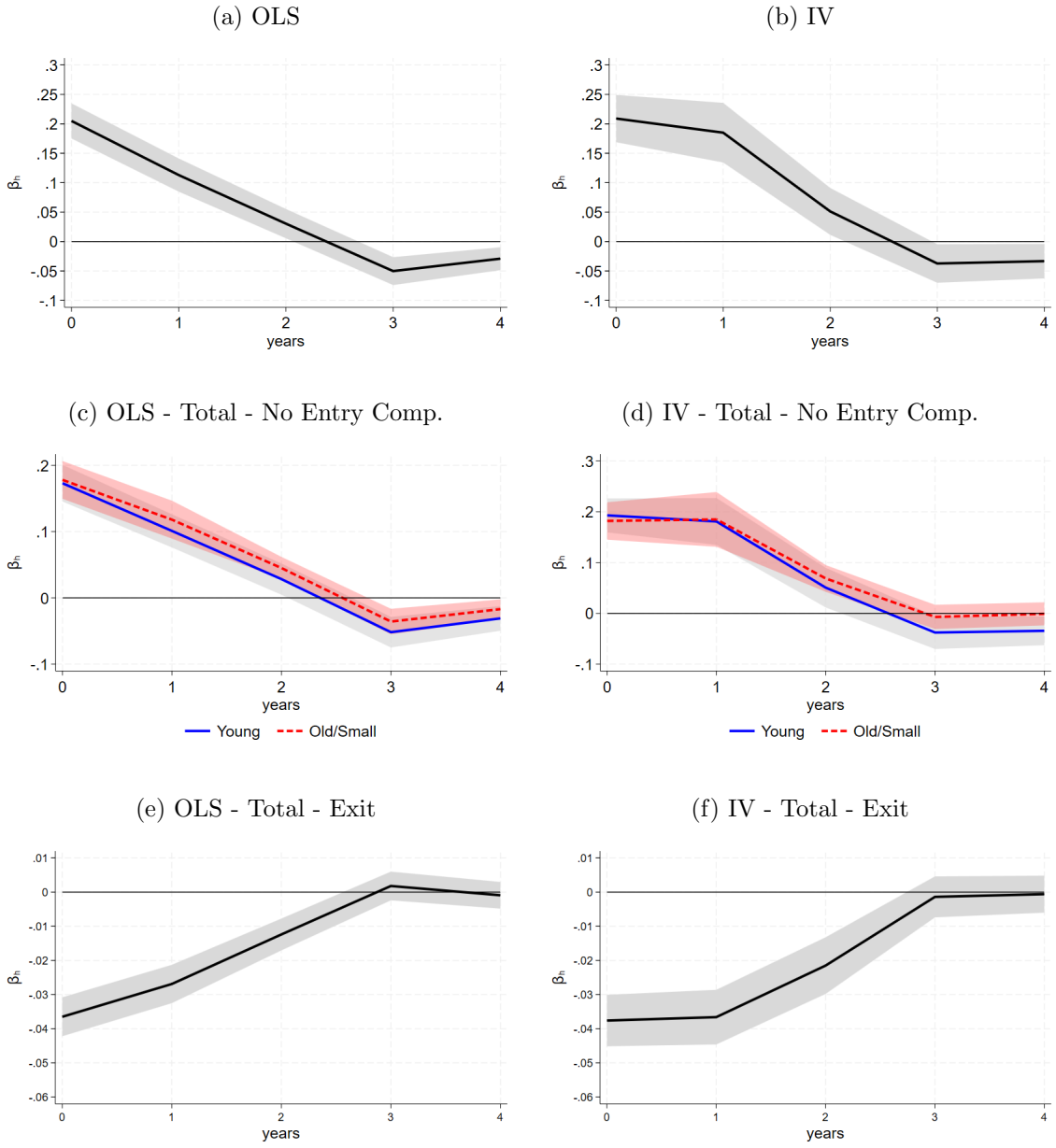
Displays β_h in equation (1)- effect on exit probability in $t+h$ of house prices in t. Dependent variable is an indicator for firm exit regressed on log changes in metro house price growth. Note: Young $<$ 5yo, Old$>=$5 yo, Small $<$500 employees; controls include log(DHS denominator), metro area effects, industry effects, and (region $\times$ year) effects. IV regressions indicate house prices instrumented by changes in national house prices interacted with metro housing supply elasticities from Guren et al. (2021). Includes entrants in each year $t+h$, where post-shock entrants are given their initial size for the DHS denominator. See appendix C for table format with observation counts, r-squared statistics, and F-statistics. Standard errors clustered at the metro level. Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

Figure OA5: Main and Cumulative Effects of House Prices by Finance Type in SBO



Displays $\beta_{q,h}+\beta_{f,h}$ where $f \in \{HLC, SVG\}$ in equation (3) - total effect house prices when indicator for home equity/personal saving financing dependent establishments. Dependent variable is the DHS employment growth rate at time $t+h$ regressed on log changes in metro house price growth and interaction of house prices at time t with finance indicators. Note: controls include log(DHS denominator), metro area effects, industry effects, and (region $\times$ year) effects. IV regressions indicate house prices instrumented by changes in national house prices interacted with metro housing supply elasticities from Guren et al. (2021). Includes entrants in each year $t+h$, where post-shock entrants are given their initial size for the DHS denominator. Standard Errors Clustered at the metro level. Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

Figure OA6: House Prices and Employment Growth Dynamics - Full Sample



Displays β_h in equation (1)- semi-elasticity of employment (in (a)-(d)) or exit ((e)-(f)) at $t+h$ on house prices in t. Dependent variable is the DHS employment growth rate regressed on log changes in metro house price growth. Note: controls include log(DHS denominator), metro area effects, industry effects, and (region $\times$ year) effects. IV regressions indicate house prices instrumented by changes in national house prices interacted with metro housing supply elasticities from Guren et al. (2021). Area bands indicate $1.645 * SE$ of the estimated effect. Includes entrants in each year $t+h$, where post-shock entrants are given their initial size for the DHS denominator. See Appendix C for table format with observation counts, r-squared statistics, and F-statistics. Standard Errors clustered at the metro level. Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

B.3 Additional Fully Interacted Tables

Table OA3: Emp. Growth Regressions by Firm Age/Size: Fully Interacted

	Boom (OLS)	**Recovery (OLS)**	**Boom (IV)**	**Recovery (IV)**
△ HPI	0.0965	0.103	-0.0459	0.121
	(0.0147)	(0.0204)	(0.0598)	(0.0377)
Young ×△ HPI	0.165	0.212	0.181	0.0874
	(0.0278)	(0.0349)	(0.055)	(0.0701)
Obs.	40020000	29270000	40020000	29270000
First-stage F-stat			67.98	50.04

Displays β_h and interactions with age categories in a fully interacted version of equation (1). Dependent variable is the DHS employment growth rate regressed on log changes in metro house price growth in the boom (1997-2006) and recovery (2013-2019) periods. Note: Young < 5yo, Old>=5 yo, Small <500 employees; includes log(DHS denominator), metro area effects, industry effects, and (region × year) effects. IV regression instruments for both house prices and interaction with mortgage share, house prices instrumented by changes in national house prices interacted with metro housing supply elasticities from Guren et al. (2021). First Stage Kleibergen-Paap rk Wald and Angrist Pischke F-statistics are reported for IV regressions. Standard error clustered at the metro level. Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

C Table Format for Figures

C.1 Figure 1

Table OA4: Panels (a)-(b) - Young

	gr_t	gr_{t+1}	gr_{t+2}	gr_{t+3}	gr_{t+4}
OLS					
$\triangle$ HPI	0.269	0.0761	-0.00537	-0.0663	-0.0385
	(0.0268)	(0.0213)	(0.0266)	(0.0221)	(0.0175)
N	6760000	7440000	8110000	8820000	9550000
Adj. R^2	0.0108	0.0407	0.0199	0.0139	0.0109
IV					
$\triangle$ HPI	0.289	0.148	-0.00649	-0.0628	-0.0587
	(0.0423)	(0.0368)	(0.0366)	(0.0291)	(0.0246)
N	6760000	7440000	8110000	8820000	9550000
K-P rk Wald F-stat	289.5	288.3	285.7	285.1	286.3
Cent. R^2	0.00204	0.0331	0.0162	0.0117	0.00865

Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

Table OA5: Panels (a)-(d) - Old/Small

	gr_t	gr_{t+1}	gr_{t+2}	gr_{t+3}	gr_{t+4}
OLS					
$\triangle$ HPI	0.164	0.129	0.0503	-0.0326	-0.0156
	(0.017)	(0.0173)	(0.011)	(0.0128)	(0.0107)
N	13880000	12740000	11790000	11030000	10370000
Adj. R^2	0.0349	0.0101	0.00467	0.00294	0.00281
IV					
$\triangle$ HPI	0.142	0.199	0.0692	0.00371	-0.00185
	(0.0202)	(0.0333)	(0.0172)	(0.0138)	(0.0145)
N	13880000	12740000	11790000	11030000	10370000
K-P rk Wald F-stat	179.1	174.7	171.3	169.5	168.4
Cent. R^2	0.0271	0.000702	0.000835	0.000333	0.00013

Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

Table OA6: Panels (c)-(d) - Young

	gr_t	gr_{t+1}	gr_{t+2}	gr_{t+3}	gr_{t+4}
OLS					
$\triangle$ HPI	0.164	0.0750	0.0117	-0.0638	-0.0383
	(0.0222)	(0.0242)	(0.0133)	(0.0151)	(0.0133)
N	5220000	4420000	3900000	3530000	3250000
Adj. R^2	0.105	0.0146	0.00579	0.00415	0.004
IV					
$\triangle$ HPI	0.175	0.138	0.021	-0.0284	-0.0241
	(0.0317)	(0.0442)	(0.0168)	(0.0211)	(0.0203)
N	5220000	4420000	3900000	3530000	3250000
K-P rk Wald F-stat	284.8	277	271.3	267.8	266.6
Cent. R^2	0.0906	0.00296	0.000308	0.000606	0.000636

Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

Table OA7: Panels (e)-(f) - Young

	exit_t	exit_{t+1}	exit_{t+2}	exit_{t+3}	exit_{t+4}
OLS					
$\triangle$ HPI	-0.0496	-0.0226	-0.00609	0.00842	0.00174
	(0.00532)	(0.00473)	(0.00391)	(0.00429)	(0.00359)
N	6760000	7440000	8110000	8820000	9550000
Adj. R^2	0.0656	0.013	0.0106	0.00944	0.00904
IV					
$\triangle$ HPI	-0.0497	-0.0316	-0.0161	0.00813	0.00617
	(0.0088)	(0.00643)	(0.00616)	(0.00576)	(0.00449)
N	6760000	7440000	8110000	8820000	9550000
K-P rk Wald F-stat	289.5	288.3	285.7	285.1	286.3
Cent. R^2	0.0554	0.00337	0.00358	0.00363	0.00369

Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

Table OA8: Panels (e)-(f) - Old/Small

	exit_t	exit_{t+1}	exit_{t+2}	exit_{t+3}	exit_{t+4}
OLS					
$\triangle$ HPI	-0.0268	-0.0295	-0.0155	-0.00265	-0.00345
	(0.00306)	(0.00328)	(0.00308)	(0.00227)	(0.00234)
N	13880000	12740000	11790000	11030000	10370000
Adj. R^2	0.0786	0.022	0.0138	0.0108	0.00923
IV					
$\triangle$ HPI	-0.0234	-0.0372	-0.0209	-0.00819	-0.00701
	(0.00328)	(0.00478)	(0.00508)	(0.00262)	(0.00337)
N	13880000	12740000	11790000	11030000	10370000
K-P rk Wald F-stat	179.1	174.7	171.3	169.5	168.4
Cent. R^2	0.0714	0.0152	0.00899	0.00694	0.00586

Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

C.2 Figure 2

Table OA9: SBO Finance Source Comparison

	gr_t	gr_{t+1}	gr_{t+2}	gr_{t+3}	gr_{t+4}
OLS					
$\triangle$ HPI	0.0807	-0.0431	-0.121	-0.226	-0.137
	(0.019)	(0.0183)	(0.0203)	(0.0223)	(0.0217)
$\triangle$ HPI*	0.239	0.451	0.355	0.398	0.268
$\mathbf{I}\{HELOC\}$	(0.0453)	(0.0334)	(0.0282)	(0.0272)	(0.0418)
$\triangle$ HPI*	0.143	0.224	0.190	0.207	0.152
$\mathbf{I}\{SAVING\}$	(0.0203)	(0.0151)	(0.0174)	(0.0174)	(0.0172)
N	5720000	5660000	5560000	5450000	5290000
Adj. R^2	0.0429	0.0388	0.022	0.0163	0.0115
IV					
$\triangle$ HPI	0.0540	-0.0335	-0.177	-0.323	-0.220
	(0.0259)	(0.0242)	(0.035)	(0.0429)	(0.0439)
$\triangle$ HPI*	0.252	0.610	0.495	0.574	0.412
$\mathbf{I}\{HELOC\}$	(0.0344)	(0.0476)	(0.0553)	(0.0796)	(0.0668)
$\triangle$ HPI*	0.183	0.353	0.319	0.359	0.276
$\mathbf{I}\{SAVING\}$	(0.0198)	(0.041)	(0.0507)	(0.0577)	(0.0438)
N	5720000	5660000	5560000	5450000	5290000
K-P rk Wald F-stat	64.83	64.66	64.27	64.39	64.35
HPI A-P F-stat	114.5	112.6	109.8	113.1	115.1
HPI*$\mathbf{I}\{HELOC\}$ A-P F-stat	131.3	140.4	134.5	125.7	117.5
HPI*$\mathbf{I}\{SAVING\}$ A-P F-stat	86.5	83.24	80.87	79.2	81.36
Cent. R^2	0.00214	0.01	0.00299	0.00203	0.000977

Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

C.3 Figure 3

Table OA10: Young - Mortgage Share Approach

	gr_t	gr_{t+1}	gr_{t+2}	gr_{t+3}	gr_{t+4}
OLS					
$\triangle$ HPI	0.351	0.129	-0.0183	-0.0746	-0.0517
	(0.0495)	(0.0296)	(0.0354)	(0.0282)	(0.0221)
$\triangle$ HPI*	0.127	0.0828	-0.0204	-0.013	-0.0209
MTG_RATIO	(0.0451)	(0.0233)	(0.0239)	(0.0206)	(0.0176)
N	6760000	7440000	8110000	8820000	9550000
Adj. R^2	0.0108	0.0407	0.0199	0.0139	0.0109
IV					
$\triangle$ HPI	0.421	0.246	0.0336	-0.0327	-0.0291
	(0.0748)	(0.0553)	(0.0495)	(0.0367)	(0.0317)
$\triangle$ HPI*	0.245	0.183	0.0753	0.0569	0.056
MTG_RATIO	(0.0826)	(0.0547)	(0.0553)	(0.0299)	(0.036)
N	6760000	7440000	8110000	8820000	9550000
K-P rk Wald F-stat	34.51	34.41	34.74	34.93	34.79
HPI A-P F-stat	161.9	161.8	160.7	160.6	161.3
HPI*MTG_RATIO A-P F-stat	42.59	42.72	43.24	43.61	43.72
Cent. R^2	0.00204	0.0331	0.0162	0.0117	0.00864

Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

Table OA11: Old/Small - Mortgage Share Approach

	gr_t	gr_{t+1}	gr_{t+2}	gr_{t+3}	gr_{t+4}
OLS					
$\triangle$ HPI	0.220	0.155	0.0452	-0.0397	-0.0224
	(0.0187)	(0.0231)	(0.0174)	(0.0147)	(0.0135)
$\triangle$ HPI*	0.0920	0.0419	-0.00851	-0.012	-0.0115
MTG_RATIO	(0.0172)	(0.0207)	(0.0148)	(0.016)	(0.0132)
N	13880000	12740000	11790000	11030000	10370000
Adj. R^2	0.0349	0.0101	0.00467	0.00294	0.00281
IV					
$\triangle$ HPI	0.209	0.228	0.0761	0.000991	0.0117
	(0.0282)	(0.0378)	(0.0206)	(0.0152)	(0.0206)
$\triangle$ HPI*	0.128	0.0567	0.0132	-0.00523	0.026
MTG_RATIO	(0.0312)	(0.0246)	(0.0183)	(0.0177)	(0.0208)
N	13880000	12740000	11790000	11030000	10370000
K-P rk Wald F-stat	38.17	38.18	38.31	38.42	38.53
HPI A-P F-stat	96.71	94.32	92.45	91.44	90.76
HPI*MTG_RATIO A-P F-stat	41.22	40.97	40.95	40.97	41.02
Cent. R^2	0.0271	0.000709	0.000834	0.000333	0.000127

Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

D Empirical Robustness Exercises

D.1 Bank Exit Approach

I consider a two-stage setup where bank health is proxied by bank exit. Bank health is a complicated multi-dimensional object, so it is difficult to develop a "bank health" index. However, I can observe bank exit over the period I am investigating. Since bank exit is endogenous to firm performance, I develop a predictive model of bank exit in my first stage with pre-crisis bank balance sheet variables interacted with house prices on the right hand side. In a sense, predicted bank exit acts as an index for the interaction of house prices with bank balance sheets. Specifically I include as a regressor the exit rate of banks weighted by total assets $BE_{s,t}$ in the state s in which county c is located, either by failure or acquisition:[34]

$$\Delta Y_{e,t+h} = \gamma \Delta UR_{m,t} + \beta_{\mathbf{1}} \mathbf{\Delta HPI_{m,t}} + \beta_{\mathbf{3}} \mathbf{BE_{s,t}} + \alpha_{m,h} + \delta_{r,h} \times \delta_{t,h} + \tau_{i,h} + \mu_h size_{e,t} + \epsilon_{e,t+h} \quad (20)$$

As before, I run this regression separately by firm age and size. Since bank exit is clearly endogenous to contemporaneous business conditions, I instrument for bank exit with interactions of house prices and bank balance sheet variables as in Cole and White (2012) (variables expressed as a percentage of assets unless otherwise noted):

- Return on Average Assets: Interest and Noninterest income
- Efficiency Ratio: (noninterest income + net interest margin)/noninterest expense
- Equity Ratio
- Core Deposit Ratio: Non-brokered deposits
- Money Market Ratio
- Security Ratio
- Mortgage Backed Security Ratio: Agency and non-Agency
- Mortgage Ratio: 1-4 family homes
- Home Equity Lines of Credit
- Non-mortgage RE Ratio
- Commercial Loan Ratio
- Consumer Loan Ratio
- Non-Performing Loan Ratio

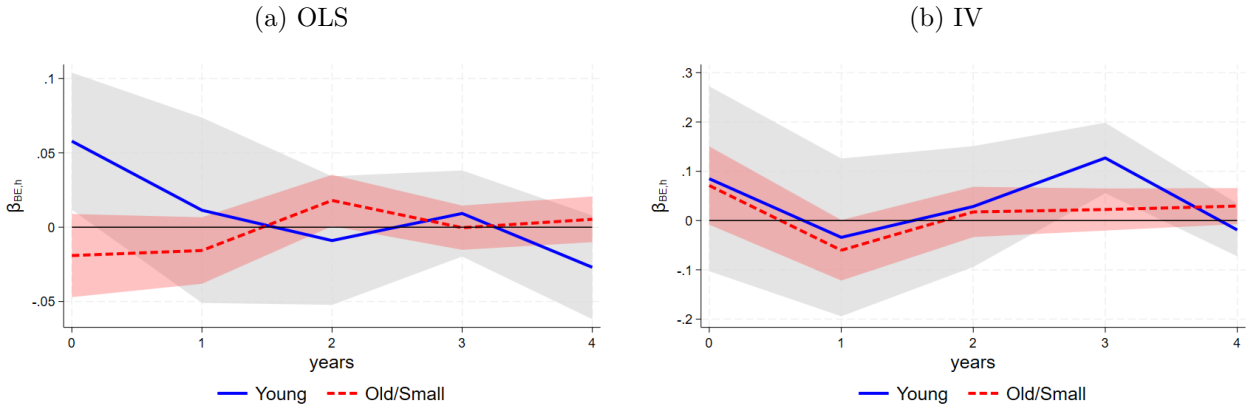
[34] While failure is a clear sign of distress, and exit in normal times is not necessarily a sign of distress, it is reasonable to assume the tremendous increases in both failures and exits are likely due to distress given the financial environment in the time frame considered.

- Loan Loss Allowance Ratio

The first stage for an instrumental variable approach is given by:

$$BE_{s,t} = \Gamma \Delta UR_{m,t} + \sum_j \zeta_j \Delta HPI_{m,t} * BBS_{j,s,2005} + a_m + d_r \times d_t + \varepsilon_{m,s,t} \tag{21}$$

Figure OA7: House Prices, Employment Growth Dynamics, and Local Credit Supply - Bank Exit Approach



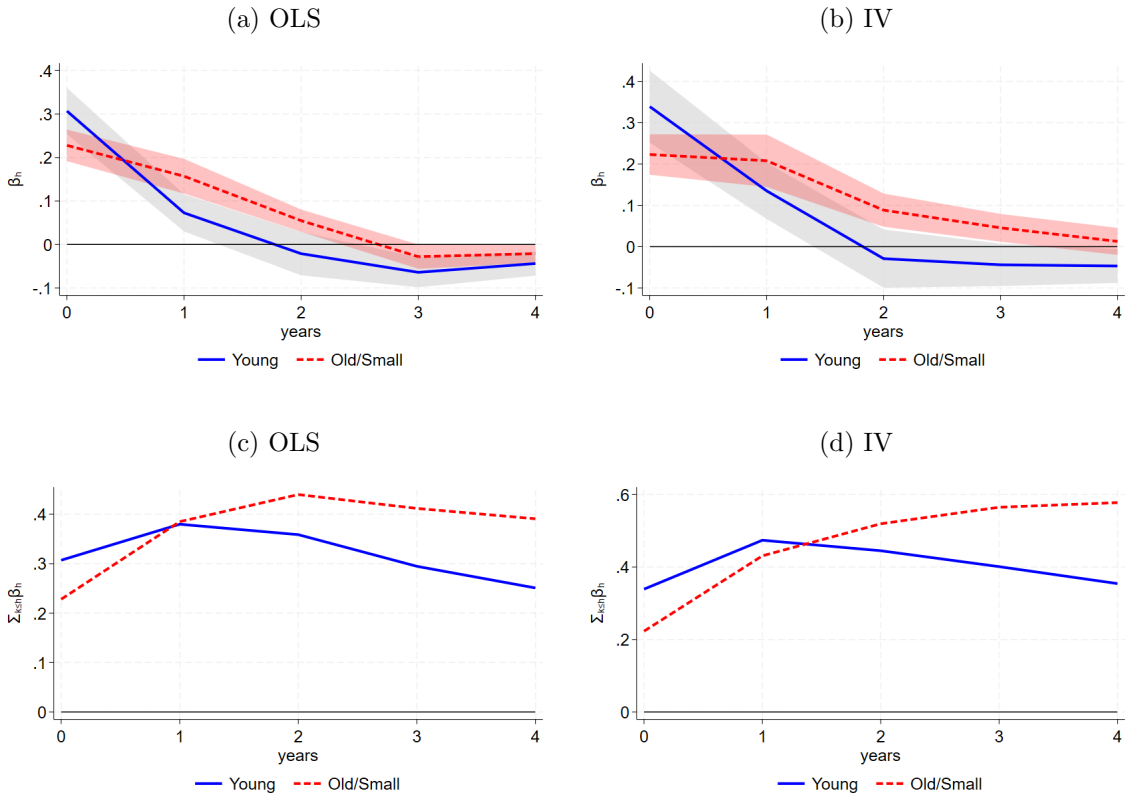
Displays $\beta_{BE,h}$ in equation (20)- semi-elasticity of employment growth at $t+h$ of share of state bank assets at exiting banks in t. Dependent variable is the DHS employment growth rate regressed on log changes in metro house price growth. Note: controls include log(DHS denominator), metro area effects, industry effects, and (region $\times$ year) effects. IV regressions indicate house prices instrumented by changes in national house prices interacted with metro housing supply elasticities from Guren et al. (2021). Area bands indicate $1.645 * SE$ of the estimated effect. Includes entrants in each year $t+h$, where post-shock entrants are given their initial size for the DHS denominator. See Appendix C for table format with observation counts, r-squared statistics, and F-statistics. Standard Errors clustered at the metro level. Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

Here, each bank balance sheet variable $BBS_{j,s,2005}$ listed in the appendix is normalized to mean zero in 2005. The identifying assumption is that $\widehat{BE}_{s,t} = \widehat{\zeta}_j \Delta HPI_{m,t} * BBS_{j,s,2005}$ is orthogonal to the disturbances in the second stage equation. I then employ the same state

unemployment housing supply elasticity instruments as before, along with interactions with balance sheet variables, to isolate plausibly exogenous house price growth interacting with balance sheet variables to predict bank exit $BE_{s,t}$.

D.2 Payroll Regressions

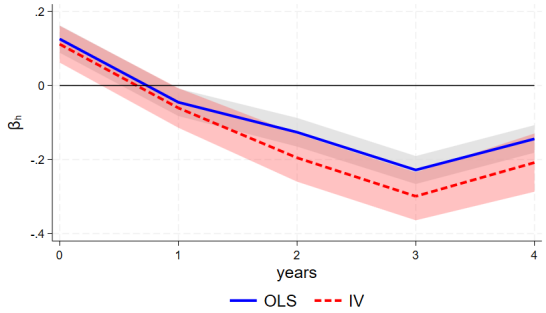
Figure OA8: House Prices and Employment Growth Dynamics - Age Comparison



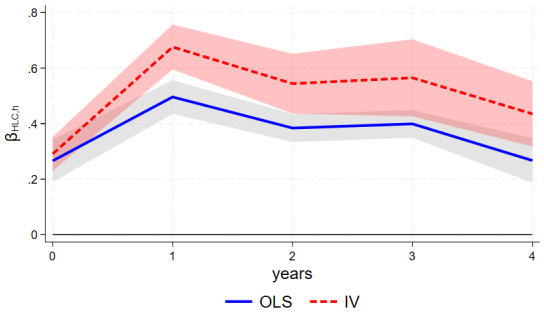
Displays β_h in equation (1)- semi-elasticity of payroll at year $t+h$ on house prices in t. Dependent variable is the DHS payroll growth rate in (a)-(d) and a firm exit indicator in (e) and (f), regressed on log changes in metro house price growth. Note: Young $<$ 5yo, Old$>$=5 yo, Small $<$500 employees; controls include log(DHS denominator), metro area effects, industry effects, and (region $\times$ year) effects. IV indicates house prices instrumented by changes in national house prices interacted with metro housing supply elasticities from Guren et al. (2021). Area bands indicate $\pm 1.645 * SE$. Includes entrants in each year $t+h$, where post-shock entrants are given the initial size for the DHS denominator. Standard error clustered at the metro level. Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

Figure OA9: House Prices and Entrepreneurial Collateral

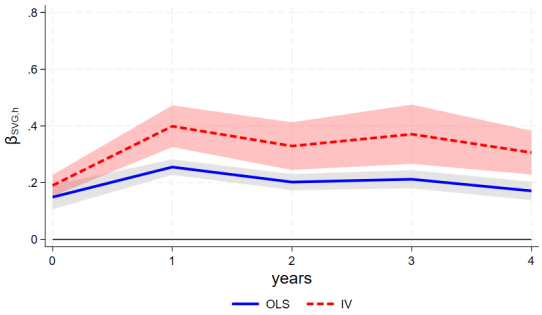
(a) House Price



(b) Home Equity



(c) Personal Saving

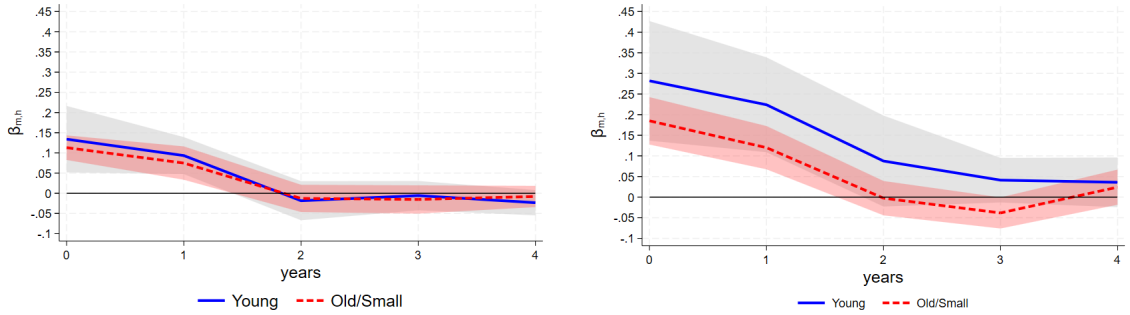


Displays $\beta_{q,h}$ and $\beta_{f,h}$ in equation (3) - marginal impact on $\beta_{q,h}$ of a indicator for home equity/personal saving financing ($f \in SVG, HLC$). Dependent variable is the DHS payroll growth rate at year $t+h$ regressed on log changes in metro house price growth and interaction of house prices at year t with finance indicators. Note: controls include log(DHS denominator), metro area effects, industry effects, and (region $\times$ year) effects. IV regressions indicate house prices instrumented by changes in national house prices interacted with metro housing supply elasticities from Guren et al. (2021). Area bands indicate $1.645 * SE$ of the estimated effect. Includes entrants in each year $t + h$, where post-shock entrants are given their initial size for the DHS denominator. Standard error clustered at the metro level. Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

Figure OA10: House Prices and Local Bank Balance Sheets

(a) OLS

(b) IV



Displays $\beta_{m,h}$ in equation (4) - marginal impact on $\beta_{q,h}$ of a 1-standard deviation increase in 2005 mortgage share. Dependent variable is the DHS employment growth rate at year $t+h$ regressed on log changes in metro house price growth and interaction of house prices at time t with standardized state-bank mortgage share in 2005. Note: controls include log(DHS denominator), metro area effects, industry effects, and (region $\times$ year) effects. IV indicates house prices instrumented by changes in national house prices interacted with metro housing supply elasticities from Guren et al. (2021). Area bands indicate $\pm 1.645 * SE$. Includes entrants in year $t+h$, where post-shock entrants are given their initial size for the DHS denominator. See Appendix C for table format with observation counts, r-squared statistics, and F-statistics. Standard error clustered at the metro level. Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

Table OA12: Payroll Growth Regressions by Firm Age/Size: Fully Interacted

	Payroll (OLS)	**Payroll (IV)**
△ HPI	0.228	0.223
	(0.022)	(0.0298)
Young ×△ HPI	0.0784	0.116
	(0.0219)	(0.0326)
Obs.	20590000	20590000
First-stage F-stat		89.47

Displays β_h and interactions with age categories in a fully interacted version of equation (1). Dependent variable is the DHS payroll growth rate regressed on log changes in metro house price growth. Note: Young < 5yo, Old>=5 yo, Small <500 employees; includes log(DHS denominator), metro area effects, industry effects, and (region × year) effects. IV regression instruments for both house prices and interaction with mortgage share, house prices instrumented by changes in national house prices interacted with metro housing supply elasticities from Guren et al. (2021). First Stage Kleibergen-Paap rk Wald and Angrist Pischke F-statistics are reported for IV regressions. Standard error clustered at the metro level. Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

D.3 Demand Channel

One potential alternative explanation for the results is a demand channel, as emphasized by Mian and Sufi (2014). I take two approaches to address this potential contributing factor. I control for local demand effects by controlling for metro-area unemployment rates, as in Davis and Haltiwanger (2024), and I restrict my analysis to "tradable" industries as in Mian and Sufi (2014).

D.3.1 Controlling for unemployment rate changes

Following Davis and Haltiwanger (2024), I control for the unemployment rate changes at the metro-level to eliminate general demand conditions in the metro area. In general, Tables OA13-OA14 and Figures OA11-OA13 show similar effects to the main analysis.

Table OA13: Employment Growth Regressions on House Prices by Firm Age/Size (with change in unemployment rate)

	Total	**Young**	**Old**	**Old/Small**	**Old/Large**
OLS					
$\triangle$ HPI	0.173	0.228	0.134	0.135	0.0543
	(0.0168)	(0.0239)	(0.0136)	(0.0147)	(0.0134)
Obs.	22110000	6760000	15350000	13880000	1470000
Adj. R^2	0.0414	0.0109	0.0312	0.035	0.0106
IV					
$\triangle\widehat{HPI}$	0.193	0.269	0.136	0.128	0.0384
	(0.0203)	(0.0338)	(0.0154)	(0.0177)	(0.0242)
Obs.	22110000	6760000	15350000	13880000	1470000
First Stage F-stat	193.2	246.3	172.6	165	292.8
Cent. R^2	0.00448	0.00209	0.0233	0.0272	0.00098

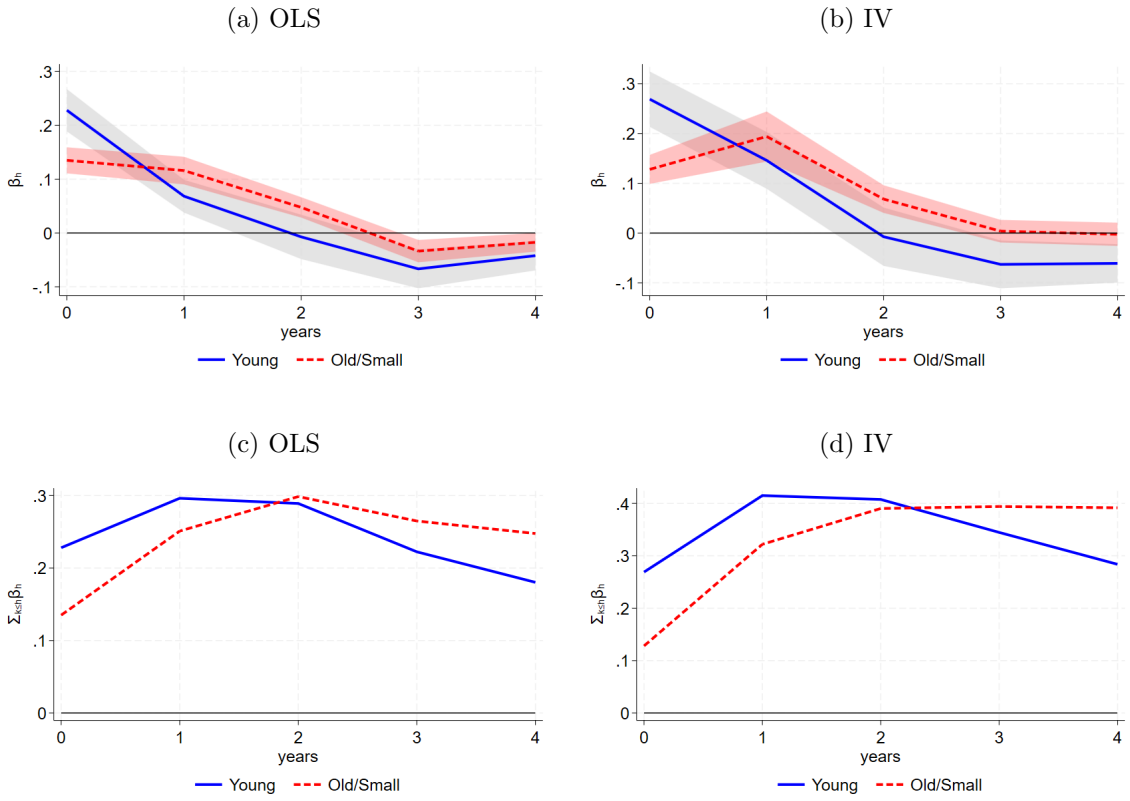
Displays β_0 in equation (1)- semi-elasticity of employment to house prices. Dependent variable is the DHS employment growth rate regressed on log changes in metro house price growth. Note: Young < 5yo, Old>=5 yo, Small <500 employees, Large >=500 employees; controls include log(DHS denominator), metro area effects, industry effects, (region $\times$ year) effects, and changes in metro area unemployment rates. $\triangle\widehat{HPI}$ indicates house prices instrumented by changes in national house prices interacted with metro housing supply elasticities from Guren et al. (2021). First Stage Kleibergen-Paap rk Wald and Angrist Pischke F-statistics are reported for IV regressions. Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

Table OA14: Emp. Growth Regressions by Firm Age/Size: Mtg. Share (with change in unemployment rate)

	Young	**Old**	**Old/Small**	**Old/Large**
OLS				
$\triangle$ HPI	0.307	0.185	0.188	0.0643
	(0.0480)	(0.0150)	(0.0176)	(0.0174)
$\triangle$ HPI*	0.121	0.0824	0.0863	0.0155
MTG_RATIO	(0.0483)	(0.0164)	(0.0189)	(0.0151)
Obs.	6760000	15350000	13880000	1470000
Adj. R^2	0.0109	0.0312	0.035	0.0106
IV				
$\triangle\widehat{HPI}$	0.386	0.184	0.183	0.0463
	(0.0640)	(0.0206)	(0.0237)	(0.0283)
$\triangle\widehat{HPI}$*	0.216	0.0921	0.105	0.0147
MTG_RATIO	(0.0804)	(0.0268)	(0.0313)	(0.0243)
Obs.	6760000	15350000	13880000	1470000
K-P F-stat	35.78	40.61	39.57	75.57
HPI A-P F-stat	130.6	90.32	86.51	144.7
$HPI * MTG_RATIO$ A-P F-stat	48.66	45.36	44.56	76.39
Cent. R^2	0.00209	0.0233	0.0272	0.00098

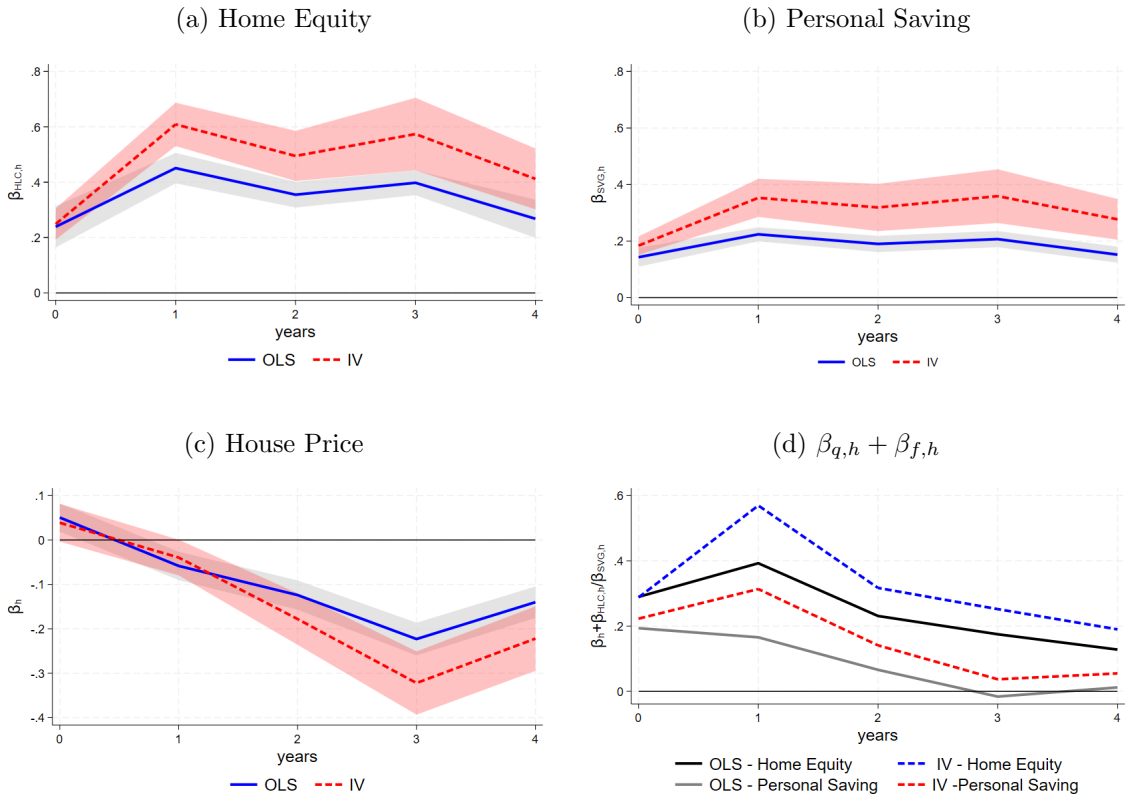
Displays $\beta_{q,0}$ and $\beta_{m,0}$ in equation (4). Dependent variable is the DHS employment growth rate regressed on log changes in metro house price growth and interaction of house prices with standardized state-bank mortgage share in 2005. Note: Young < 5yo, Old>=5 yo, Small <500 employees, Large >=500 employees; includes log(DHS denominator), metro area effects, industry effects, (region × year) effects, and changes in metro area unemployment rates. IV regression instruments for both house prices and interaction with mortgage share, house prices instrumented by changes in national house prices interacted with metro housing supply elasticities from Guren et al. (2021). First Stage Kleibergen-Paap rk Wald and Angrist Pischke F-statistics are reported for IV regressions. Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

Figure OA11: House Prices and Employment Growth Dynamics - Age Comparison - with unemployment rate control



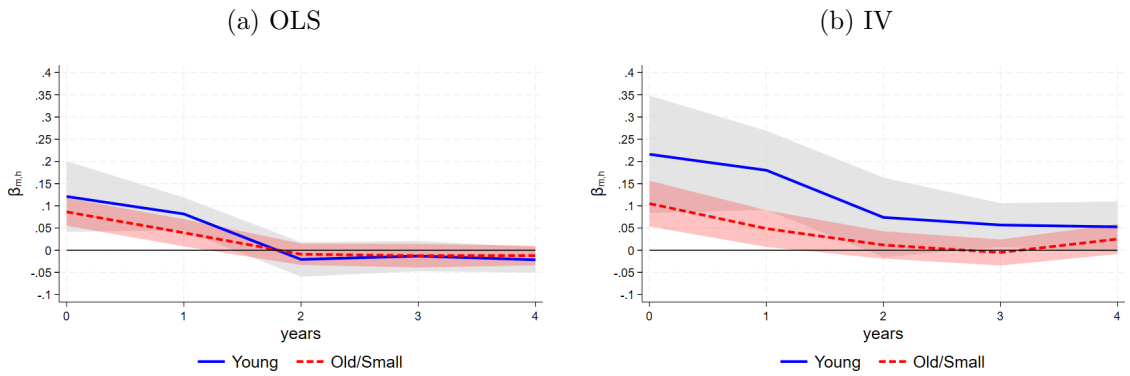
Displays β_h in equation (1)- semi-elasticity of employment at year $t+h$ on house prices in t. Dependent variable is the DHS employment growth rate regressed on log changes in metro house price growth. Note: Young < 5yo, Old>=5 yo, Small <500 employees; controls include log(DHS denominator), metro area effects, industry effects, (region × year) effects, and change in metro unemployment rate. IV indicates house prices instrumented by changes in national house prices interacted with metro housing supply elasticities from Guren et al. (2021). Area bands indicate $\pm 1.645 * SE$. Includes entrants in each year $t+h$, where post-shock entrants are given the initial size for the DHS denominator. See Appendix C for table with observation counts, r-squared statistics, and F-statistics. Standard error clustered at the metro level. Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

Figure OA12: House Prices and Entrepreneurial Collateral - with unemployment rate control



Displays $\beta_{q,h}$ and $\beta_{f,h}$ in equation (3) - marginal impact on $\beta_{q,h}$ of a indicator for home equity/personal saving financing ($f \in SVG, HLC$). Dependent variable is the DHS employment growth rate at year $t+h$ regressed on log changes in metro house price growth and interaction of house prices at year t with finance indicators. Note: controls include log(DHS denominator), metro area effects, industry effects, (region $\times$ year) effects, and changes in metro area unemployment rates. IV regressions indicate house prices instrumented by changes in national house prices interacted with metro housing supply elasticities from Guren et al. (2021). Area bands indicate $1.645 * SE$ of the estimated effect. Includes entrants in each year $t+h$, where post-shock entrants are given their initial size for the DHS denominator. See Appendix C for table format with observation counts, r-squared statistics, and F-statistics. Standard error clustered at the metro level. Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

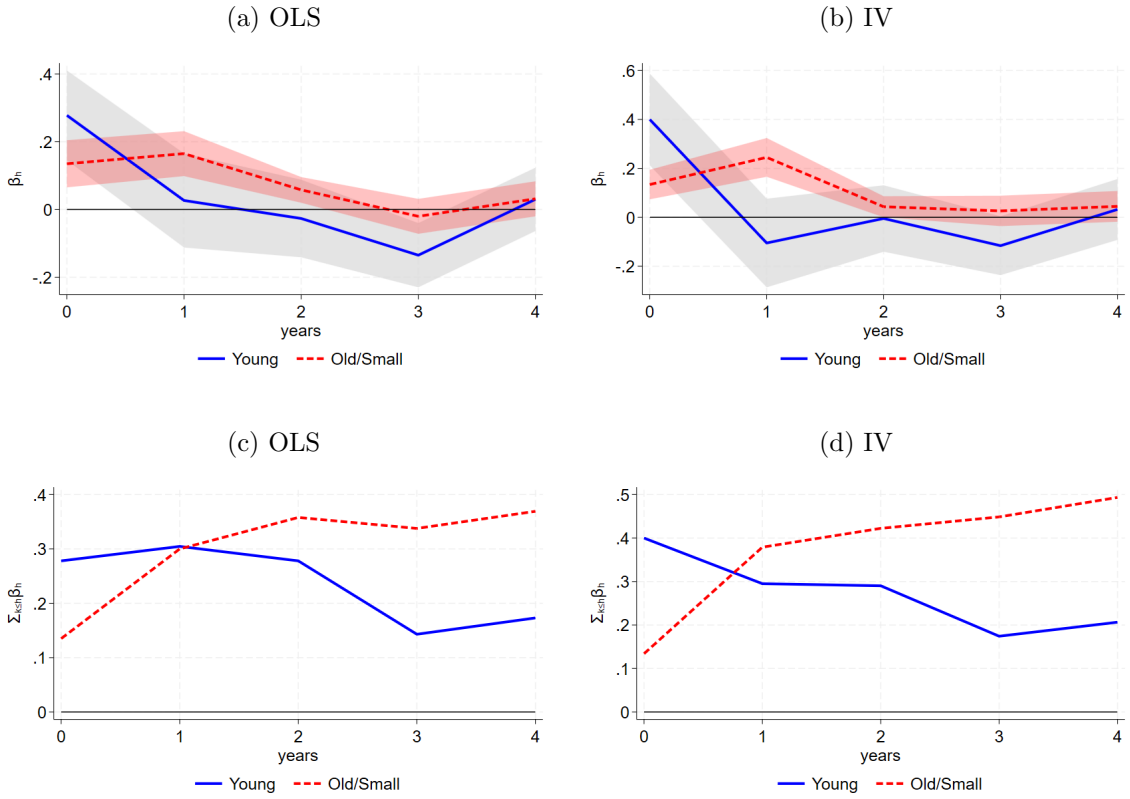
Figure OA13: House Prices and Local Bank Balance Sheets - with unemp. rate control



Displays $\beta_{m,h}$ in equation (4) - marginal impact on $\beta_{q,h}$ of a 1-standard deviation increase in 2005 mortgage share. Dependent variable is the DHS employment growth rate at year $t+h$ regressed on log changes in metro house price growth and interaction of house prices at time t with standardized state-bank mortgage share in 2005. Note: controls include log(DHS denominator), metro area effects, industry effects, (region $\times$ year) effects, and change is metro area unemployment rates. IV indicates house prices instrumented by changes in national house prices interacted with metro housing supply elasticities from Guren et al. (2021). Area bands indicate $\pm 1.645 * SE$. Includes entrants in year $t+h$, where post-shock entrants are given their initial size for the DHS denominator. See Appendix C for table format with observation counts, r-squared statistics, and F-statistics. Standard error clustered at the metro level. Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

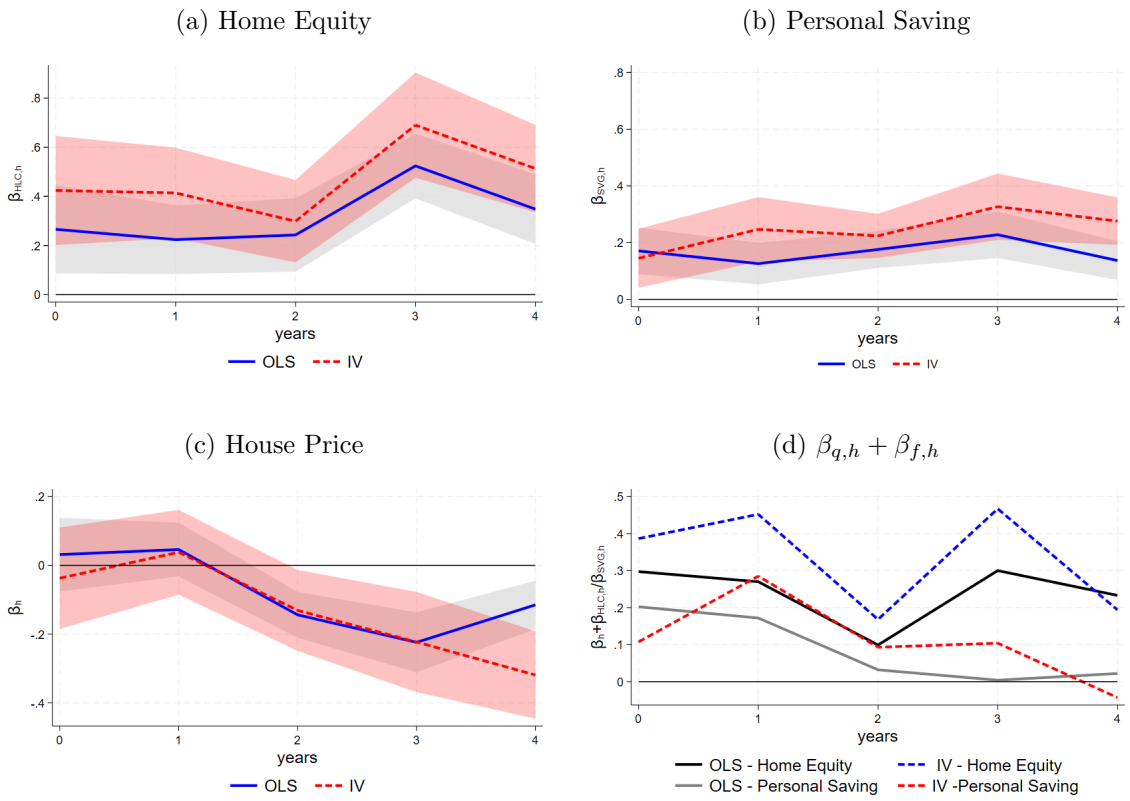
D.3.2 Tradable Sectors

Figure OA14: House Prices and Employment Growth Dynamics - Age Comparison (Tradable sectors)



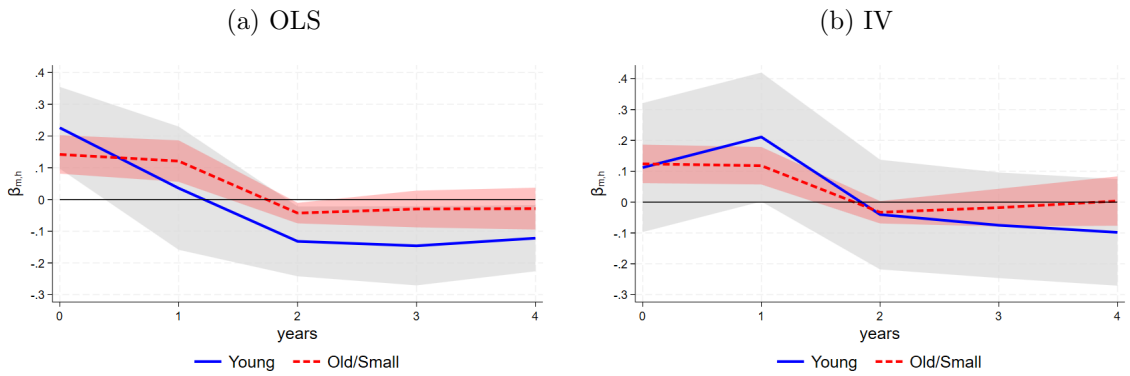
Displays β_h in equation (1)- semi-elasticity of employment at $t+h$ on house prices in t. Dependent variable is the DHS employment growth rate regressed on log changes in metro house price growth. Note: Young $<$ 5yo, Old$>$=5 yo, Small $<$500 employees; controls include log(DHS denominator), metro area effects, industry effects, (region $\times$ year) effects, and changes in metro area unemployment rates. IV regressions indicate house prices instrumented by changes in national house prices interacted with metro housing supply elasticities from Guren et al. (2021). Area bands indicate $1.645 * SE$ of the estimated effect. Includes entrants in each year $t+h$, where post-shock entrants are given their initial size for the DHS denominator. See Appendix C for table format with observation counts, r-squared statistics, and F-statistics. Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

Figure OA15: House Prices and Entrepreneurial Collateral



Displays $\beta_{q,h}$ and $\beta_{f,h}$ in equation (3) - marginal impact on $\beta_{q,h}$ of a indicator for home equity/personal saving financing ($f \in SVG, HLC$). Dependent variable is the DHS employment growth rate at year $t+h$ regressed on log changes in metro house price growth and interaction of house prices at year t with finance indicators. Note: controls include log(DHS denominator), metro area effects, industry effects, (region $\times$ year) effects, and changes in metro area unemployment rates. IV regressions indicate house prices instrumented by changes in national house prices interacted with metro housing supply elasticities from Guren et al. (2021). Area bands indicate $1.645 * SE$ of the estimated effect. Includes entrants in each year $t+h$, where post-shock entrants are given their initial size for the DHS denominator. See Appendix C for table format with observation counts, r-squared statistics, and F-statistics. Standard error clustered at the metro level. Tradable sectors only. Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

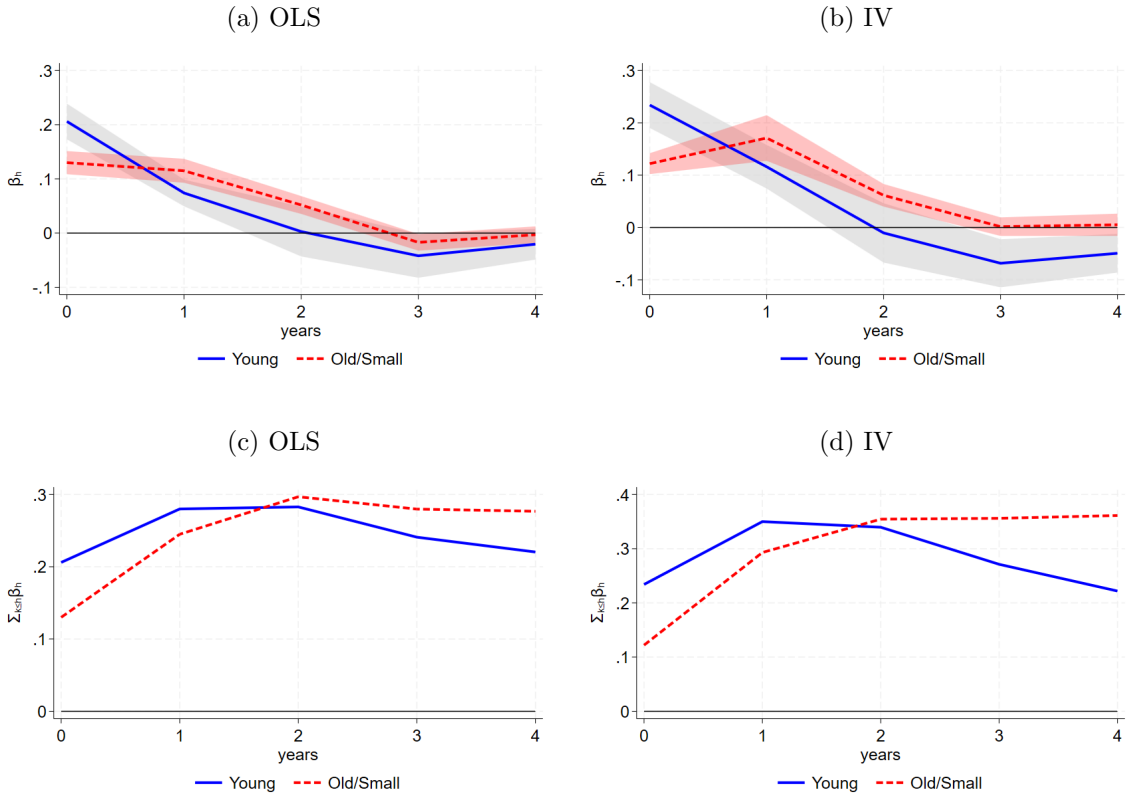
Figure OA16: House Prices and Local Bank Balance Sheets



Displays $\beta_{m,h}$ in equation (4) - marginal impact on $\beta_{q,h}$ of a 1-standard deviation increase in 2005 mortgage share. Dependent variable is the DHS employment growth rate at year $t+h$ regressed on log changes in metro house price growth and interaction of house prices at time t with standardized state-bank mortgage share in 2005. Note: controls include log(DHS denominator), metro area effects, industry effects, (region $\times$ year) effects, and change is metro area unemployment rates. IV indicates house prices instrumented by changes in national house prices interacted with metro housing supply elasticities from Guren et al. (2021). Area bands indicate $\pm 1.645 * SE$. Includes entrants in year $t+h$, where post-shock entrants are given their initial size for the DHS denominator. See Appendix C for table format with observation counts, r-squared statistics, and F-statistics. Standard error clustered at the metro level. Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

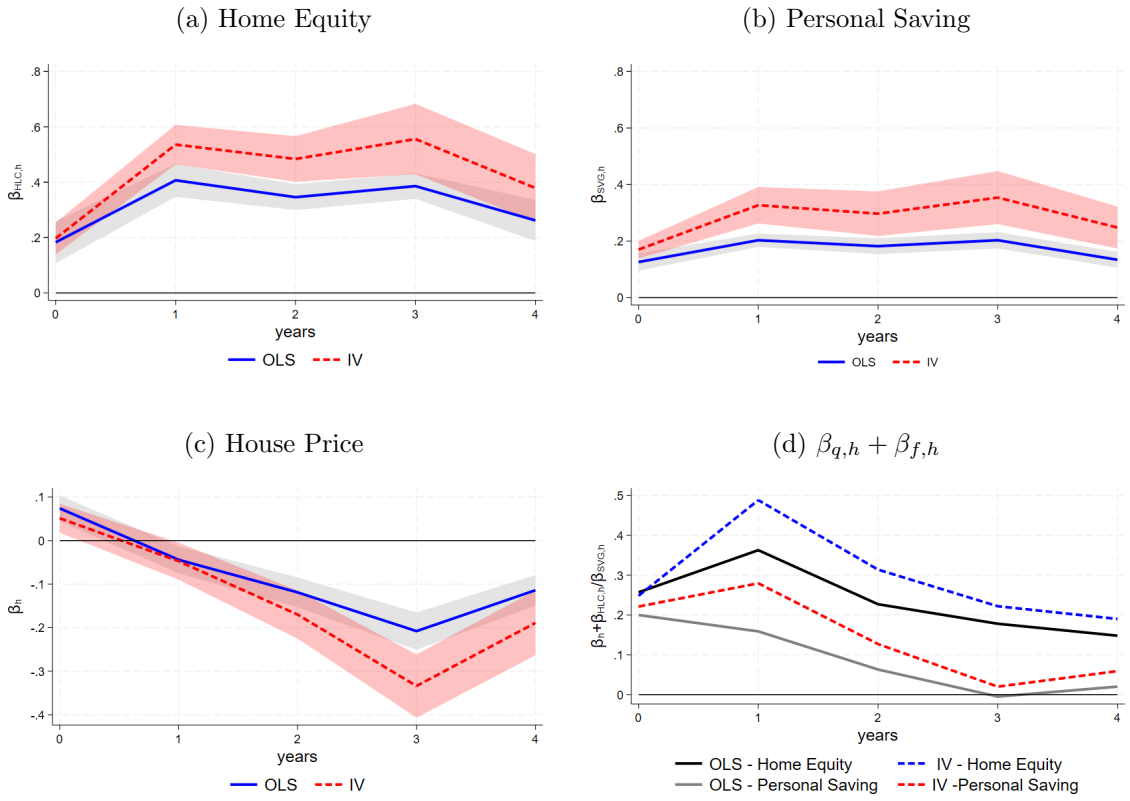
D.4 No Construction Sector

Figure OA17: House Prices and Employment Growth Dynamics - Age Comparison (Tradable sectors)



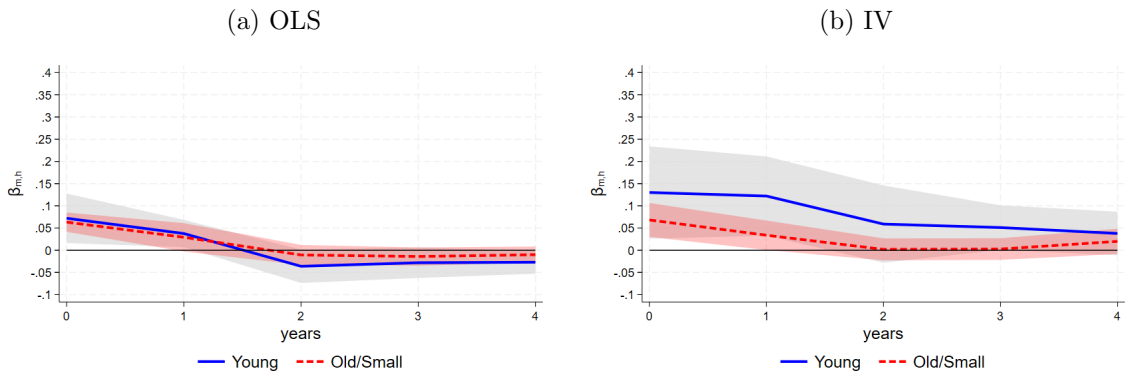
Displays β_h in equation (1)- semi-elasticity of employment at $t+h$ on house prices in t. Dependent variable is the DHS employment growth rate regressed on log changes in metro house price growth. Note: Young $<$ 5yo, Old$>$=5 yo, Small $<$500 employees; controls include log(DHS denominator), metro area effects, industry effects, (region $\times$ year) effects, and changes in metro area unemployment rates. IV regressions indicate house prices instrumented by changes in national house prices interacted with metro housing supply elasticities from Guren et al. (2021). Area bands indicate $1.645 * SE$ of the estimated effect. Includes entrants in each year $t+h$, where post-shock entrants are given their initial size for the DHS denominator. See Appendix C for table format with observation counts, r-squared statistics, and F-statistics. Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

Figure OA18: House Prices and Entrepreneurial Collateral



Displays $\beta_{q,h}$ and $\beta_{f,h}$ in equation (3) - marginal impact on $\beta_{q,h}$ of a indicator for home equity/personal saving financing ($f \in SVG, HLC$). Dependent variable is the DHS employment growth rate at year $t+h$ regressed on log changes in metro house price growth and interaction of house prices at year t with finance indicators. Note: controls include log(DHS denominator), metro area effects, industry effects, (region $\times$ year) effects, and changes in metro area unemployment rates. IV regressions indicate house prices instrumented by changes in national house prices interacted with metro housing supply elasticities from Guren et al. (2021). Area bands indicate $1.645 * SE$ of the estimated effect. Includes entrants in each year $t+h$, where post-shock entrants are given their initial size for the DHS denominator. See Appendix C for table format with observation counts, r-squared statistics, and F-statistics. Standard error clustered at the metro level. Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

Figure OA19: House Prices and Local Bank Balance Sheets



Displays $\beta_{m,h}$ in equation (4) - marginal impact on $\beta_{q,h}$ of a 1-standard deviation increase in 2005 mortgage share. Dependent variable is the DHS employment growth rate at year $t+h$ regressed on log changes in metro house price growth and interaction of house prices at time t with standardized state-bank mortgage share in 2005. Note: controls include log(DHS denominator), metro area effects, industry effects, (region $\times$ year) effects, and change is metro area unemployment rates. IV indicates house prices instrumented by changes in national house prices interacted with metro housing supply elasticities from Guren et al. (2021). Area bands indicate $\pm 1.645 * SE$. Includes entrants in year $t+h$, where post-shock entrants are given their initial size for the DHS denominator. See Appendix C for table format with observation counts, r-squared statistics, and F-statistics. Standard error clustered at the metro level. Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

D.5 Saiz (2010) Instrument

The results from this section contain results from a previous version of the LBD, FHFA house prices, and the Saiz (2010) instrument. As such, the results could differ due to a difference in sample as well as external data. Nonetheless, results are comparable, so I present them here as a robustness check.

I begin with baseline regressions of establishment (within-industry/county) employment growth on house price growth, as in equation (1). Results show a strongly significant relationship between house prices and employment growth for establishments of young, old, and old/small firms. The result is also consistent with the notion that large businesses are less sensitive to local conditions, as the magnitude of the coefficient on house prices for employment growth at large businesses is much smaller. The IV results, at the bottom of Table OA15, show an increased point estimate on the house price coefficient that is still significant for both young and old. However, the point estimate for old businesses is driven by small firms. Overall, the results are economically meaningful for old/small and young firms, with a 1 percent increase in house prices leading to a roughly 0.19-0.26 percentage point increase in young employment growth and a 0.11-0.17 percentage point increase in old/small employment growth. In the pooled sample, the elasticity is between 0.15 and 0.25. By comparison, this is somewhat higher than what was found for only retail employment in Guren et al. (2021), who estimate elasticities between 0.05 and 0.16. These point estimates suggest that the national decline of roughly 18 percent can account for 2.7-4.5 percentage points of the decline in young employment (of about a 25 percent decline) and about 2.0-3.0 percentage points of the overall decline in small business employment.[35]

D.5.1 Mortgage Share Approach

Table OA16 presents the results from the regression. As before, house prices appear to have a large effect on young businesses in both the OLS and IV specification. Point estimates are still consistent with Table OA15, implying an elasticity of 0.28 for young firms and 0.17 for establishments at old/small firms. Although the interaction with mortgage shares

[35]Interestingly, while there is a larger elasticity for young firms, it explains less of the overall decline.

Table OA15: Establishment-Level Employment Growth Regressions on House Prices by Firm age/size (with change in metro area unemployment rate)

	Total	Young	Old	Old/Small	Old/Large
OLS					
△ HPI	0.1509	0.1903	0.1164	0.1143	0.0485
	(0.0251)	(0.0333)	(0.0212)	(0.0250)	(0.0128)
N	27,230,000	7,810,000	19,420,000	16,510,000	2,910,000
IV					
$\triangle\widehat{HPI}$	0.2453	0.2646	0.1872	0.1672	0.0570
	(0.0437)	(0.0603)	(0.0434)	(0.0507)	(0.0403)
N	19,000,000	5,670,000	13,330,000	11,430,000	1,900,000

Employment Growth Regressions on House Prices Note: Young < 5yo, Old>=5 yo, Small <500 employees, Large >=500 employees; includes log(DHS denominator), county effects, industry effects, time effects, and changes in county and state unemployment rates. $\triangle\widehat{HPI}$ indicates house prices instrumented by changes in state unemployment interacted with Saiz (2010) elasticities. See appendix for R^2 and first-stage tests. Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

is significant for young firms in the OLS specification, I cannot reject the null under the IV specification. On the other hand, there is a significant response of employment among establishments of old businesses, in particular old/small firms, in response to house prices and their interaction with mortgage shares in both specifications. This supports the notion that a higher mortgage share among state banks impacts older firms through those banks, who could be more sensitive to house price shocks. As before, old/large businesses are affected by house price shocks to a lesser extent than young and old/small firms.

D.5.2 Bank Exit Approach

One would expect that increasing house prices would generally be associated with increasing employment, and an increased bank exit share should negatively impact employment. The first two rows of Table (OA17) confirm this result, with employment strongly positively correlated with house prices negatively correlated with bank exit. The impact of house prices and bank exit vary across size/age categories. Establishments of young firms appear more sensitive to the direct effect of house prices. The bank exit effect seems to impact

Table OA16: Emp. Growth Regressions by Firm Age/Size: Mtg. Share

	Young	Old	Old/Small	Old/Large
OLS				
$\triangle$ HPI	0.2800	0.1649	0.1676	0.0420
	(0.0341)	(0.0155)	(0.0182)	(0.0178)
$\triangle$ HPI*	0.1116	0.0633	0.0689	-0.0090
MTG_RATIO	(0.0325)	(0.0169)	(0.0210)	(0.0175)
N	7,810,000	19,420,000	16,510,000	2,910,000
Total IV				
$\triangle\widehat{HPI}$	0.3034	0.3199	0.2999	0.0808
	(0.1037)	(0.0799)	(0.0919)	(0.0830)
$\triangle\widehat{HPI}$*	0.0588	0.1605	0.1579	0.0326
MTG_RATIO	(0.0933)	(0.0627)	(0.0762)	(0.0738)
N	5,670,000	13,330,000	11,430,000	1,900,000

Employment Growth Regressions on House Prices and Mortgage Share Interactions Note: Young < 5yo, Old>=5 yo, Small <500 employees, Large >=500 employees; includes log(DHS denominator), county effects, industry effects, time effects, and changes in county and state unemployment rates. IV regression instruments for both house prices and interaction with mortgage share. See appendix for R^2 and first-stage tests. Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

establishments of old/small firms, but the coefficient for establishments of young firms is not significant. For larger firms, only the direct effect of house prices is significant in the OLS specification. I then instrument for house prices, as before, using the county-level housing price elasticities to instrument for house prices in the second section of Table (OA17). Results are broadly consistent with the OLS estimates, although the main effect of house prices on old/large businesses is no longer significant. Since bank exit is endogenous to firm performance, I also instrument bank exit with the interaction of house prices with pre-crisis bank balance sheet variables. These results are displayed in the bottom of Table (OA17). In these specifications, it is still the case that bank exit has a sizable and significant effect on old/small firms, but a smaller and generally insignificant effect on young firms.

Overall, the point estimates on bank exit, or predicted bank exit, for old/small firms suggest that a one percent increase in bank exit yields a 0.03 to 0.23 percent decline in employment at young firms. In the sample, the average asset-weighted bank exit rate was 2.2

percent across states-year observations, but there is substantial heterogeneity. For example, the interdecile range is about 5.9 percentage points, so establishments in a state-year with the 90th percentile bank exit rate would exhibit between 0.18 and a more substantial 1.33 percent decline in employment relative to identical establishments in the 10th percentile state-year observation.

Table OA17: Emp. Growth Regressions by Firm Age/Size: Bank Exit

	Young	Old	Old/Small	Old/Large
OLS				
$\triangle$ HPI	0.1905	0.1152	0.1128	0.0480
	(0.0334)	(0.0211)	(0.0249)	(0.0132)
Bank Exit	0.0043	-0.0333	-0.0390	-0.0201
	(0.0271)	(0.0121)	(0.0127)	(0.0189)
N	7,810,000	19,420,000	16,510,000	2,910,000
Instrumented HPI				
$\triangle\widehat{HPI}$	0.2646	0.1875	0.1676	0.0570
	(0.0603)	(0.0432)	(0.0505)	(0.0402)
Bank Exit	-0.0154	-0.0317	-0.0387	-0.0205
	(0.0297)	(0.0129)	(0.0140)	(0.0233)
N	5,670,000	13,330,000	11,430,000	1,900,000
Instrumented Bank Exit				
$\triangle$ HPI	0.1847	0.1088	0.1055	0.0472
	(0.0348)	(0.0216)	(0.0257)	(0.0125)
$\widehat{BE}$	-0.1288	-0.2140	-0.2305	-0.0530
	(0.1217)	(0.0633)	(0.0727)	(0.0814)
N	7,810,000	19,420,000	16,510,000	2,910,000
Instrumented HPI and Bank Exit				
$\triangle\widehat{HPI}$	0.2341	0.1800	0.1698	0.0277
	(0.0449)	(0.0342)	(0.0402)	(0.0387)
$\widehat{BE}$	0.0599	-0.1907	-0.1976	-0.2630
	(0.1592)	(0.0791)	(0.0923)	(0.0942)
N	5,670,000	13,330,000	11,430,000	1,900,000

Employment Growth Regressions on House Prices and State Bank Exit Share. Note: Young < 5yo, Old>=5 yo, Small <500 employees, Large >=500 employees; includes log(DHS denominator), county effects, industry effects, time effects, and changes in county and state unemployment rates.$\triangle\widehat{HPI}$ indicates house prices instrumented by changes in state unemployment interacted with Saiz (2010) elasticities. See appendix for R^2 and first-stage tests. Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

D.5.3 Full Bank Balance Sheet Approach

In this appendix, rather than restricting to a single variable or channeling the balance sheet effect through an outcome like bank exit, I turn to a specification where the full set of house price and bank balance sheet interactions is included in the main regression equation. The advantage here is that it captures a broader effect than bank exit and is more comprehensive than relying on the mortgage ratio. However, the output is difficult to interpret, especially in table form.

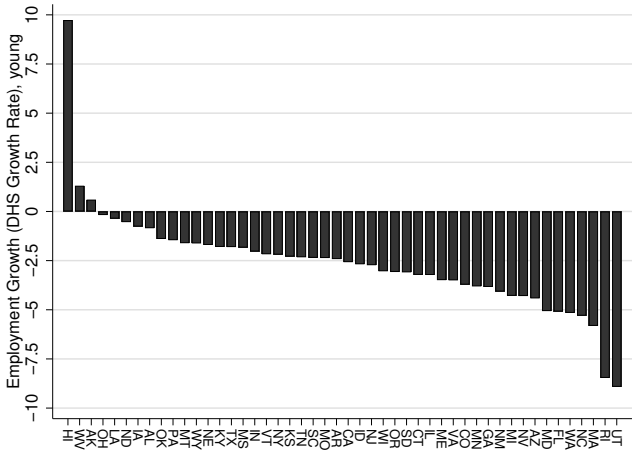
Instead, I plot the total effect of house prices and their interaction with bank balance sheets for each state. I first plot firm DHS growth rate responses for each individual state to a common house price decline of 5%, which is roughly in line with annual national house price declines during the Great Recession. This isolates the variation due to differences in each state's bank balance sheet conditions, as the house price change and "direct" elasticity to this change are the same across states. Figure OA20 shows the impact of the this decline effect on the young, old/small, and old/large groups.

In general, these graphs show substantial impacts of a credit supply channel working through bank balance sheets for all categories of firms. However, they appear qualitatively different. Focusing on the old/small group, the figure shows that although the coefficient on house prices indicates that employment is predicted to fall by 0.2% in response to the 5% decline in house prices, bank balance sheets vary enough to produce notable differences across states: some states' bank balance sheets create an "amplification" effect as predicted employment declines by an additional percentage point, and some state bank balance sheets mitigated the effect of house price declines by as much as 2 percent. Furthermore, some of the states with the largest declines in house prices (Nevada, Arizona, and Florida) seem to have bank balance sheets that contribute to the shock.

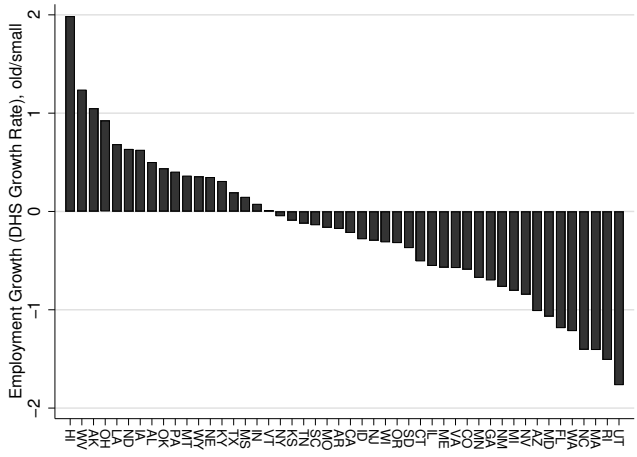
Among young firms, the range is even larger. The elasticity with respect to house prices creates a decline of about 2.5 percent, but overall predicted responses range from almost a 10 percent increase to an 8 percent decrease. The vast majority of responses are negative, however, with predicted increases in only three states, compared to 18 states with predicted employment increases in response to the decline in house prices for old/small firms.

Figure OA20: Effect of 5% HPI Change on DHS Employment Growth

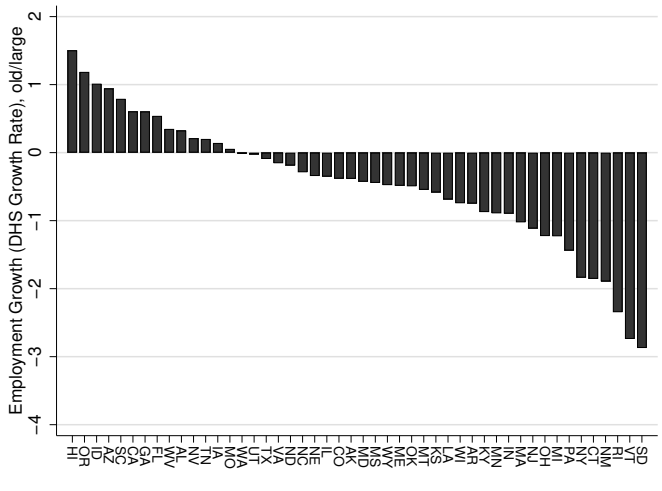
(a) Young



(b) Old/Small



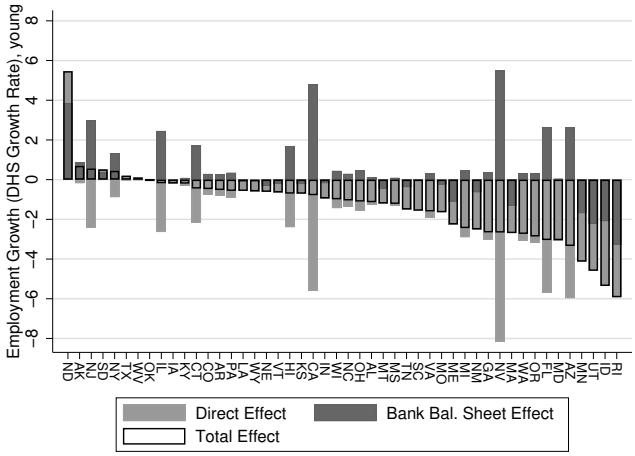
(c) Old/Large



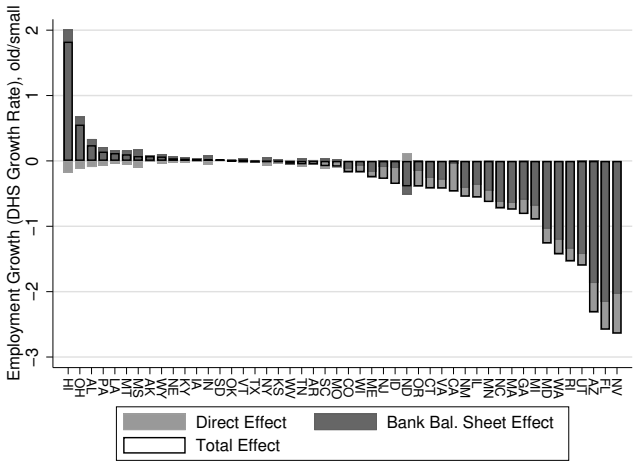
Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

Figure OA21: Effect of Average State-Level HPI Change on DHS Employment Growth

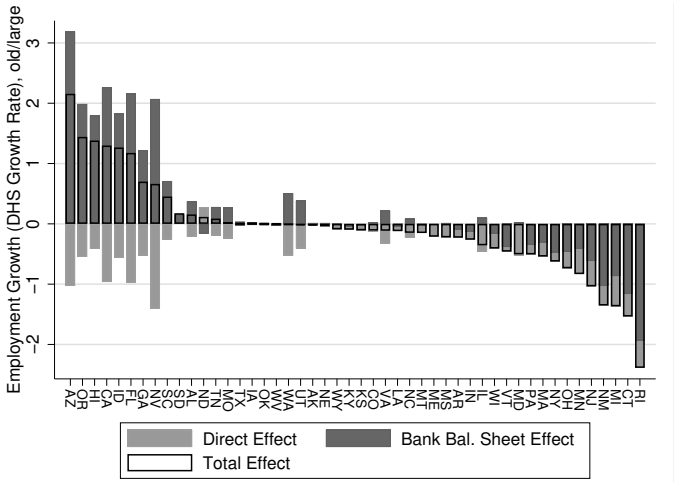
(a) Young



(b) Old/Small



(c) Old/Large



Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

The response of establishments at old/large firms looks similar to establishments of old/small firms in terms of magnitudes. The 5% decline in house prices generates a predicted decline of 0.5% for old/large firms directly. This means the average effect is slightly lower, but bank effects are generating similar variation to old/small businesses.

To demonstrate further the decomposition of house prices into direct effects and balance sheet effects, I show in Figure OA21 the effect of the change in house prices via the balance sheet interaction terms (called the Bank Balance Sheet effect), the main effect of the house price decline (called the Direct Effect), and the cumulative effect of the interaction and the direct effect of house prices. In these charts, I use the average house price decline in each state rather than a common house price decline, so each state's predicted values are the result of "actual" house price declines in the state and interactions with state bank balance sheets.[36]

From these figures, it is more clear that credit supply, as proxied by local bank balance sheets, seems to be contributing a substantial amount to the overall decline in employment among old/small firms due to house price declines. For young firms, although the overall and local bank credit supply effects of house prices are larger, the effect captured by bank balance sheets appears to have less explanatory power for overall growth in employment among young firms due to house price declines over the Great Recession and subsequent recovery. In particular, for some of the states with the most severe housing collapses (Arizona, California, Florida, and Nevada), bank balance sheets seem to have mitigated their impact on employment. Among establishments at old/large firms, the effect of house prices is largely transmitted through the bank credit supply channel, but again it does not seem to be contributing to declines in the hardest hit states.

[36]To establish the significance of the results, I perform F-tests on the joint significance of the sum of all coefficients, properly signed for the direction of the point estimate (that is, I add positive coefficients and subtract negative coefficients). This is a more restrictive test than simply testing the hypothesis that all coefficients are not significantly different from zero. Inclusive of the main effect of house prices, the joint test F-stat is 20.0 (p-value of 0.000) for young, 37.2 (0.000) for old/small, and 26.7 (0.000) for old/large. Considering only the interactions, the joint F-stat is 13.0 (0.000) for young, 33.5 (0.000) for old/small, and 23.5 (0.000) for old/large.

D.5.4 Firm-Level Productivity Analysis

The results in previous sections suggest important differences in the effect of local house price changes and credit conditions on firms depending on their size and age. Another concern for macroeconomists is the effect of the housing crisis on productivity. Are more productive firms more or less likely to experience adverse effects from local house price drops and deteriorating financial conditions? Related literature discusses "cleansing" and "sullying" effects of recessions (e.g. Barlevy, 2002), and Foster, Grim, and Haltiwanger (2016) find that the Great Recession did not "cleanse" in the way previous recessions had in the United States. Such effects could matter for aggregate productivity dynamics as entry, exit, and reallocation are potentially important margins for determining productivity growth.

To explore the potential impacts of house price growth on productivity, I use revenue data from the LBD at the firm level. I then create a labor productivity measure using firm revenue divided by total firm employment. This measure captures the existing productivity of the firm as a type of an average revenue product, and and could serve as a proxy for both underlying fundamental total factor productivity and variation in non-labor factor usage, like capital. If house price declines influence firms differently based on their productivity, this could indicate either cleansing effects that mitigate the effects of the housing crisis or sullying effects that amplify the crisis.

However, under the standard assumptions of Cobb-Douglas production and CES utility, there should be no variation in output per unit of labor.[37] Thus, variation in revenue per worker could suggest distortions preventing efficiency from perfect optimization on the part of firms, as Hsieh and Klenow (2009) and the following literature explores. Some financial frictions distort the labor margin, and so variation in the employment response to house prices based on productivity could reflect the presence of financial fricitions. The interpretation of this analysis then depends on whether the theoretical framework implies that output per worker is reflective fundamentals or distortions. We return to these considerations in the theory section as we compare model implications to the following results.[38]

[37]Under these assumptions, average revenue products like output per unit of input are proportional to marginal revenue products, which are in turn equal to the flow cost of the input by first order conditions.

[38]In practice, measures of revenue productivity similar to revenue per worker (like revenue TFP) tend to

I then use the lag of the natural log of this labor productivity measure as an additional regressor that varies at the firm level in similar regressions to those described previously. Furthermore, I am interested in how the effect of house price growth (measured using FHFA HPI) on employment outcomes varies with productivity, so I include an interaction of the house price growth with the lag labor productivity measure.

$$\Delta Y_{e,t} = \gamma \Delta UR_{c,t} + \gamma \Delta UR_{s,t} + \beta_1 \Delta HPI_{c,t} + \beta_2 LLP_{f,t-1} + \beta_3 \mathbf{\Delta HPI_{c,t}} \times \mathbf{LLP_{f,t-1}} + \alpha_c + \delta_t + \tau_i + size_{e,t} + \epsilon_{e,t} \tag{22}$$

I then run the OLS bank exit regression, including an interaction between bank exit and log labor productivity. This captures differential effects of banking stress across the productivity distribution. Similarly, I consider an additional specification where I include higher order interactions with mortgage shares to capture differential effects of the bank credit supply channel.

About 80% of firms in the LBD have revenue data, so I use propensity score weights in each of the regressions below. I split the results up by young, old, old/small, and old/large according to the definitions used previously. Results from the first two columns of Table OA18 are consistent with previous results, as house prices are highly correlated with employment growth at establishments at young firms, as well as old and old/small firms, in both the OLS and IV specifications. Theory suggests that more productivity firms should grow, so we expect labor productivity to be highly correlated with growth. Columns III and IV in Table OA18 shows this is the case, especially for establishments at young firms. Ultimately, our coefficient of interest is the interaction of house prices with productivity. Here we see a key divergence between young and old/small businesses: young firms are impacted by house prices in a similar fashion regardless of productivity, while establishments at old/small firms are disproportionately impacted by house prices if they have low productivity. That is, the negative coefficient on the interaction term suggests a "cleansing" selection mechanism that is present for old/small firms, but not for young firms.

That such an effect does not show up for young firms could then be the result of an offsetting sullying effect. However, on the view that higher productivity firms are more constrained, it is puzzling that more constrained firms are not necessarily more susceptible

be positively correlated for underlying fundamentals (Blackwood et al. (2021)).

Table OA18: Firm-Level Emp. Growth Regressions on FHFA HPI, grouped by firm age/size

	(I)	(II)	(III)	(IV)	(V)
		Young			
△ HPI (OLS)	0.1364		0.1359	0.1359	0.1086
	(0.0215)		(0.0210)	(0.0210)	(0.0255)
△ HPI (IV)		0.1436			
		(0.0421)			
log labor prod. (LLP)	0.2722	0.2680	0.2710	0.2737	0.2749
	(0.0024)	(0.0029)	(0.0027)	(0.0027)	(0.0022)
△ HPI * LLP			-0.0233	-0.0340	0.0550
			(0.0283)	(0.0281)	(0.0373)
Bank Exit				-0.0210	
				(0.0145)	
Bank Exit * LLP				-0.1072	
				(0.0228)	
Mortgage Share					-0.0378
* △ HPI					(0.0223)
Mortgage Share					0.0141
*LLP					(0.0019)
Mortgage Share					0.1452
△ HPI LLP					(0.0282)
N	3,820,000	2,800,000	3,820,000	3,820,000	3,820,000
		Old/Small			
△ HPI (OLS)	0.1047		0.1086	0.1077	0.1247
	(0.0208)		(0.0203)	(0.0201)	(0.0190)
△ HPI (IV)		0.1080			
		(0.0356)			
log labor prod. (LLP)	0.1853	0.1834	0.1820	0.1827	0.1826
	(0.0012)	(0.0015)	(0.0012)	(0.0011)	(0.0011)
△ HPI * LLP			-0.0736	-0.0766	-0.0569
			(0.0179)	(0.0174)	(0.0220)
Bank Exit				-0.0336	
				(0.0095)	
Bank Exit * LLP				-0.0279	
				(0.0145)	
Mortgage Share					0.0195
* △ HPI					(0.0176)
Mortgage Share					0.0029
*LLP					(0.0012)
Mortgage Share					0.0327
△ HPI LLP					(0.0176)
N	12,910,000	8,980,000	12,910,000	12,910,000	12,910,000
		Old/Large			
△ HPI (OLS)	0.0271		0.0243	0.0222	0.0162
	(0.0140)		(0.0140)	(0.0140)	(0.0220)
△ HPI (IV)		0.0089			
		(0.0442)			
log labor prod. (LLP)	0.0132	0.0150	0.0139	0.0129	0.0143
	(0.0006)	(0.0008)	(0.0007)	(0.0008)	(0.0007)
△ HPI * LLP			0.0156	0.0211	0.0334
			(0.0101)	(0.0174)	(0.0115)
Bank Exit				-0.0464	
				(0.0182)	
Bank Exit * LLP				0.0443	
				(0.0150)	
Mortgage Share					-0.0112
* △ HPI					(0.0204)
Mortgage Share					-0.0007
*LLP					(0.0008)
Mortgage Share					0.0217
△ HPI LLP					(0.0134)
N	1,790,000	1,200,000	1,790,000	1,790,000	1,790,000

Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

to house price shocks.

Columns (VI) and (VII) produce some qualifying results to this main finding on productivity. In these specifications, the main effects of financial channels used in previous analysis are not significant for young firms, and thus consistent with the establishment analysis. However, interactions between these indicators, bank exit and the mortgage share of local banks, and labor productivity suggest that whatever influence financial channels have on young firms falls disproportionately on productive firms. Thus, to the extent that the financial channel matters, it disproportionately impacts firms with higher labor productivity. Similarly, higher productivity old/small firms appear to be more impacted by these channels, as the interactions amplify the sign of the main coefficient in the second panel.

Again, interpretation of these results depend on what generates productivity variation. Higher sensitivity to house prices among more productive firms could be the result of credit channels producing a sullying effect, but it also could be the result of constrained firms experiencing increased tightening while unconstrained firms are less affected.

D.5.5 Firm Productivity in the Quantitative Model

Is the model consistent with the evidence on differential effects based on productivity? There are ambiguous effects on productivity within the model, as TFP and labor productivity rises within firm type, but TFP eventually falls overall as selection effects take hold. This eventual decline in TFP appears to be largely down to the collateral channel, as TFP is negatively impacted by the collateral channel and positively impacted by the interest rate channel. Since the collateral channel matters more for young firms in the model, this result is consistent with the weaker cleansing effect found in the data for young firms compared to old/small firms.

However, it is primarily the bank channel in the data that leads to sullying effects, both for young and old/small firms. Thus, the consistency of the model with data is varied. House prices create a disproportionate sullying effect through collateral channels, which primarily affect young firms, but sullying effects appear strongest through the bank channel in the data. Thus, future work could look to allow banking conditions to vary across firms rather than through a single interest rate, providing an additional role for banks to play in misallocation.

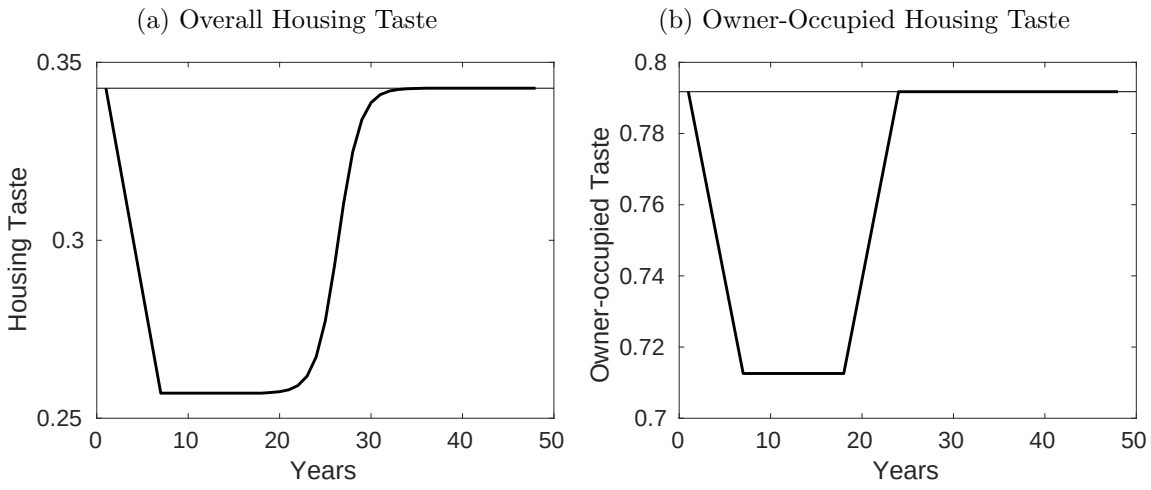
Further, there is no real role for fundamental productivity to influence labor productivity variation in the model. Recent research has shown that fundamental productivity is closely associated with revenue productivity, suggesting that distortions might not be the whole story. Future research could explore model environments where fundamentals have a greater role to play in productivity variation.

E Model Details

E.1 Model: Parameterization and Housing Taste Shock

Figure OA22 displays the housing taste shock.

Figure OA22: Housing Crisis: Simultaneous shock to Overall Housing Taste and Owner-Occupied Preference



E.2 Entrepreneur Problem

Using Ito's Lemma to write this as a Bellman equation, and specifying a quadratic cost function $f(i,k) = \chi\frac{(i-\delta k)^2}{2k}$, we have:

$$(\rho+\zeta)\, v(z,k,b) = \max_{c,i,h} u\,(c) +$$

$$v_b(z,k,b)\left(\gamma z^{\gamma} k^{\gamma\alpha} h^{\gamma(1-\alpha)} + r(b)b - wh - r_\ell \min\left\{\max\left\{wh-b,0\right\},wh\right\} - i - f(i,k) - c\right)$$

$$+v_k(z,k,b)\,(i-\delta_k) + v_z\mu_z z + \frac{1}{2}\sigma_z v_{zz}(z,k,b)$$

Standard first order conditions imply:

$$h = \begin{cases} \left(\frac{\gamma(1-\alpha)z^{\gamma}k^{\gamma\alpha}}{w(1+r_\ell)}\right)^{\frac{1}{1-\gamma(1-\alpha)}}, & \left(\frac{\gamma(1-\alpha)z^{\gamma}k^{\gamma\alpha}}{w(1+r_\ell)}\right)^{\frac{1}{1-\gamma(1-\alpha)}} > b \\ \left(\frac{\gamma(1-\alpha)z^{\gamma}k^{\gamma\alpha}}{w}\right)^{\frac{1}{1-\gamma(1-\alpha)}}, & \left(\frac{\gamma(1-\alpha)z^{\gamma}k^{\gamma\alpha}}{w}\right)^{\frac{1}{1-\gamma(1-\alpha)}} < b \\ b, \quad o.w. \end{cases} \tag{23}$$

$$u'(c) = v_b \tag{24}$$

$$i = \frac{k}{\chi}\frac{v_k}{v_b} + \delta k \tag{25}$$

One additional feature of the algorithm is to define v_b at the boundary by using the last two equations above, the budget constraint when $\dot{b} = 0$, and the the guess for v_k (implied by the guess, or updated guess, for v).[39]

E.3 Market Clearing Conditions

Market clearing conditions in equilibrium for all t:

$$Y(t) = \int \left(zk^{\alpha}h(t)^{1-\alpha}\right)^{\gamma} d\mu_t \tag{26}$$

$$Y(t) = C(t) + \int_i \left(c(z,k,b,t) + i(z,k,b,t) + f(i(z,k,b,t),k)\right) d\mu_t + \left[\left(r_\ell(t) - \psi - r(t)\right) A(t) + r(t)N(t)\right] + q_h(t)\delta_H \tag{27}$$

$$L(t) = \int h(z,k,b,t) d\mu_t \tag{28}$$

$$H_R(t) + H(t) = 1 \tag{29}$$

$$A(t) = M(t) + q_h(t)H_R(t) - \int \mathbb{1}\left(b(t) < 0\right) b(t) d\mu_t = D(t) + \int \mathbb{1}\left(b(t) > 0\right) b(t) d\mu_t + N(t) \tag{30}$$

I relax the last constraint and consider a "small open economy" with a perfectly elastic deposit rate in the main results.

[39] This allows for the possibility of movement *along* the boundary, since entrepreneurs can still invest while just maintaining the borrowing constraint with strict equality.

E.4 Guess and Verify Solution to the Banking problem

Note that ℓ, m, and H_R are state variables, and thus the relevant choice variables for the bank are $\dot{\ell}$, $\dot{m}$, and $\dot{H_R}$, along with $\dot{d} = \frac{d}{dt}(q_h h_R + \ell + m) - \dot{n}$. So the Bellman equation can be written as (denoting $V_x = V_x(\ell, m, h_R, d)$:

$$(\rho+\sigma) V(\ell, m, h_R, d) = \max_{\dot{l}, \dot{m} \dot{H_R}} \left(1 - e^{-\sigma}\right) n + V_\ell \dot{\ell} + V_m \dot{m} + V_H \dot{h_R} \tag{31}$$

$$+V_d \left(\dot{q_h} H_R + q_h \dot{h_R} + \dot{\ell} + \dot{m} - \left(\frac{\dot{q_h} + R_h - \delta_h}{q_h} - \psi \right) q_h h_R - (r_\ell - \psi)\ell - (r_m - \psi) m + rd \right) + V_t$$
$$+\lambda \left(V(\ell, m, h_R, d) - \theta (q_h h_R + \ell + m) \right)$$

The first order conditions of this problem can be combined to show that rates of return are equalized across assets, so that there is a single rate of return R_a:

$$R_a = r_\ell = r_m = \frac{\dot{q_h} + R_h - \delta_h}{q_h} \tag{32}$$

And thus the gross return on total assets can be written as $(R_a - \psi) a = \left(\frac{\dot{q_h} + R_h - \delta_h}{q_h} - \psi \right) q_h h_R + (r_\ell - \psi)\ell + (r_m - \psi) m$.

We can also show the following holds for $x \in \{H, \ell, m\}$:

$$(R_a - \psi - r) V_x = \lambda \theta \tag{33}$$

Thus, we see that the spread $(R_a - r)$ is strictly positive, but the spread is greater than ψ if and only if $\lambda > 0$, i.e. when the constraint binds. The constraint is what generates the spread between the lending and the deposit rate.

Guessing the value function is linear in deposits and loans, so $V = \gamma_{at} a + \gamma_{dt} d$ where γ_{at} and γ_{dt} are time-varying constants, with appropriate substitution of constraints we can rewrite the Bellman equation as:

$$(\rho+\sigma) V(a, d, t) = \max_{\dot{a}} \left(1 - e^{-\sigma}\right)(a - d) + \gamma_{at} \dot{a} + \gamma_{dt} (\dot{a} - (R_a - \psi) a + rd) + \dot{\gamma_{at}} a + \dot{\gamma_{dt}} d + \lambda (\gamma_{at} a + \gamma_{dt} d - \theta a) \tag{34}$$

Where γ_{xt} is the time-derivative of the constant associated with the state variable x. Recognizing that first order conditions imply $\gamma_{at} = -\gamma_{dt}$, and likewise $\frac{d}{dt}(\gamma_{at} - \gamma_{dt}) = 0$, we can

re-arrange the above equation to obtain:

$$V(a,d,t) = \frac{(1-e^{-\sigma}) + (R_a - \psi)\gamma_{at} + \dot{\gamma_{at}} + \lambda(\gamma_{at} - \theta)}{\rho + \sigma} a + \frac{-(1-e^{-\sigma}) - r\gamma_{at} - \dot{\gamma_{at}} - \lambda\gamma_{at})}{\rho + \sigma} d \tag{35}$$

Since λ is also an endogenous variable, we use equation (26) to show neither constant depends on current choices:

$$V(a,d,t) = \underbrace{\frac{(1-e^{-\sigma}) + (R_a - \psi)\gamma_{at} + \dot{\gamma_{at}} + \frac{(R_a-\psi-r)\gamma_{at}}{\theta}(\gamma_{at} - \theta)}{\rho + \sigma}}_{\gamma_{at}} a \tag{36}$$

$$+ \underbrace{\frac{-(1-e^{-\sigma}) - r\gamma_{at} - \dot{\gamma_{at}} - \frac{(R_a-\psi-r)\gamma_{at}}{\theta}\gamma_{at})}{\rho + \sigma}}_{\gamma_{dt}} d$$

Since we know $\gamma_{at} = -\gamma_{dt} = \gamma_t$, we can simplify:

$$V(a,d,t) = \gamma_t(a-d) = \gamma_t n \tag{37}$$

Where the constant multiplying net worth is the solution to the differential equation

$$\dot{\gamma_t} = \left(\rho + \sigma - r - \frac{(R_a - \psi - r)\gamma_t}{\theta}\right)\gamma_t - \left(1 - e^{-\sigma}\right) \tag{38}$$

Note that when the constraint binds, and $V = \theta a$ and $V = \gamma_t n$ imply the following expression for bank leverage φ_t:

$$\varphi_t = \frac{a}{n} = \frac{\gamma_t}{\theta} \tag{39}$$

This implies the following differential equation for bank leverage when the constraint is binding:

$$\dot{\varphi_t} = (\rho + \sigma - r - (R_a - \psi - r)\varphi_t)\varphi_t - \frac{(1-e^{-\sigma})}{\theta} \tag{40}$$

In a stationary distribution with constant prices, γ_t, and φ_t, leverage is given by the quadratic implied by (21) when $\dot{\varphi_t} = 0$:

$$\varphi = \frac{(\rho + \sigma - r) - \left((\rho + \sigma - r)^2 - \frac{4(R_a-\psi-r)\left(1-e^{-\sigma}\right)}{\theta}\right)^{\frac{1}{2}}}{2(R_a - \psi - r)} \tag{41}$$

In general, under relevant parameterizations, this implies leverage ratios in steady state are positively associated with credit spreads.

E.4.1 Steady State for the Banking Sector

The deposit rate and spread are targeted directly, and σ are calibrated, and the deposit rate r is derived from the household problem. So, we need to pick ψ and θ to target leverage φ and the spread $R_a - r$. Given this result, we find the w_b that is consistent with an aggregate stationary steady state. That is, we have no net worth growth in the aggregate: $\frac{\dot{N}}{N} = 0$. This implies:

$$w_b = 1 - \frac{e^{-\sigma}}{1 - e^{-\sigma}} \left[\left(R_a - \psi - r \right) \Phi + r \right] \tag{42}$$

Noting that aggregate leverage is equal to bank leverage, set equal to the expression for bank leverage to obtain the steady state entrant wealth as a fraction of total bank net worth w_b as a function of parameters, the deposit rate (which in steady state is given by the household problem and $\rho = r$), and the spread:

$$w_b = 1 - \frac{e^{-\sigma}}{1 - e^{-\sigma}} \left[\left(R_a - \psi - r \right) \frac{\sigma - \left(\sigma^2 - \frac{4(R_a - \psi - r)\left(1 - e^{-\sigma}\right)}{\theta} \right)^{\frac{1}{2}}}{2 \left(R_a - \psi - r \right)} + r \right] \tag{43}$$

E.5 Algorithm for Entrepreneurial Sector

The algorithm for the Entrepreneurial Sector largely follows the steps and methods discussed in Kaplan, Moll, and Violante (2018).[40] One significant update is that I transform the endogenous variables to a new state-space to help incorporate the fact that bond holdings are limited by the amount of housing (a constant constraint) and capital (a variable constraint that depends on the other asset). Rather than solve the model on (z, k, b) space, I create a new "asset" $a = b + \xi \left(k + q_h H \right)$. The value function is then adjusted to be:

$$\left(\rho + \zeta \right) v(z, k, a) = \max_{c, i, h} u \left(c \right)$$

$$+ v_a(z, k, a) \left(z^{\gamma} k^{\gamma\alpha} h^{\gamma(1-\alpha)} + r(b) b - w(1 + r_\ell) h - \left(1 - \xi \right) \left(i - \delta k \right) - \delta k - \phi \frac{\left(i - \delta k \right)^2}{2k} - c \right)$$

$$+ v_k(z, k, a) \left(i - \delta_k \right) + v_z \mu_z z + \frac{1}{2} \sigma_z v_{zz}(z, k, a),$$

[40] See: https://benjaminmoll.com/wp-content/uploads/2020/06/two_asset_kinked.pdf.

where $b = a - \xi\left(k + q_h H\right)$. Optimality conditions are adjusted accordingly. The benefit here is that a has a natural boundary of zero, and thus the gridspace is constant and matrices defined on this space can be readily inverted using the algorithm referenced above.

E.6 Solution Algorithm for Household

E.6.1 Steady State

Given the above parameters and the target for the price-to-rent ratio, the bank's problem implies that in steady state we have:

$$\frac{R_h - \delta_H}{q_h} = r_m \implies \frac{1}{12} = r_m + \frac{\delta_h}{q_h} \implies q_h = \frac{\delta_h}{1/12 - r_m} \implies R_h = \frac{\delta_h}{1 - 12 r_m}$$

With q_h, we can now solve the entrepreneur's problem, which involves searching for ν_l such that $L = 0.67$. and

For the household, we have the following preferences:

$$U\left(C, 1-L, H, H_R\right) = ln\left(C - \nu_l L\right) + \nu_H \left(\epsilon H^{1-\sigma_h} + \left(1-\epsilon\right) H_R^{1-\sigma_h}\right)^{\frac{1}{1-\sigma_h}}$$

From the Euler equation, in steady state we have:

$$r = \rho$$

Given calibration targets, we can then find:

$$M = q_h H_R * \frac{1}{3}$$

$$d_M = r_m M + q_h \delta_h H = \left(r_m \frac{1}{3} + \delta_h\right) q_h H$$

From the FOC for d_M and envelope for M, we have:

$$\left(r_m - \rho\right)\left(1 + \phi \frac{d_M}{M}\right) = \frac{\phi}{2}\left(\frac{d_M}{M}\right)^2$$

Re-arranging, I can express the calibrated value for ϕ as a function of now known variables given targets:

$$\phi = \frac{r_m - \rho}{\frac{1}{2}\left(\frac{d_M}{M}\right)^2 - \left(r_m - \rho\right)\frac{d_M}{M}}$$

From first order and envelope conditions, we have:

$$(\rho + \delta_h) \frac{q_h}{R_h} \left(1 + \phi \frac{d_M}{M}\right) U_{H_R} = U_H$$

Using the assumed functional form and re-arranging, we have:

$$(\rho + \delta_h) \frac{q_h}{R_h} \left(1 + \phi \frac{d_M}{M}\right) = \frac{\epsilon}{1-\epsilon} \left(\frac{H}{H_R}\right)^{-\sigma_h}$$

Re-arranging again, I can show that the calibrated value for ϵ is given by

$$\epsilon = \frac{(\rho + \delta_h) \left(1 + \phi \frac{d_M}{M} \frac{R_h}{q_h} \left(\frac{H}{1-H}\right)^{\sigma_h}\right)}{1 + (\rho + \delta_h) \left(1 + \phi \frac{d_M}{M} \frac{R_h}{q_h} \left(\frac{H}{1-H}\right)^{\sigma_h}\right)}$$

Note that the right hand side contains all known values either from a calibration target, assumption, or previously derivation. Turning to the bank problem, since production sector liabilities L_B are known, M is known, and $q_h H_R$ is known, and furthermore we have the expression for bank leverage:

$$\Phi = \varphi = \frac{(\rho + \delta - r) - \left((\rho + \delta - r)^2 - \frac{4(r_m - \psi - r)\left(1 - e^{-\delta}\right)}{\theta}\right)^{\frac{1}{2}}}{2\left(r_m - \psi - r\right)}$$

We then have bank leverage by definition:

$$\Phi = \frac{q_h H_R + M + L_B}{N}$$

We have $N = \frac{(q_h H_R + M + L_B)}{\Phi}$. Since the assets not funded by net worth come from deposits, then:

$$D = (\Phi - 1) N = \frac{\Phi - 1}{\Phi} (q_h H_R + M + L_B)$$

From the household budget constraint, then, we have:

$$C = rD + wL + \Pi - d_m - \phi \frac{d_m^2}{2M} - R_h H_R$$

Given the solution to the firm's problem is consistent with the target L:

$$\nu_H = \frac{R_h (1 - H)^{\sigma_h} \left(\epsilon H^{1-\sigma_H} + (1 - \epsilon)(1 - H)^{1-\sigma_H}\right)^{\frac{-\sigma_h}{1-\sigma_h}}}{(1 - \epsilon)(C - \nu_l L)}$$

E.6.2 Transition

Our goal is to find the appropriate shock to generate a path of house prices by what was observed in the United States during the great recession and after. Thus we are targeting a house price path $\{q_h(t)\}$ that falls by 20 percent over 4 years, then rises back to its steady state level after 5 years.[41] The main results use a shock to the path $\{\epsilon(t)\}$ to match this house price path, so our goal is to find the appropriate path of this parameter to generate the observed house price movement. The algorithm in simple terms consists of the following steps:

1. Guess $\{r(t), r_m(t)\}_{t=0}^{T}$ for some large T where $r(T) = r$, and $r_m(T) = r_m$ meaning we have converged back to steady state.

2. Given our targeted path $\{q_h(t)\}_{t=0}^{T}$, we can solve for the rental rate path from the necessary conditions for equilibrium in the bank problem: $R_h(t) = r_m(t)q_h(t) + \delta_h - \frac{q_h(t+\Delta)-q_h(t)}{\Delta}$.

3. Solve transition path for firm, including aggregate production sector quantities: $\{B^+(t), B^-(t), L(t)\}$, where $B^+(t) = \int_i \mathbb{1}(b_i(t) \geq 0)\, b_i(t)di$, and $B^-(t) = \int_i \mathbb{1}(b_i(t) < 0)\, b_i(t)di$

   (a) Use backward induction to find path of value function, value function derivatives, and decision rules for producers in each period t.

   (b) Calculate mass of entrants in each period t.

   (c) Use Kolgomorov forward equation to solve for evolution of distribution.

Step 4: From FOC for HH problem, we have the time derivative approximation of the Euler equation (suppressing arguments for choice variables):

$$(\rho - r(t))\, U_c(t) = \frac{U_c(t+\Delta) - U_c(t)}{\Delta}$$

[41] According to the FHFA, house prices fell by roughly 18%, whereas the S&P Case-Shiller indicates a larger decline of roughly 26%. It's debatable whether the appropriate recovery period is the time frame for house prices to reach pre crisis levels or trends. I use the former. The timing of the cycle in terms of peak/trough/recovery is roughly the same across the two house price measures.

Given $U_c(T)$, we can then solve backward:

$$U_c(t) = \frac{U_c(t+\Delta)}{(1+\Delta(\rho - r(t)))}$$

Given the functional form, and the solution to the firm's problem which yields $L(t)$, we then have:

$$C(t) = \nu_l L(t) + (C(t+\Delta) - \nu L(t+\Delta))(1+\Delta(\rho - r(t)))$$

So the path of consumption can be determined by backwards induction. Likewise, we can use the following FOC and envelope conditions to determine the ratio of mortgage payments to outstanding mortgage balances $\frac{d_m(t)}{M(t)}$. From the FOCs:

$$U_c(t) = V_d(t) \quad ; \quad V_d(t)\left(1+\phi\left(\frac{d_m(t)}{M(t)}\right)\right) = -V_M(t)$$

The envelope condition for M:

$$\rho V_M(t) = V_d(t)\frac{\phi}{2}\left(\frac{d_m(t)}{M(t)}\right)^2 + r_m(t)V_M(t) + \frac{V_M(t+\Delta) - V_M(t)}{\Delta}$$

Together these imply:

$$(\Delta(r_m(t)-\rho)-1)\left(1+\phi\left(\frac{d_m(t)}{M(t)}\right)\right)U_c(t) = \Delta U_c(t)\frac{\phi}{2}\left(\frac{d_m(t)}{M(t)}\right)^2 - \left(1+\phi\left(\frac{d_m(t+\Delta)}{M(t+\Delta)}\right)\right)U_c(t+\Delta)$$

Applying the time derivative of Euler equation, dividing by $U_c(t)$, and re-arranging, we have the following quadratic:

$$\Delta\frac{\phi}{2}\left(\frac{d_m(t)}{M(t)}\right)^2 + (1-\Delta(r_m(t)-\rho))\left(1+\phi\left(\frac{d_m(t)}{M(t)}\right)\right) - \left(1+\phi\left(\frac{d_m(t+\Delta)}{M(t+\Delta)}\right)\right)(1+\Delta(\rho - r(t))) = 0$$

$$\frac{d_m(t)}{M(t)} = \frac{-\phi(1-\Delta(\rho - r_m(t)))}{\Delta\phi} +$$

$$\frac{\left(\phi^2(1-\Delta(\rho-r_m(t)))^2 - 2\Delta\phi\left((1-\Delta(\rho-r_m(t))) - (1+\Delta(\rho-r(t)))\left(1+\phi\frac{d_m(t+\Delta)}{M(t+\Delta)}\right)\right)\right)^{\frac{1}{2}}}{\Delta\phi}$$

Where the positive root is used since deposits are non-negative.

Step 5: Given initial bank states $\{N(0), A(0), D_B(0)\}$ where $D_B(0)$ are bank deposits, solve for path of bank deposits and household deposits, taking the solution to production sector deposits and liabilities $\{D_P(t), L_P(t)\}$ from the aggregated solutions to the firms'

problems, where $D_P(t) = \int \mathbb{1}(b(t) \geq 0)\, b(t) d\mu_t$ and $L_P(t) = wL(t) + \int \mathbb{1}(b(t) < 0)\, b(t) d\mu_t$. From the bank problem, we know:

$$N(t+\Delta) = N(t)\left(1 + \Delta\left(1 - e^{-\sigma}\left(1 - r_d(t)\right)\right)\right) + \Delta e^{-\sigma}\left(r_\ell(t) - r_d(t)\right) A(t) + \Delta W_B$$

Assuming the enforcement constraints bind, individual bank assets follow the law of motion $\dot{a} = (\rho + \delta)\, a - (1 - e^{-\sigma})\, \frac{n}{\theta}$, so aggregate assets evolve as follows:

$$A(t+\Delta) = A(t) + \Delta e^{-\sigma}\left((\rho + \sigma) A(t) - \left(1 - e^{-\sigma}\right) \frac{N(t)}{\theta}\right) - \Delta\left(1 - e^{-\sigma}\right) A(t) + \Delta W_B \frac{A(t)}{N(t)}$$

Then, we can find the evolution of household deposits:

$$D(t) = D_B(t) - D_P(t) = A(t) - N(t) - D_P(t)$$

Step 6: Given a path of interest rates, decisions for $\{L(t), C(t), \frac{d_m(t)}{M(t)}\}$, a path of deposits $\{D(t)\}$, and initial conditions $\{D(0), M(0), H(0)\}$, we can solve for the evolution of assets and the rental housing taste parameters as follows. The envelope condition for housing gives:

$$\left(1 + \Delta\left(\rho + \delta_h\right)\right) V_H(t) = \Delta U_H(t) + V_H(t + \Delta)$$

From the FOCs, we have:

$$V_H(t) = -q_h(t) V_M(t) = q_h(t)\left(1 + \phi \frac{d_m(t)}{M(t)}\right) V_D(t) = q_h(t)\left(1 + \phi \frac{d_m(t)}{M(t)}\right) U_c(t)$$

Plugging into the envelope condition, and using the intertemporal Euler equation, we have:

$$\left(1 + \Delta\left(\rho + \delta_h\right)\right) q_h(t)\left(1 + \phi \frac{d_m(t)}{M(t)}\right) U_c(t) = \Delta U_H(t) + q_h(t+\Delta)\left(1 + \phi \frac{d_m(t+\Delta)}{M(t+\Delta)}\right)\left(1 + \Delta\left(\rho - r(t)\right)\right) U_c(t)$$

Dividing both sides by $U_c(t)$ and recognizing $\frac{U_{H_r}(t)}{R_h(t)} = U_c(t)$

$$\left(1 + \Delta\left(\rho + \delta_h\right)\right)\left(1 + \phi \frac{d_m(t)}{M(t)}\right) q_h(t) = \Delta R_h(t) \frac{U_H(t)}{U_{H_R}(t)} + q_h(t+\Delta)\left(1 + \phi \frac{d_m(t+\Delta)}{M(t+\Delta)}\right)\left(1 + \Delta\left(\rho - r(t)\right)\right)$$

Plugging in the functional form for housing and re-arranging, we have:

$$\frac{\epsilon(t)}{1 - \epsilon(t)} = \frac{\left[\left(1 + \Delta\left(\rho + \delta_h\right)\right)\left(1 + \phi \frac{d_m(t)}{M(t)}\right) q_h(t) - \left(1 + \phi \frac{d_m(t+\Delta)}{M(t+\Delta}\right)\left(1 + \Delta\left(\rho - r(t)\right)\right) q_H\left(t + \Delta\right)\right]\left(\frac{H(t)}{1 - H(t)}\right)^{\sigma_h}}{\Delta R_h(t)}$$

Defining the left hand side as $\epsilon_{ratio}(t)$, we have:

$$\epsilon(t) = \frac{\epsilon_{ratio}(t)}{1 + \epsilon_{ratio}(t)}$$

Re-arranging the evolution of mortgages, we have:

$$M(t+\Delta) = \left(1 + \Delta\left(r_m(t) - \frac{d_m(t)}{M(t)}\right)\right) M(t) + q_h\left(t\right)\left(H_R(t) - H_R\left(t+\Delta\right)\right)$$

Define $NI(t) = r(t)D(t) + wL(t) + \pi(t)$, and plugging into the budget constraint:

$$H_R(t+\Delta) =$$

$$\frac{NI(t+\Delta) - C(t+\Delta) - \frac{B(t+2\Delta)-B(t+\Delta)}{\Delta} - \left(\frac{d_m(t+\Delta)}{M(t+\Delta)} + \frac{\phi}{2}\frac{d_m(t+\Delta)^2}{M(t+\Delta)^2}\right)\left(\left(1 + \Delta\left(r_m(t) - \frac{d_m(t)}{M(t)}\right)\right) M(t) + q_h(t)H_R(t)\right)}{R_H(t+\Delta) - q_h(t)\frac{d_m(t+\Delta)}{M(t+\Delta)}\left(1 + \frac{\phi}{2}\frac{d_m(t+\Delta)}{M(t+\Delta)}\right)}$$

This then gives $H(t+\Delta) = 1 - H_R(t+\Delta)$, and mortgages:

$$M(t+\Delta) = \left(1 + \Delta\left(r_m(t) - \frac{d_m(t)}{M(t)}\right)\right) M(t) + q_h(t)\left(H(t+\Delta) - H(t)\right)$$

Finally, $\epsilon(t+\Delta)$ can be found as before. Repeating each step above, one can find a path of household assets $\{H(t), M(t), B(t)\}_{t=0}^{T}$ and parameter path $\{\epsilon(t)\}_{t=0}^{T}$ that is consistent with the bank problem and targeted path of house prices, given the interest rate guesses.

Step 7: Check asset market clearing conditions in each time period. Specifically, and asset markets clear $A(t) = M(t) + B(t) + wL(t) + \int \mathbb{1}\left(b(t) < 0\right) b(t) d\mu_t$. If asset markets clear, proceed to step 8. If not, and implied bank assets $A(t)$ are greater than combined household/production sector liabilities, then lower lending rates, if bank assets are less than implied liabilities, raise the lending rate in that time period. Then go back to step 2.

Step 8: Check that output markets clear:

$$C(t) + q_h\delta_h + \int_i \left(c(z,k,b,t) + i(z,k,b,t) + f(i(z,k,b,t),k)\right) d\mu_t + \left[\left(r_\ell(t) - \psi - r(t)\right) A(t) + r(t)N(t)\right] = \int y_i^\rho di$$

If markets clear, then we have a solution. If not, and output is greater than consumption, lower the deposit rate. If output is less than consumption, then raise the interest rate. Return to step 2.

E.7 Extension: Endogenous Choice of Entreprenerial Housing

I consider an extension of the model where housing H_{ent} is explicitly chosen by entrepreneurs at entry, and maintained throughout the life of their business. In other words, owner-occupied is a one-time choice that cannot be adjusted after entry. However, the entrepreneur can also

rent additional housing, which can be adjusted over time. The entrepreneur has similar preferences to the household, which reflects that they are essentially a type of household that has the opportunity to start a business:

$$U(c, H_{ent}, H_R) = ln\left(c\right) + \nu_h \left(\epsilon_h H_{ent}^{1-\sigma_h} + \left(1-\epsilon_h\right) H_R^{1-\sigma_h}\right)^{\frac{1}{1-\sigma_h}}. \tag{44}$$

Here, note preferences are identical except that there is no utility flow from leisure for the entrepreneur (they live to work). After entry, the problem is exactly as discussed in 3.2 except that housing collateral is now potentially different from the household owner-occupied housing stock and the budget constraint is now:

$$\dot{b} = z^{\gamma} k^{\gamma\alpha} h^{\gamma(1-\alpha)} + r(b)b - w\left(1+r_{\ell}\right)h - i - f(i,k) - c - \delta_h H_{ent} - R_h H_R.$$

Here, the entrepreneur also has to replace depreciated housing $\delta_H H_e nt$, but this is just an additional flow cost to the entrepreneur–one that cannot be avoided once chosen, and so this is taken into account at entry. Additionally, the entrepreneur must pay for rental housing, which can be adjusted in spot markets.[42]

Thus, the main difference in this extension is at the point of entry. Rather than endowing entrepreneurs with zero liquid assets and startup physical capital k_{ent}, I now provide them with and endowment of equity d_{ent}, which the entrepreneur has a choice to split between three assets: capital (k_{ent}), liquid saving/debt (b_{ent}), and housing (H_{ent}). Since they can borrow, I require that entry level liquid asset be such that $b_{ent} > -d_{ent}$. That is, they cannot lever their initial equity and any asset purchases must be made through equal parts equity and debt. To capture that this is a venture with seed money, I further require the entrepreneur to hit this limit.[43] Thus, the choice for the entrepreneur is to choose housing H_{ent} and capital k_{ent} given startup funds $-b_{ent} + d_{ent} = 2 * d_{ent}$. When making this choice,

[42]Additionally, the exit cost c_{exit} is calibrated to be the flow value of a hand-to-mouth worker with only the wage to consume and owner-occupied and rental housing identical to an entering entrperneur. That is, $c_{exit} = ln(w) + \nu_h \left(\epsilon_h H_{ent}^{\sigma_h} + (1-\epsilon_h) H_{R,i,ent}^{\sigma_h}\right)$. This captures the effect of bankruptcy on an entreprneur, who is forced to lvie hand to mouth after discharging their debt, but gets to maintain their initial levels of housing.

[43]In practice, this is not a necessary assumption, since it was generally optimal in early versions of the model. It simplifies the problem a great deal to eliminate this choice.

the entrant does not yet know productivity which is revealed after asset allocation choices, and so all entrants make this choice based on the expected value, and thus choose the same level of housing. We can formulate the problem for the entering firm as follows:

$$V_{entry} = \max_{k_{ent}, H_{ent}} E\left[V(z, k_{ent}, b_{ent}, H_{ent},)\right],$$

$$s.t. \quad k_{ent} + q_H H_{ent} = -b_{ent} + d_{ent},$$

$$-b_{ent} = d_{ent}.$$

The first order conditions for the problem imply (after substituting in the fact that $b_{ent} = -d_{ent}$ into the value function):

$$\frac{E\left[V_H(z, k_{ent}, -d_{ent}, H_{ent})\right]}{E\left[V_K(z, k_{ent}, -d_{ent}, H_{ent})\right]} = q_H.$$

One method to solving this problem would be to discretize the housing choice set, guess values for V_H along with the guesses for V_K and V_B in the current algorithm, then update through value function iteration. However, this would be computationally intractable. Instead, the fact that it is a one time choice allows me to exploit the envelope condition. Modifying the Hamilton-Jacobi-Bellman equation from E.2, we have:

$$(\rho + \zeta) V(z, k, b, H_{ent}) = \max_{c,i,h,H_R} U\left(c, H_{ent}, H_R\right)$$

$$+V_b(z,k,b,H_{ent}) \underbrace{\left(\gamma z^{\gamma} k^{\gamma\alpha} h^{\gamma(1-\alpha)} + r(b)b - w(1+r_\ell)h - i - \phi\frac{(i-\delta k)^2}{2k} - c - \delta_h q_h H_{ent} - R_h H_R\right)}_{\dot{b}}$$

$$+V_k(z,k,b,H_{ent}) \underbrace{(i-\delta_k)}_{\dot{k}} + V_H(z,k,b,H_{ent})\dot{H} + V_z(z,k,b,H_{ent})\mu_z z + \frac{1}{2}\sigma_z V_{zz}(z,k,b,H_{ent})$$

$$+\lambda\left(\xi_h q_h H_{ent} + \xi k + b - w\left(1+r\right)h\right)$$

where λ is the Lagrange multiplier associated with the collateral constraint. Here, by assumption $\dot{H} = 0$. The envelope condition for housing is then (suppressing arguments):

$$(\rho + \zeta) V_H = U_H + q_h\left(\lambda\xi_h - V_b\delta_h\right) + \underbrace{V_{bH}\dot{b} + V_{kH}\dot{k} + V_{zH}\mu_z z + V_{zzH}\frac{1}{2}\sigma_z}_{\dot{V}_H}.$$

From the first order condition for h, we have:

$$V_b\left(\gamma\alpha\gamma z^{\gamma}k^{\gamma\alpha}h^{\gamma(1-\alpha)-1} - w(1+r)\right) - \lambda w(1+r)$$
$$\implies \lambda = \frac{V_b\left(\gamma\alpha\gamma z^{\gamma}k^{\gamma\alpha}h^{\gamma(1-\alpha)-1} - w(1+r)\right)}{w(1+r)}.$$

With this in hand, we have all we need for our algorithm. The goal is to find a optimal allocation of k_{ent} and H_{ent} given inital capital d_{ent}. The algorithm proceeds as follows:

1. Guess a value of H_{ent}

2. Set grid of b and k such that $b_{ent} = -d_{ent}$ and the implied value of $k_{ent} = 2d_{ent} - q_h H_{ent}$ is a point on the grid. Adjust grid spacing within parameters (trying to keep the grid from being too fine with a small range, or too big of a range with too sparse spacing.)

3. Given H_ent, use continuous time finite difference HJB method to solve for V_k and V_b on the grid space for capital and liquid assets.

4. Initialize a guess for V_H.

5. Use finite difference equation to update guess:

$$(\rho+\zeta)\,V_H(t) = U_H(t) + q_h V_b\left(\xi_h\frac{\left(\gamma\alpha\gamma z^{\gamma}k^{\gamma\alpha}h^{\gamma(1-\alpha)-1} - w(1+r)\right)}{w(1+r)} - \delta_h\right) + \frac{V_h(t+\Delta) - V_H(t)}{\Delta},$$

$$\implies V_H(t) = \frac{\Delta\left(U_H(t) + q_h V_b\left(\xi_h\frac{\left(\gamma\alpha\gamma z^{\gamma}k^{\gamma\alpha}h^{\gamma(1-\alpha)-1} - w(1+r)\right)}{w(1+r)} - \delta_h\right)\right) + V_h(t+\Delta)}{1+\Delta\,(\rho+\zeta)}$$

   where δ is the time step.

6. If $|V_H(t) - V_H(t+\Delta)| < tol$, for some tolerance level tol, then move to step 6, if not return to step 4.

7. Having found V_H at every gridpoint in (z,k,b), we can now calculate $E\left[V_H(z,k,b,H_{ent})\right]$ and $E\left[V_k(z,k,b,H_{ent})\right]$.

8. Check ratio of $\frac{E[V_H(z,k,b,H_{ent})]}{E[V_k(z,k,b,H_{ent})]}$ relative to q_h.

   (a) If $|\frac{E[V_H(z,k,b,H_{ent})]}{E[V_k(z,k,b,H_{ent})]} - q_h| < tol2$ for some tolerance level $tol2$, then the optimal H_{ent} is found (given prices and other parameters). End.

(b) Elseif $\frac{E[V_H(z,k,b,H_{ent})]}{E[V_k(z,k,b,H_{ent})]} > q_h$, increase H_{ent} and return to step 2.

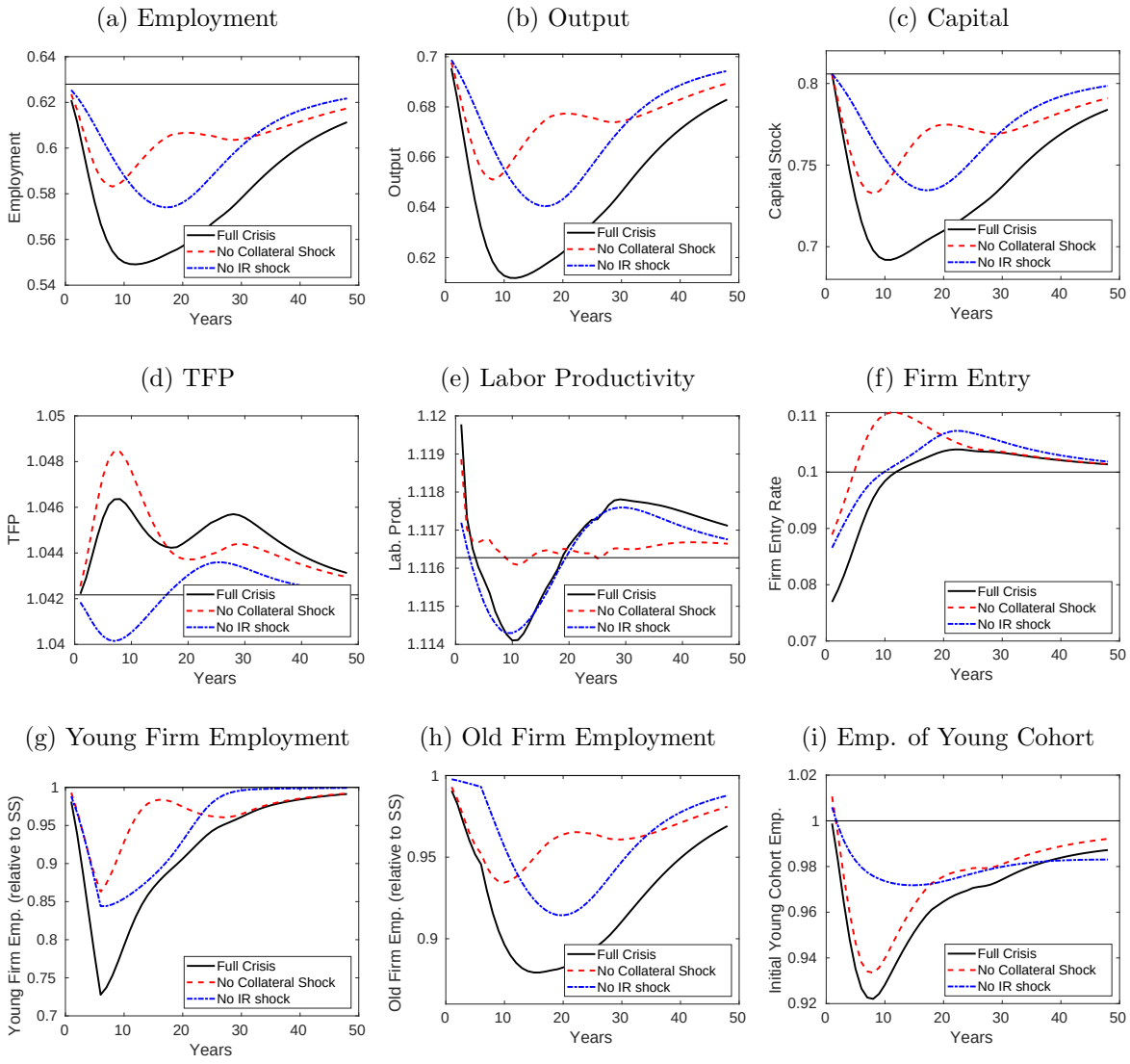
(c) Elseif $\frac{E[V_H(z,k,b,H_{ent})]}{E[V_k(z,k,b,H_{ent})]} < q_h$, decrease H_{ent} and return to step 2.

F Model Extensions and Robustness

F.1 Partial Equilibrium Experiments

This section explores the counterfactual response of the entrepreneurial sector to each of the two channels, without considering endogenous responses of prices. The first experiment consider responses when interest rates are held constant but house prrices drop by as much as in the full crisis. The second holds collateral values constant while feeding in the response of interest rates in the full crisis.

Figure OA23: Counterfactual Experiments: Collateral vs. Credit Channels



Note: Measures are relative to steady state values in (d) and (f).

Results are similar to Section 5.2. Notable exceptions include a medium-term fluctuation in the "no collateral shock" counterfactual that is not present in the corresponding panels of Figure 6. Additionally, the entry rate responds more quickly in the "no interest rate" shock. Part of what could be driving this difference is that a more muted house price decline in Figure 6 not only impacts entry, but also exit. Thus, the denominator in the entry rate is larger in the "no interest rate" counterfactual.

F.2 Endogenous Entrepreneurial Housing

I consider an extension of the model where housing H_{ent} is explicitly chosen by entrepreneurs at entry, and maintained throughout the life of their business. In other words, owner-occupied is a one-time choice that cannot be adjusted after entry. However, the entrepreneur can also rent additional housing, which can be adjusted over time. The entrepreneur has similar preferences to the household, which reflects that they are essentially a type of household that has the opportunity to start a business:

$$U(c, H_{ent}, H_R) = ln\left(c\right) + \nu_h \left(\epsilon_h H_{ent}^{1-\sigma_h} + \left(1-\epsilon_h\right) H_R^{1-\sigma_h}\right)^{\frac{1}{1-\sigma_h}}. \tag{45}$$

Here, note preferences are identical except that there is no utility flow from leisure for the entrepreneur (they live to work). After entry, the problem is exactly as discussed in 3.2 except that housing collateral is now potentially different from the household owner-occupied housing stock and the budget constraint is now:

$$\dot{b} = z^{\gamma} k^{\gamma\alpha} h^{\gamma(1-\alpha)} + r(b)b - w\left(1+r_{\ell}\right)h - i - f(i,k) - c - \delta_h H_{ent} - R_h H_R.$$

Here, the entrepreneur also has to replace depreciated housing $\delta_H H_e nt$, but this is just an additional flow cost to the entrepreneur–one that cannot be avoided once chosen, and so this is taken into account at entry. Additionally, the entrepreneur must pay for rental housing, which can be adjusted in spot markets.[44]

Thus, the main difference in this extension is at the point of entry. Rather than endowing entrepreneurs with zero liquid assets and startup physical capital k_{ent}, I now provide them with and endowment of equity d_{ent}, which the entrepreneur has a choice to split between three assets: capital (k_{ent}), liquid saving/debt (b_{ent}), and housing (H_{ent}). Since they can borrow, I require that entry level liquid asset be such that $b_{ent} > -\xi d_{ent}$. That is, they cannot lever their initial equity and any asset purchases must be made through equal parts

[44] Additionally, the exit cost c_{exit} is calibrated to be the flow value of a hand-to-mouth worker with only the wage to consume and owner-occupied and rental housing identical to an entering entrperneur. That is, $c_{exit} = ln(w) + \nu_h \left(\epsilon_h H_{ent}^{\sigma_h} + (1-\epsilon_h) H_{R,i,ent}^{\sigma_h}\right)$. This captures the effect of bankruptcy on an entreprneur, who is forced to lvie hand to mouth after discharging their debt, but gets to maintain their initial levels of housing.

equity and debt. To capture that this is a venture with seed money, I further require the entrepreneur to hit this limit.[45] Thus, the choice for the entrepreneur is to choose housing H_{ent} and capital k_{ent} given startup funds $-b_{ent} + d_{ent} = (1 + \xi) * d_{ent}$. When making this choice, the entrant does not yet know productivity which is revealed after asset allocation choices, and so all entrants make this choice based on the expected value, and thus choose the same level of housing. We can formulate the problem for the entering firm as follows:

$$V_{entry} = \max_{k_{ent}, H_{ent}} E\left[V(z, k_{ent}, b_{ent}, H_{ent},)\right],$$

$$s.t. \quad k_{ent} + q_H H_{ent} = -b_{ent} + d_{ent},$$

$$-b_{ent} = d_{ent}.$$

The first order conditions for the problem imply (after substituting in the fact that $b_{ent} = -d_{ent}$ into the value function):

$$\frac{E\left[V_H(z, k_{ent}, -d_{ent}, H_{ent})\right]}{E\left[V_K(z, k_{ent}, -d_{ent}, H_{ent})\right]} = q_H.$$

One method to solving this problem would be to discretize the housing choice set, guess values for V_H along with the guesses for V_K and V_B in the current algorithm, then update through value function iteration. However, this would be computationally intractable. Instead, the fact that it is a one time choice allows me to exploit the envelope condition. Modifying the Hamilton-Jacobi-Bellman equation from E.2, we have:

$$(\rho + \zeta) V(z, k, b, H_{ent}) = \max_{c, i, h, H_R} U\left(c, H_{ent}, H_R\right)$$

$$+V_b(z, k, b, H_{ent}) \underbrace{\left(\gamma z^{\gamma} k^{\gamma\alpha} h^{\gamma(1-\alpha)} + r(b)b - w(1 + r_\ell)h - i - \phi\frac{(i - \delta k)^2}{2k} - c - \delta_h q_h H_{ent} - R_h H_R\right)}_{\dot{b}}$$

$$+V_k(z, k, b, H_{ent}) \underbrace{(i - \delta_k)}_{\dot{k}} + V_H(z, k, b, H_{ent})\dot{H} + V_z(z, k, b, H_{ent})\mu_z z + \frac{1}{2}\sigma_z V_{zz}(z, k, b, H_{ent})$$

$$+\lambda\left(\xi_h q_h H_{ent} + \xi k + b - w\left(1 + r\right)h\right)$$

[45] In practice, this is not a necessary assumption, since it was generally optimal in early versions of the model. It simplifies the problem a great deal to eliminate this choice.

where λ is the Lagrange multiplier associated with the collateral constraint. Here, by assumption $\dot{H} = 0$. The envelope condition for housing is then (suppressing arguments):

$$(\rho + \zeta) V_H = U_H + q_h (\lambda \xi_h - V_b \delta_h) + \underbrace{V_{bH}\dot{b} + V_{kH}\dot{k} + V_{zH}\mu_z z + V_{zzH}\frac{1}{2}\sigma_z}_{\dot{V}_H}.$$

From the first order condition for h, we have:

$$V_b \left(\gamma\alpha\gamma z^\gamma k^{\gamma\alpha} h^{\gamma(1-\alpha)-1} - w(1+r)\right) - \lambda w(1+r)$$

$$\implies \lambda = \frac{V_b \left(\gamma\alpha\gamma z^\gamma k^{\gamma\alpha} h^{\gamma(1-\alpha)-1} - w(1+r)\right)}{w(1+r)}.$$

With this in hand, we have all we need for our algorithm. The goal is to find a optimal allocation of k_{ent} and H_{ent} given inital capital d_{ent}. The algorithm proceeds as follows:

1. Guess a value of H_{ent}

2. Set grid of b and k such that $b_{ent} = -d_{ent}$ and the implied value of $k_{ent} = 2d_{ent} - q_h H_{ent}$ is a point on the grid. Adjust grid spacing within parameters (trying to keep the grid from being too fine with a small range, or too big of a range with too sparse spacing.)

3. Given H_ent, use continuous time finite difference HJB method to solve for V_k and V_b on the grid space for capital and liquid assets.

4. Initialize a guess for V_H.

5. Use finite difference equation to update guess:

$$(\rho + \zeta) V_H(t) = U_H(t) + q_h V_b \left(\xi_h \frac{\left(\gamma\alpha\gamma z^\gamma k^{\gamma\alpha} h^{\gamma(1-\alpha)-1} - w(1+r)\right)}{w(1+r)} - \delta_h\right) + \frac{V_h(t+\Delta) - V_H(t)}{\Delta},$$

$$\implies V_H(t) = \frac{\Delta\left(U_H(t) + q_h V_b \left(\xi_h \frac{\left(\gamma\alpha\gamma z^\gamma k^{\gamma\alpha} h^{\gamma(1-\alpha)-1} - w(1+r)\right)}{w(1+r)} - \delta_h\right)\right) + V_h(t+\Delta)}{1 + \Delta(\rho + \zeta)}$$

   where δ is the time step.

6. If $|V_H(t) - V_H(t+\Delta)| < tol$, for some tolerance level tol, then move to step 6, if not return to step 4.

7. Having found V_H at every gridpoint in (z, k, b), we can now calculate $E\left[V_H(z, k, b, H_{ent})\right]$ and $E\left[V_k(z, k, b, H_{ent})\right]$.

8. Check ratio of $\frac{E[V_H(z,k,b,H_{ent})]}{E[V_k(z,k,b,H_{ent})]}$ relative to q_h.

   (a) If $|\frac{E[V_H(z,k,b,H_{ent})]}{E[V_k(z,k,b,H_{ent})]} - q_h| < tol2$ for some tolerance level $tol2$, then the optimal H_{ent} is found (given prices and other parameters). End.

   (b) Elseif $\frac{E[V_H(z,k,b,H_{ent})]}{E[V_k(z,k,b,H_{ent})]} > q_h$, increase H_{ent} and return to step 2.

   (c) Elseif $\frac{E[V_H(z,k,b,H_{ent})]}{E[V_k(z,k,b,H_{ent})]} < q_h$, decrease H_{ent} and return to step 2.

F.2.1 Results

To discipline this model, I limit the changes to the calibration to a few necessary parameters given the new structure. First, I fix the total financial capital to that in the original experiment (that is $(1 + \xi)d_{ent} = \kappa_e + \xi\overline{q_h}\overline{H}$, and then find the ν_h that ensures housing markets clear. I also adjust c_{exit} and ξ to better match average size and debt ratios, respectively. Most other targets seem relatively robust, see Table **??**. One notable exception is the employment-weighted entry, which is over a percentage point lower. This understandable since firms are dedicating more of their initial funding to capital, rather than housing.

The resulting H_{ent} tends to be relatively large: more than 2.2 times that in the original model. For comparison, the 2005 Survey of Income and Program Participation (SIPP) estimates from the US Census Bureau show that the median home equity valuation for households in the prime business formation age range (45-54) was 10 percent higher, and 25 percent higher for the age bracket above that (55-65). Bachelor's degree holders (education may indicate higher propensity to start a business) hold 26 percent more housing equity at the median. If income is an indicator of entrepreneurial capability, then the median of the fourth and fifth quintile of income hold 8 percent *less* and 60 percent more housing equity. Household home eqity wealth also tends to be very skewed, as those with over $500,000 in net worth tended to have home equity of 224 percent higher, and those with net worth of $250,000-$499,999 had home equity that was about twice that of the median. Finally, the mean was about 75 percent higher than the median. On balance, it seems likely that wealth effects of housing (Hurst and Pugsley, 2015) will lead to a positive correlation between

housing and entrepreneurship, but most of the comparisons above suggest the 2.2 multiple is on the high side.

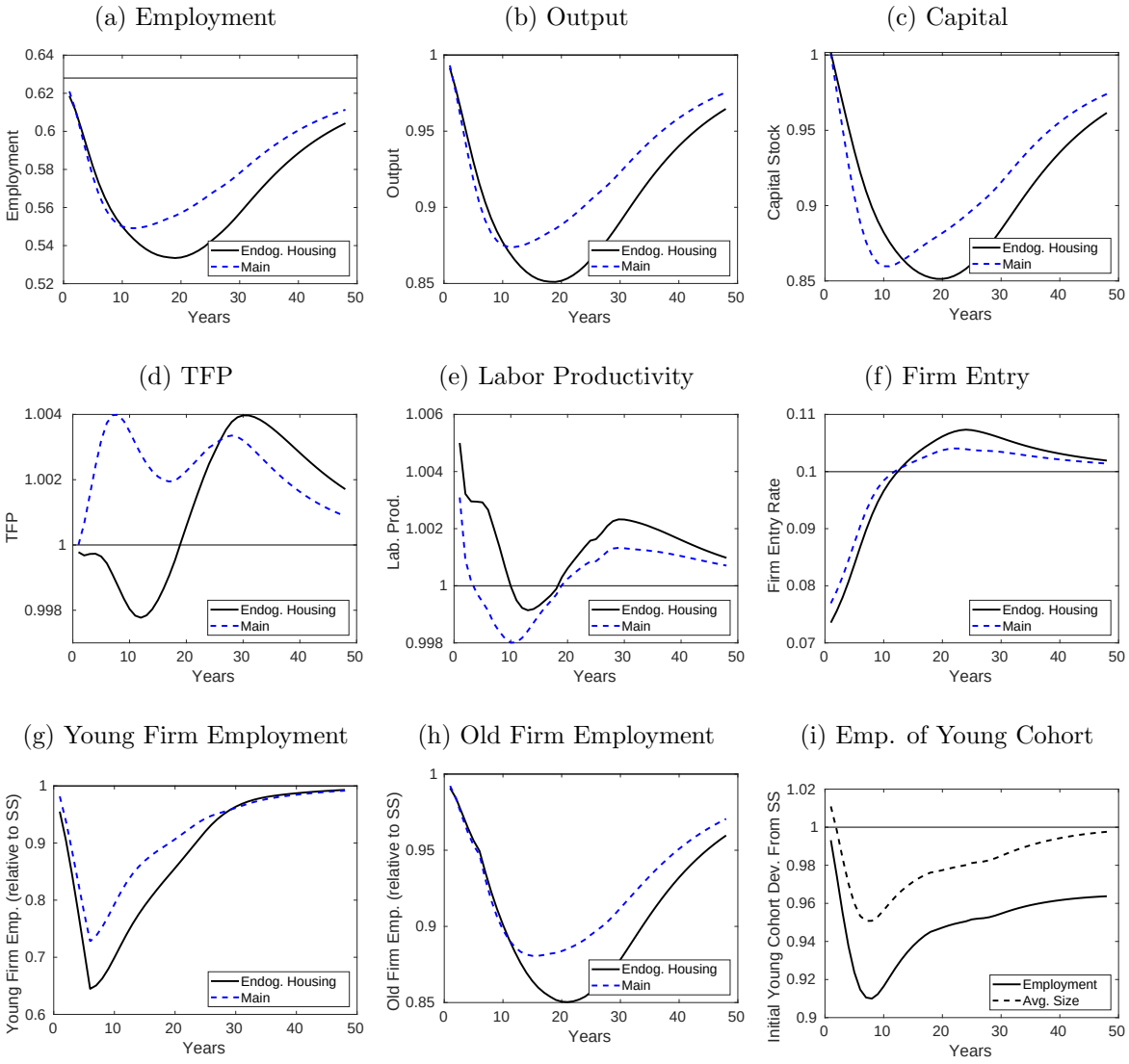
To evaluate the effect of the housing crisis in this model, I make two more simplifications. First, I fix housing choices to the steady state, rather than tracking the endogenous evolution of housing choices over the business cycle. Second, I feed in the house price and interest rate paths from the main experiment, making this a partial equilibrium exercise. I also not that I adjust η to better capture the entry response–entry is more sensitive in this model, in part because margins are tighter for entrepreneurs that have additional expenses (rental housing and replacement of depreciated housing).

Employment and output track the main experiment fairly well during the early stages of the recession, but the downturn is ultimately deeper and more persistent. The most interesting deviation is productivity, which falls on impact due to a stronger distortionary effect working through the more active collateral constraint. This difference is subtle, as all deviations are less than half a percentage point of TFP.

Calibration: Endogenous Housing					
Parameter	**Value**	**Target Description**	**Data**	**Model**	**Source**
Household					
Targeted					
ν_l	0.551	Entry Rate	0.1	0.1	BDS (2004-2006 average)
ϕ_d	2.89	Mtg. debt-to-housing	0.39	0.42	FF (2004-2006 average)
ν_h	1.55	Owner-occupied share	0.685	0.685	Census (2002-2007 average)
ϵ_h	0.792	q_h/R_h	12	12	Garner and Verbrugge (2009)
Selected					
ρ	0.04	Deposit Rate	4%	4%	–
δ_h	0.03	Housing Depreciation	–	–	BEA
Production					
Targeted					
α	0.370	Entrep. Sector Labor Share	0.61	0.55	BEA (2004-2006 average)
γ	0.91	Markups	0.1	0.1	Basu and Fernald (1997)
σ_z	0.035	IQR of TFPR	0.5	0.51	Chow et al. (2021)
ϕ	0.49	S.D. of (I/K)	0.337	0.339	Cooper and Haltiwanger (2006)
ξ	1.18	Corporate Debt-to-GDP	0.418	0.417	FF/ BEA (2004-2006 average)
d_{ent}	1.82	Emp. Entry Rate	0.024	0.012	BDS (2004-2006 average)
$\overline{m}$	0.05	Entrep. to Pop. Share	0.05	0.05	GEM (2004-2006 average)
c_{exit}	0.11	Mean Emp.	22.3	23.4	BDS (2005)
c_{ent}	26.4	EPOP	0.628	0.628	BLS (2004-2006 average)
Selected					
μ_z	0.151	Prod. Persistence	0.151	0.151	Asker, Collard-Wexler, and Loecker (2014)
δ	0.069	Capital Depreciation	0.069	0.069	Cooper and Haltiwanger (2006)
ζ	0.067	Nondistressed Exit	0.67	0.67	Headd (2003)
σ_h	1.5	Elasticity of Housing to price	2/3	2/3	Albouy, Ehrlich, and Liu (2016)
η	0.2	Elasticity of Entry	0.076	0.074	See Section 5
Banking					
Targeted					
θ	0.165	Leverage	10	10	FDIC (2005)
ψ	0.0147	Credit Spread	1%-4%	1.8%	ICE BofA
Selected					
σ	0.08	Death Rate	8%	8%	BDS (2005-2006 average)
ω_b	0.1	Entrant size/Mean Size	–	–	–

Note: Data sourced from U.S. Bureau of Economic Analysis (2004-2006), Board of Governors of the Federal Reserve System (US) (2004-2006a), Board of Governors of the Federal Reserve System (US) (2004-2006b), US Census Bureau (2005-2012), and Bureau of Labor Statistics (2007-2012) are denoted as such. Data from the BEA, FF, and BLS were accessed through FRED. FDIC targets are based on my tabulations from state bank data Federal Deposit Insurance Corporation, 2005-2012. ICE Data Indices, LLC (2004-2006a) and ICE Data Indices, LLC (2004-2006b) data from FRED displays range from "Investment Grade" options-adjusted spreads to "high-yield." Flow of Funds mortgages reflect all single-family mortgages, and housing from the flow of funds includes all household owner-occupied real estate.

Figure OA24: Aggregate Results: Endogenous Housing



Note: Measures are relative to steady state values.

F.3 Elastic Housing Supply

In the main section, we considered housing supply that was perfectly inelastic. In this appendix, I consider a case where $\gamma_H = 0$ and $\Upsilon = 2.43$ is picked so that the elasticity of housing supply is 0.7, roughly the mean of the housing supply elasticities in my data.[46] $\kappa_H = 189$ is picked so that $H_{new} = \delta_h$ in steady state.

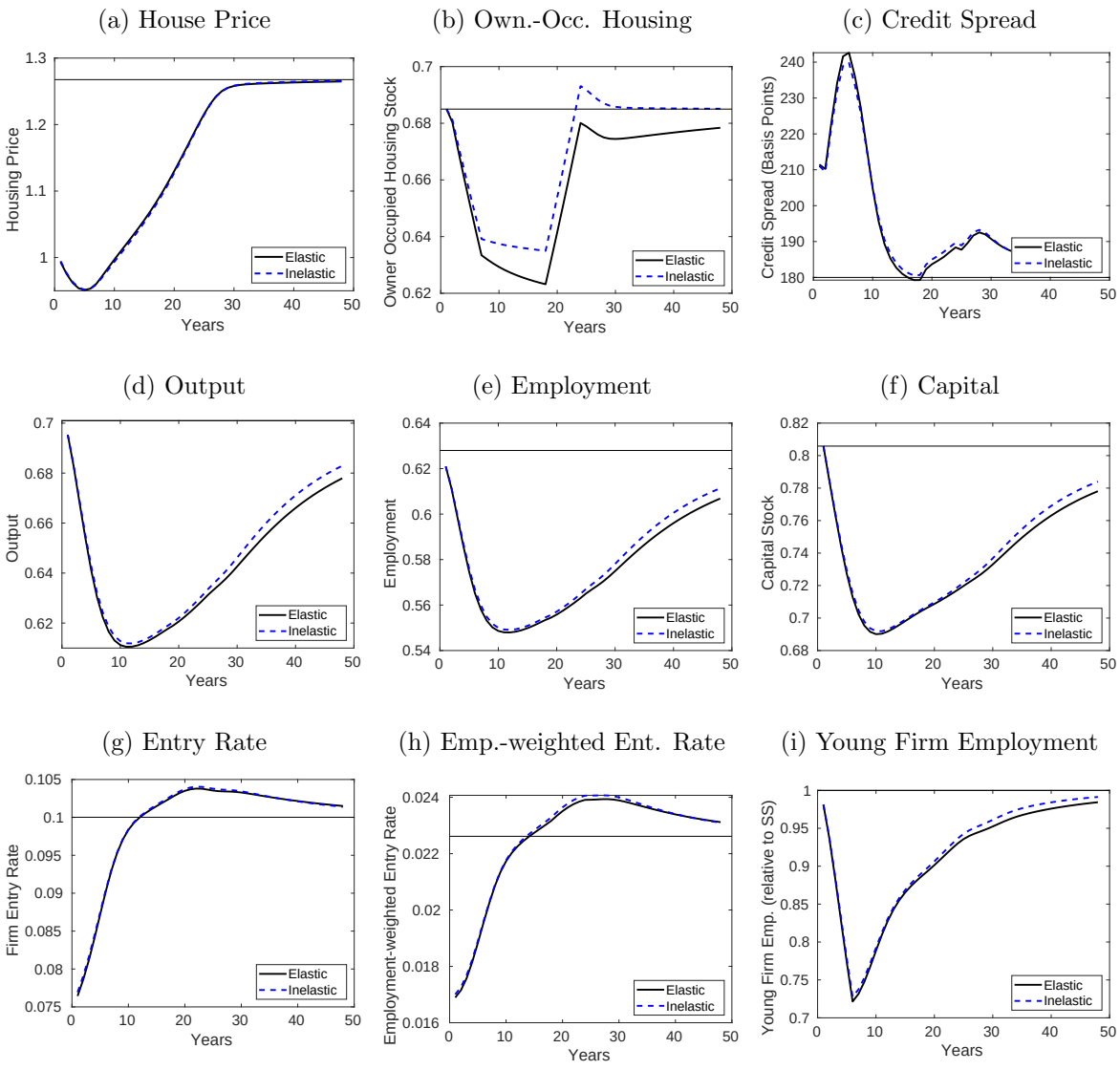
The results are very similar to the main analysis. One difference here is that the owner-occupied hosuing impulse response falls more sharply and remains low in panel (b) of Figure OA25 since overall housing supply has responded. Overall housing supply falls more than 12 percent in the model. This likely overstates the impact on housing. Dividing BEA data on total housing units by the Census Population estimate (since this model is detrended) suggests a decline of around 2 percent in the housing stock per capita.[47]

This decline in housing constricts housing collateral values more substantially, and leads to a more persistent effect. However, the effects are not substatially different on house prices or employment outcomes.

[46] An alternative would be to calibrate Υ so the implied decline in housing stock is equivalent to that in the US during the great recession.

[47] U.S. Census Bureau, Housing Inventory Estimate: Total Housing Units in the United States [ETOTALUSQ176N], retrieved from FRED, Federal Reserve Bank of St. Louis; https://fred.stlouisfed.org/series/ETOTALUSQ176N. U.S. Bureau of Economic Analysis, Population [POPTHM], retrieved from FRED, Federal Reserve Bank of St. Louis; https://fred.stlouisfed.org/series/POPTHM.

Figure OA25: Housing Crisis: Elastic Housing Supply

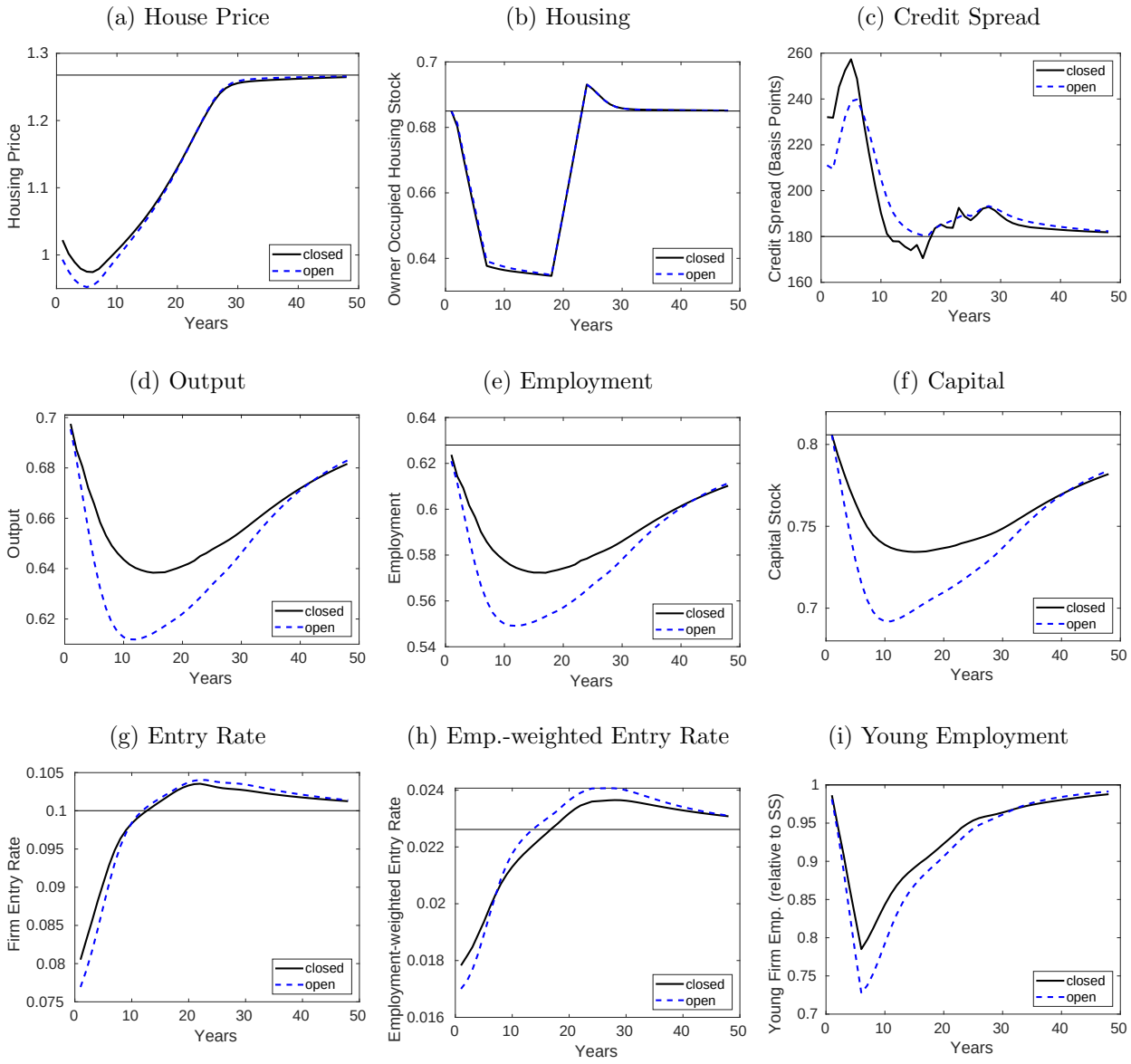


F.4 Closed Economy

In this section, I solve a version where the deposit rate is determined endogenously. Due to increased computational requirments, I solve it on a sparser grid, so comparisions are necessarily general and qualitative in nature.

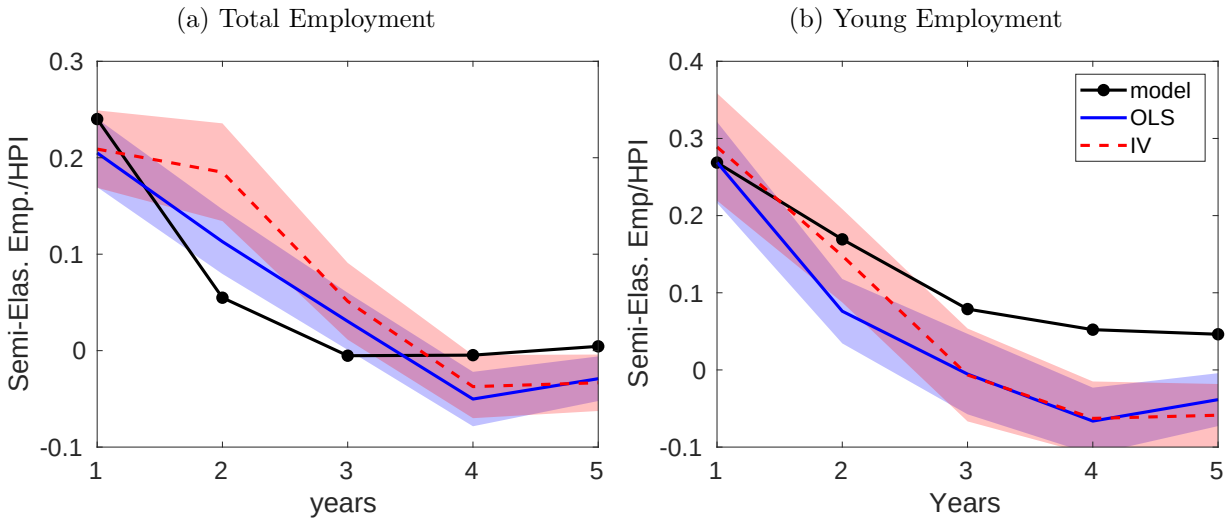
In the closed economy version, displayed in Figure OA26, the result of the same shock to housing preferences is similar along many dimensions. House prices fall roughly equivalently (although perhaps a bit more sharply), and the financial crisis is similar in that leverage rises sharply, although rates do not rise by as much since deposit rates fall. Overall, this translates into a similar performance of overall economic outcomes.

Figure OA26: Housing Crisis: Aggregate Impulse Responses (Closed Economy)



F.5 Model Evaluation

Figure OA27: Model and Data: Semi-elasticity of Employment to House Prices



Displays model semielasticity of employment to house prices ($gr_{t+h}/\%\Delta HPI_t$) where gr_{t+h} is the DHS growth rate, and β_h in equation (1)- semi-elasticity of employment at $t+h$ on house prices in t. Dependent variable is the DHS employment growth rate regressed on log changes in metro house price growth. Note: Young < 5yo, Old>=5 yo, Small <500 employees; controls include log(DHS denominator), metro area effects, industry effects, (region × year) effects, and changes in metro area unemployment rates. IV regressions indicate house prices instrumented by changes in national house prices interacted with metro housing supply elasticities from Guren et al. (2021). Area bands indicate $1.645 * SE$ of the estimated effect. Includes entrants in each year $t+h$, where post-shock entrants are given their initial size for the DHS denominator. Standard error clustered at the metro level. Disclaimer: Any opinions or conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7518031: CBDRB-FY22-CES004-039, CBDRB-FY22-CES004-044, CBDRB-FY24-CED006-0022, CBDRB-FY25-CED006-013).

To evaluate the model in light of my empirical work, I explore a shock to house prices in line with the variation observed in the data used to inform the empirical results. The standard deviation of house price changes in the Freddie Mac HPI during this period is around 5.5 percent, which also happens to be roughly equivalent to the average annual decline in house prices from 2007-2012. Thus, I feed a shock into the model that generates this initial annual change in house prices, and then compare the semi-elasticity of employment

to house prices.[48]

Figure OA27 shows this comparison. Generally, the model performs well relative to the OLS estimates, lying within the 90 percent confidence intervals for the first two years, although it is somewhat more persistent than the empirical results. In particular, the initial response matches well, and lies within the 90 percent confidence interval for the IV as well. Likewise, young employment matches the initial response well, and has a similar shape, but is more persistent than in the data. This is perhaps due to the persistence of the shock in the model which, for example, keeps entry rates relatively low.

[48] As with the shock described in Section 3.6, I shock overall housing tastes ν_h and owner-occupied housing tastes ϵ_h. In this experiment, the shock to housing tastes is a one-time jump that decays over the course of 10 years before reverting to the steady state value. As before, interest rates also rise in response, so Figure OA27 reflects interest rates and collateral effects.